Book Crunch- 2

Physics

Problems in General Physics

MECHANICS

1. Two particles, 1 and 2, move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their radius vectors are equal to $\vec{r}_1$ and $\vec{r}_2$. How must these four vectors be interrelated for the particles to collide?

2. Two boats, *A* and *B*, move away from a buoy anchored at the middle of a river along the mutually perpendicular straight lines: the boat *A* along the river, and the boat *B* across the river. Having moved off an equal distance from the buoy the boats returned. Find the ratio of times of motion of boats τ_A/τ_B if the velocity of each boat with respect to water is $\eta = 1.2$ times greater than the stream velocity.

3. Three points are located at the vertices of an equilateral triangle whose side equals *a*. They all start moving simultaneously with velocity *v* constant in modulus, with the first point heading continually for the second, the second for the third, and the third for the first. How soon will the points converge?

4. Point *A* moves uniformly with velocity *v* so that the vector $\vec{v}$ is continually "aimed" at point *B* which in its turn moves rectilinearly and uniformly with velocity $u < v$. At the initial moment of time $\vec{v} \perp \vec{u}$ and the points are separated by a distance ℓ. How soon will the points converge?

5. From point *A* located on a highway (Fig. 1) one has to get by car as soon as possible to point *B* located in the field at a distance ℓ from the highway. It is known that the car moves in the field η times slower than on the highway. At what distance from point *D* one must turn off the highway?

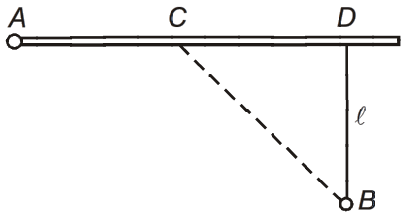


Fig. 1

6. A particle moves in the plane *xy* with velocity $\vec{v} = a\hat{i} + bx\hat{j}$, where $\hat{i}$ and $\hat{j}$ are the unit vectors of the *x* and *y* axes, and *a* and *b* are constants. At the initial moment of time the particle was located at the point $x = y = 0$. Find:

   (a) the equation of the particle's trajectory $y(x)$;

   (b) the curvature radius of trajectory as a function of *x*.

7. A particle moves along the plane trajectory $y(x)$ with velocity *v* whose modulus is constant. Find the acceleration of the particle at the point $x = 0$ and the curvature radius of the trajectory at that point if the trajectory has the form

   (a) of a parabola $y = ax^2$;

   (b) of an ellipse $(x/a)^2 + (y/b)^2 = 1$; *a* and *b* are constants here.

8. A rotating disc (Fig. 2) moves in the positive direction of the x axis. Find the equation $y(x)$ describing the position of the instantaneous axis of rotation, if at the initial moment the axis C of the disc was located at the point O after which it moved.

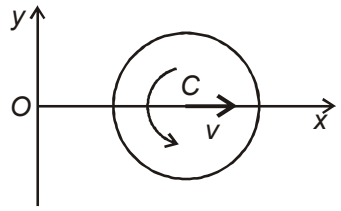


Fig. 2

(a) with a constant velocity v, while the disc started rotating counterclockwise with a constant angular acceleration β (the initial angular velocity is equal to zero);

(b) with a constant acceleration w (and the zero initial velocity), while the disc rotates counterclockwise with a constant angular velocity ω.

9. A solid body rotates with angular velocity $\vec{\omega} = at\hat{i}\ bt^2\hat{j}$, where a = 0.50 rad/s^2, b = 0.060 rad/s^3, and $\hat{i}$ and $\hat{j}$ are the unit vectors of the x and y axes. Find:

(a) the moduli of the angular velocity and the angular acceleration at the moment t = 10.0 s;

(b) the angle between the vectors of the angular velocity and the angular acceleration at that moment.

10. A solid body rotates with a constant angular velocity ω_0 = 0.50 rad/s about a horizontal axis AB. At the moment t = 0 the axis AB starts turning about the vertical with a constant angular acceleration β_0 = 0.10 rad/s^2. Find the angular velocity and angular acceleration of the body after t = 3.5 s.

11. A plank of mass m_1 with a bar of mass m_2 placed on it lies on a smooth horizontal plane. A horizontal force growing with time t as $F = at$ (a is constant) is applied to the bar. Find how the accelerations of the plank w_1 and of the bar w_2 depend on t, if the coefficient of friction between the plank and the bar is equal to k. Draw the approximate plots of these dependences.

12. In the arrangement shown in Fig. 3 the bodies have masses m_0, m_1, m_2, the friction is absent, the masses of the pulleys and the threads are negligible. Find the acceleration of the body m_1. Look into possible cases.

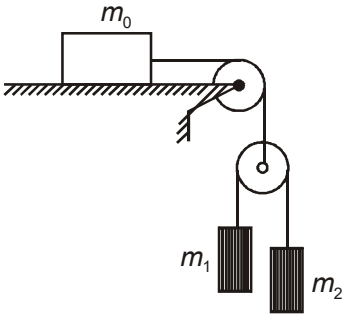


Fig. 3

13. What is the minimum acceleration with which bar A (Fig. 4) should be shifted horizontally to keep bodies 1 and 2 stationary relative to the bar? The masses of the bodies are equal, and the coefficient of friction between the bar and the bodies is equal to k. The masses of the pulley and the threads are negligible, the friction in the pulley is absent.

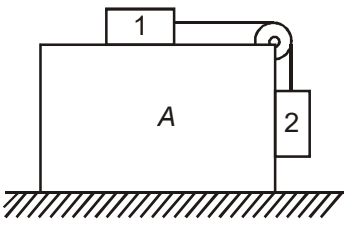


Fig. 4

14. In the arrangement shown in Fig. 5 the masses m of the bar and M of the wedge, as well as the wedge angle α, are known.

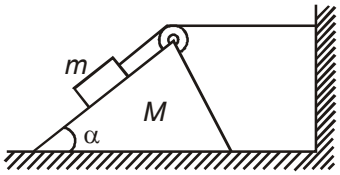


Fig. 5

The masses of the pulley and the thread are negligible. The friction is absent. Find the acceleration of the wedge M.

15. A device (Fig. 6) consists of a smooth L-shaped rod located in a horizontal plane and a sleeve A of mass m attached by a weightless spring to a point B. The spring stiffness is equal to χ. The whole system rotates with a constant angular velocity ω about a vertical axis passing through the point O. Find the elongation of the spring. How is the result affected by the rotation direction?

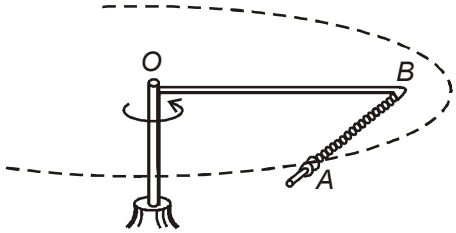


Fig. 6

16. At the moment $t = 0$ a particle of mass m starts moving due to a force $\vec{F} = \vec{F}_0 \cos \omega t$, where $\vec{F}_0$ and ω are constants. How long will it be moving until it stops for the first time? What distance will it traverse during that time? What is the maximum velocity of the particle over this distance?

17. A small disc A is placed on an inclined plane forming an angle α with the horizontal (Fig. 7) and is imparted an initial velocity v_0. Find how the velocity of the disc depends on the angle φ if the friction coefficient $k = \tan \alpha$ and at the initial moment $\varphi_0 = \frac{\pi}{2}$.

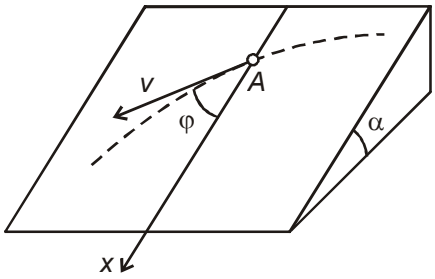


Fig. 7

18. A particle moves in a plane under the action of a force which is always perpendicular to the particle's velocity and depends on a distance to a certain point on the plane as $1/r^n$, where n is a constant. At what value of n will the motion of the particle along the circle be *steady*?

19. A horizontal plane supports a stationary vertical cylinder of radius R and a disc A attached to the cylinder by a horizontal thread AB of length ℓ_0 (Fig. 8, top view). An initial velocity v_0 is imparted to the disc as shown in the figure. How long will it move along the plane until it strikes against the cylinder? The friction is assumed to be absent.

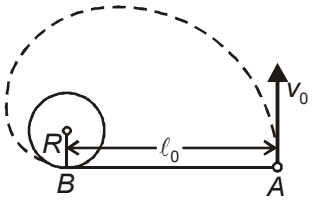


Fig. 8

20. A horizontal plane supports a plank with a bar of mass m = 1.0 kg placed on it and attached by a light elastic non-deformed cord of length ℓ_0 = 40 cm to a point O (Fig. 9). The coefficient of friction between the bar and the plank equals k = 0.20. The plank is slowly shifted to the right until the bar starts sliding over it. It occurs at the moment when the cord deviates from the vertical by an angle θ = 30°. Find the work that has been performed by that moment by the friction force acting on the bar in the reference frame fixed to the plane.

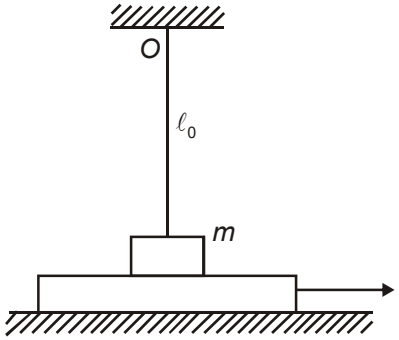


Fig. 9

21. A closed chain A of mass m = 0.36 kg is attached to a vertical rotating shaft by means of a thread (Fig. 10), and rotates with a constant angular velocity ω = 35 rad/s. The thread forms an angle θ = 45° with the vertical. Find the distance between the chain's centre of gravity and the rotation axis, and the tension of the thread.

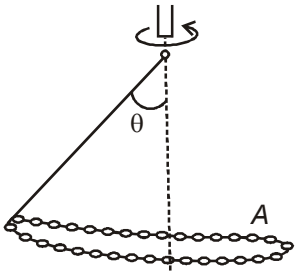


Fig. 10

22. A system consists of two identical cubes, each of mass m, linked together by the compressed weightless spring of stiffness χ (Fig. 11). The cubes are also connected by a thread which is burned through at a certain moment. Find:

    (a) at what values of $\Delta\ell$, the initial compression of the spring, the lower cube will bounce up after the thread has been burned through:

    (b) to what height h the centre of gravity of this system will rise if the initial compression of the spring $\Delta\ell = 7\ mg/\chi$.

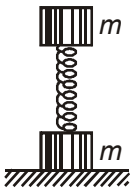


Fig. 11

23. A small disc of mass m slides down a smooth hill of height h without initial velocity and gets onto a plank of mass M lying on the horizontal plane at the base of the hill (Fig. 12). Due to friction between the disc and the plank the disc slows down and, beginning with a certain moment, moves in one piece with the plank.

    (a) Find the total work performed by the friction forces in this process.

    (b) Can it be stated that the result obtained does not depend on the choice of the reference frame?

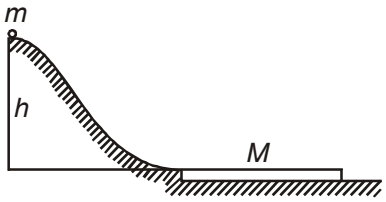


Fig. 12

24. A particle of mass m_1 collides elastically with a stationary particle of mass m_2 ($m_1 > m_2$). Find the maximum angle through which the striking particle may deviate as a result of the collision.

25. A molecule collides with another stationary molecule of the same mass. Demonstrate that the angle of divergence

    (a) equals 90° when the collision is ideally elastic;

    (b) differs from 90° when the collision is inelastic.

26. A chain AB of length ℓ is located in a smooth horizontal tube so that its fraction of length h hangs freely and touches the surface of the table with its end B (Fig. 13). At a certain moment the end A of the chain is set free. With what velocity will this end of the chain slip out of the tube?

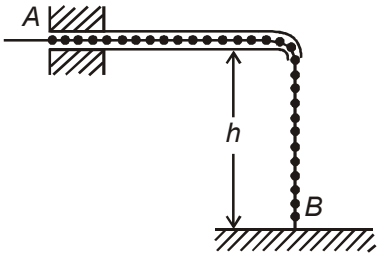


Fig. 13

27. A disc A of mass m sliding over a smooth horizontal surface with velocity v experiences a perfectly elastic collision with a smooth stationary wall at a point O (Fig. 14). The angle between the motion direction of the disc and the normal of the wall is equal to α. Find:

    (a) the points relative to which the angular momentum $\vec{M}$ of the disc remains constant in this process;

    (b) the magnitude of the increment of the vector of the disc's angular momentum relative to the point O' which is located in the plane of the disc's motion at the distance ℓ from the point O.

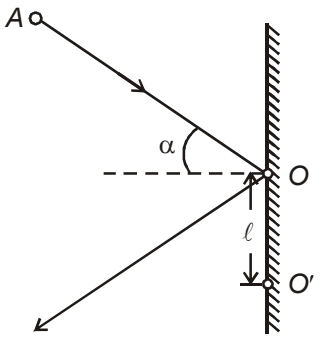


Fig. 14

28. A small body of mass m tied to a non-stretchable thread moves over a smooth horizontal plane. The other end of the thread is being drawn into a hole O (Fig. 15) with a constant velocity. Find the thread tension as a function of the distance r between the body and the hole if at $r = r_0$ the angular velocity of the thread is equal to ω_0.

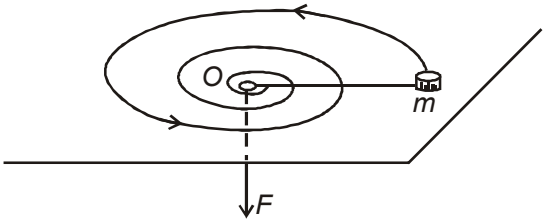


Fig. 15

29. A cosmic body A moves to the Sun with velocity v_0 (when far from the Sun) and aiming parameter ℓ the arm of the vector $\vec{v}_0$ relative to the centre of the Sun (Fig. 16). Find the minimum distance by which this body will get to the Sun.

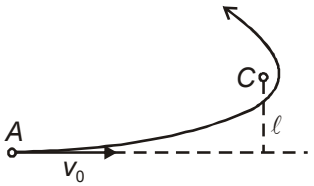


Fig. 16

30. Two Earth's satellites move in a common plane along circular orbits. The orbital radius of one satellite r = 7000 km while that of the other satellite is Δr = 70 km less. What time interval separates the periodic approaches of the satellites to each other over the minimum distance?

31. An artificial satellite of the Moon revolves in a circular orbit whose radius exceeds the radius of the Moon η times. In the process of motion the satellite experiences a slight resistance due to cosmic dust. Assuming the resistance force to depend on the velocity of the satellite as $F = \alpha v^2$, where α is a constant, find how long the satellite will stay in orbit until it falls onto the Moon's surface.

32. Find approximately the third cosmic velocity v_3, i.e. the minimum velocity that has to be imparted to a body relative to the Earth's surface to drive it out of the Solar system. The rotation of the Earth about its own axis is to be neglected.

33. A thin horizontal uniform rod AB of mass m and length ℓ can rotate freely about a vertical axis passing through its end A. At a certain moment the end B starts experiencing a constant force F which is always perpendicular to the original position of the stationary rod and directed in a horizontal plane. Find the angular velocity of the rod as a function of its rotation angle φ counted relative to the initial position.

34. A uniform cylinder of radius R is spinned about its axis to the angular velocity ω_0 and then placed into a corner (Fig. 17).

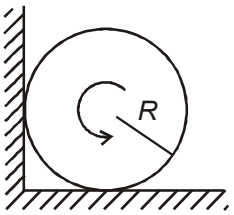


Fig. 17

The coefficient of friction between the corner walls and the cylinder is equal to k. How many turns will the cylinder accomplish before it stops?

35. A uniform solid cylinder of mass m rests on two horizontal planks. A thread is wound on the cylinder. The hanging end of the thread is pulled vertically down with a constant force F (Fig. 18).

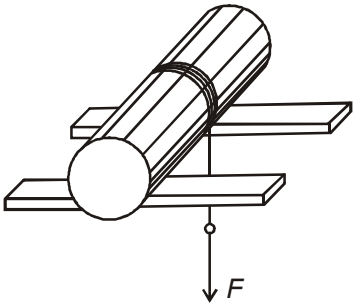


Fig. 18

Find the maximum magnitude of the force F which still does not bring about any sliding of the cylinder, if the coefficient of friction between the cylinder and the planks is equal to k. What is the acceleration w_{max} of the axis of the cylinder rolling down the inclined plane?

36. A spool with thread wound on it, of mass m, rests on a rough horizontal surface. Its moment of inertia relative to its own axis is equal to $I = \gamma mR^2$, where γ is a numerical factor, and R is the outside radius of the spool. The radius of the wound thread layer is equal to r. The spool is pulled without sliding by the thread with a constant force $\vec{F}$ directed at an angle α to the horizontal (Fig. 19). Find:

(a) the projection of the acceleration vector of the spool axis on the x-axis;

(b) the work performed by the force $\vec{F}$ during the first t seconds after the beginning of motion.

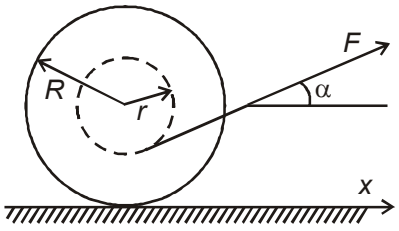


Fig. 19

37. In the arrangement shown in Fig. 20 a weight A possesses mass m, a pulley B possesses mass M. Also known are the moment of inertia I of the pulley relative to its axis and the radii of the pulley R and $2R$. The mass of the threads is negligible. Find the acceleration of the weight A after the system is set free.

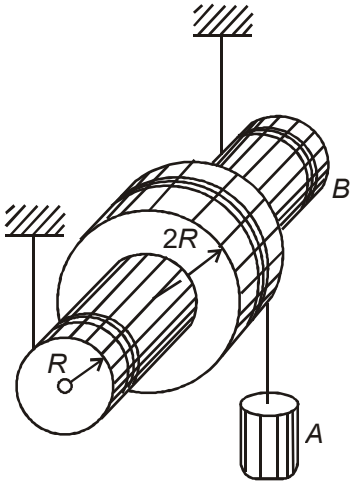


Fig. 20

38. A uniform solid cylinder of radius R = 15 cm rolls over a horizontal plane passing into an inclined plane forming an angle $\alpha = 30°$ with the horizontal (Fig. 21). Find the maximum value of the velocity v_0 which still permits the cylinder to roll onto the inclined plane section without a jump. The sliding is assumed to be absent.

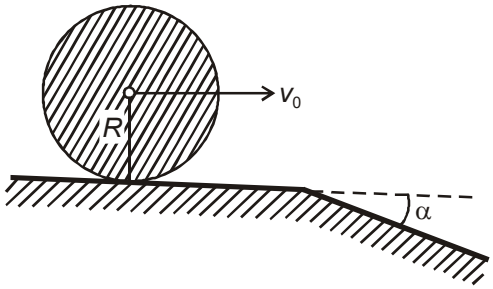


Fig. 21

39. A uniform sphere of mass m and radius r rolls without sliding over a horizontal plane, rotating about a horizontal axle OA (Fig. 22). In the process, the centre of the sphere moves with velocity v along a circle of radius R. Find the kinetic energy of the sphere.

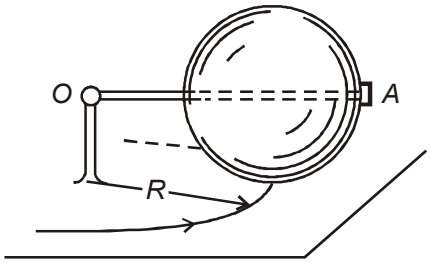


Fig. 22

40. A midpoint of a thin uniform rod AB of mass m and length ℓ is rigidly fixed to a rotation axle OO' as shown in Fig. 23. The rod is set into rotation with a constant angular velocity ω. Find the resultant moment of the centrifugal forces of inertia relative to the point C in the reference frame fixed to the axle OO' and to the rod.

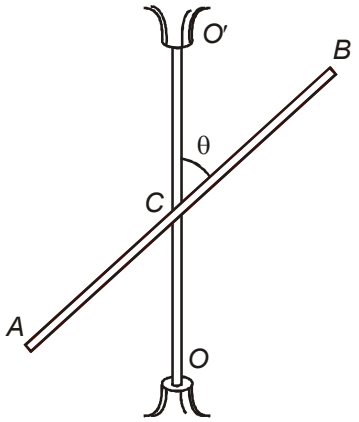


Fig. 23

41. Determine the relationship between the torque N and the torsion angle φ for

    (a) the tube whose wall thickness Δr is considerably less than the tube radius;

    (b) for the solid rod of circular cross-section. Their length ℓ, radius r, and shear modulus G are supposed to be known.

42. Calculate the torque N twisting a steel tube of length ℓ = 3.0 m through an angle φ = 2.0° about its axis, if the inside and outside diameters of the tube are equal to d_1 = 30 mm and d_2 = 50 mm.

43. Ideal fluid flows along a flat tube of constant cross-section, located in a horizontal plane and bent as shown in Fig. 24 (top view). The flow is steady. Are the pressures and velocities of the fluid equal at points 1 and 2? What is the shape of the streamlines?

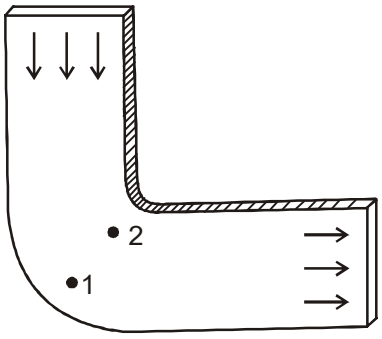


Fig. 24

44. A side wall of a wide open tank is provided with a narrowing tube (Fig. 25) through which water flows out. The cross-sectional area of the tube decreases from S = 3.0 cm^2 to s = 1.0 cm^2. The water level in the tank is h = 4.6 m higher than that in the tube.

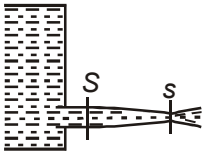


Fig. 25

Neglecting the viscosity of the water, find the horizontal component of the force tending to pull the tube out of the tank.

45. A fluid with viscosity η fills the space between two long co-axial cylinders of radii R_1 and R_2, with $R_1 < R_2$. The inner cylinder is stationary while the outer one is rotated with a constant angular velocity ω_2. The fluid flow is laminar. Taking into account that the friction force acting on a unit area of a cylindrical surface of radius r is defined by the formula $\sigma = \eta r\,(\partial\omega/\partial r)$, find:

(a) the angular velocity of the rotating fluid as a function of radius r;

(b) the moment of the friction forces acting on a unit length of the outer cylinder.

46. The cross-sectional radius of a pipeline decreases gradually as $r = r_0 e^{-\alpha x}$, where $\alpha = 0.50\ m^{-1}$, x is the distance from the pipeline inlet. Find the ratio of Reynolds numbers for two cross-sections separated by $\Delta x = 3.2$ m.

THERMODYNAMICS

1. A smooth vertical tube having two different sections is open from both ends and equipped with two pistons of different areas (Fig. 1). Each piston slides within a respective tube section. One mole of ideal gas is enclosed between the pistons tied with a non-stretchable thread. The cross-sectional area of the upper piston is $\Delta S = 10\ cm^2$ greater than that of the lower one. The combined mass of the two pistons is equal to $m = 5.0$ kg. The outside air pressure is $p_0 = 1.0$ atm. By how many kelvins must the gas between the pistons be heated to shift the pistons through $\ell = 5.0$ cm?

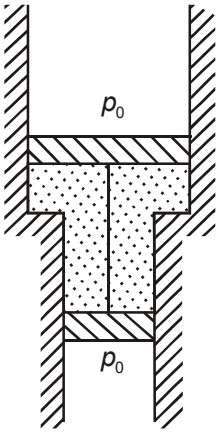


Fig. 1

2. Find the minimum attainable pressure of ideal gas in the process $T = T_0 + \alpha V^2$, where T_0 and α are positive constants, and V is the volume of one mole of gas. Draw the approximate p vs V plot of this process.

3. An ideal gas of molar mass M is located in the uniform gravitational field in which the free-fall acceleration is equal to g. Find the gas pressure as a function of height h, if $p = p_0$ at $h = 0$, and the temperature varies with height as

   (a) $T = T_0 (1 - ah)$; (b) $T = T_0 (1 + ah)$,

   where a is a positive constant.

4. Find the isothermal compressibility χ of a van der Waals gas as a function of volume V at temperature T.

   Note: By definition, $\chi = -\frac{1}{V}\frac{\partial V}{\partial p}$

5. Making use of the result obtained in the foregoing problem, find at what temperature the isothermal compressibility χ of a van der Waals gas is greater than that of an ideal gas. Examine the case when the molar volume is much greater than the parameter b.

6. Two moles of a certain ideal gas at a temperature $T_0 = 300$ K were cooled isochorically so that the gas pressure reduced $n = 2.0$ times. Then, as a result of the isobaric process, the gas expanded till its temperature got back to the initial value. Find the total amount of heat absorbed by the gas in this process.

7. Draw the approximate plots of isochoric, isobaric, isothermal, and adiabatic processes for the case of an ideal gas, using the following variables:

   (a) p, T; (b) V, T.

8. Find the rate v with which helium flows out of a thermally insulated vessel into vacuum through a small hole. The flow rate of the gas inside the vessel is assumed to be negligible under these conditions. The temperature of helium in the vessel is T = 1,000 K.

9. An ideal gas has a molar heat capacity C_V at constant volume. Find the molar heat capacity of this gas as a function of its volume V, if the gas undergoes the following process:

   (a) $T = T_0 e^{\alpha V}$;

   (b) $p = p_0 e^{\alpha V}$

   where T_0, p_0, and α are constants.

10. An ideal gas has an adiabatic exponent γ. In some process its molar heat capacity varies as $C = \frac{\alpha}{T}$, where α is a constant. Find:

    (a) the work performed by one mole of the gas during its heating from the temperature T_0 to the temperature η times higher;

    (b) the equation of the process in the variables p, V.

11. A tall vertical vessel contains a gas composed of two kinds of molecules of masses m_1 and m_2, with $m_2 > m_1$. The concentrations of these molecules at the bottom of the vessel are equal to n_1 and n_2 respectively, with $n_2 > n_1$. Assuming the temperature T and the free-fall acceleration g to be independent of the height, find the height at which the concentrations of these kinds of molecules are equal.

12. A horizontal tube with closed ends is rotated with a constant angular velocity ω about a vertical axis passing through one of its ends. The tube contains carbon dioxide at a temperature T = 300 K. The length of the tube is ℓ = 100 cm. Find the value ω at which the ratio of molecular concentrations at the opposite ends of the tube is equal to η = 2.0.

13. An ideal gas goes through a cycle consisting of alternate isothermal and adiabatic curves (Fig. 2). The isothermal processes proceed at the temperatures T_1, T_2, and T_3. Find the efficiency of such a cycle, if in each isothermal expansion the gas volume increases in the same proportion.

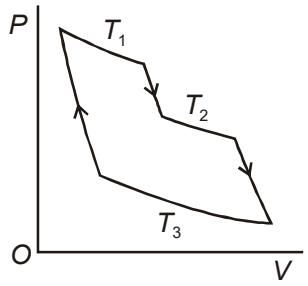


Fig. 2

14. An ideal gas goes through a cycle consisting of

    (a) isochoric, adiabatic, and isothermal lines;

    (b) isobaric, adiabatic, and isothermal lines,

    with the isothermal process proceeding at the minimum temperature of the whole cycle. Find the efficiency of each cycle if the absolute temperature varies n-fold within the cycle.

15. Find the efficiency of a cycle consisting of two isobaric and two isothermal lines if the pressure varies n-fold and the absolute temperature τ-fold within the cycle. The working substance is an ideal gas with the adiabatic exponent γ.

16. An ideal gas with the adiabatic exponent γ goes through a cycle (Fig. 3) within which the absolute temperature varies τ-fold. Find the efficiency of this cycle.

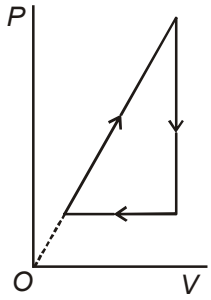


Fig. 3

17. One mole of an ideal gas with heat capacity C_V goes through a process in which its entropy S depends on T as $S = \frac{\alpha}{T}$, where α is a constant. The gas temperature varies from T_1 to T_2. Find:

    (a) the molar heat capacity of the gas as a function of its temperature;

    (b) the amount of heat transferred to the gas;

    (c) the work performed by the gas.

18. When a vertical capillary of length ℓ with the sealed upper end was brought in contact with the surface of a liquid, the level of this liquid rose to the height h. The liquid density is ρ, the inside diameter of the capillary is d, the contact angle is θ, the atmospheric pressure is p_0. Find the surface tension of the liquid.

19. Two vertical plates submerged partially in a wetting liquid form a wedge with a very small angle $\delta\varphi$. The edge of this wedge is vertical. The density of the liquid is ρ, its surface tension is α, the contact angle is θ. Find the height h, to which the liquid rises, as a function of the distance x from the edge.

20. Find the work to be performed in order to blow a soap bubble of radius R if the outside air pressure is equal to p_0 and the surface tension of the soap water solution is equal to α.

ELECTRODYNAMICS

1. A thin non-conducting ring of radius R has a linear charge density $\lambda = \lambda_0 \cos\varphi$, where λ_0 is a constant, φ is the azimuthal angle. Find the magnitude of the electric field strength

    (a) at the centre of the ring;

    (b) on the axis of the ring as a function of the distance x from its centre. Investigate the obtained function at x >> R.

2. Suppose the surface charge density over a sphere of radius R depends on a polar angle θ as $\sigma = \sigma_0 \cos\theta$, where σ_0 is a positive constant. Show that such a charge distribution can be represented as a result of a small relative shift of two uniformly charged balls of radius R whose charges are equal in magnitude and opposite in sign. Resorting to this representation, find the electric field strength vector inside the given sphere.

3. A very thin round plate of radius R carrying a uniform surface charge density σ is located in vacuum. Find the electric field potential and strength along the plate's axis as a function of a distance ℓ from its centre. Investigate the obtained expression at $\ell \to 0$ and $\ell >> R$.

4. The potential of a certain electrostatic field has the form $\varphi = a(x^2 + y^2) + bz^2$, where a and b are constants. Find the magnitude and direction of the electric field strength vector. What shape have the equipotential surfaces in the following cases:

    (a) $a > 0$, $b > 0$; (b) $a > 0$, $b < 0$?

5. Find the potential $\varphi(x, y)$ of an electrostatic field $\vec{E} = 2axy\hat{i} + a\left(x^2 - y^2\right)\hat{j}$, where a is a constant, $\hat{i}$ and $\hat{j}$ are the unit vectors of the x and y axes.

6. The field potential inside a charged ball depends only on the distance from its centre as $\varphi = ar^2 + b$, where a and b are constants. Find the space charge distribution $\rho(r)$ inside the ball.

7. A very long straight thread is oriented at right angles to an infinite conducting plane; its end is separated from the plane by a distance ℓ. The thread carries a uniform charge of linear density λ. Suppose the point O is the trace of the thread on the plane. Find the surface density of the induced charge on the plane

   (a) at the point O;

   (b) as a function of a distance r from the point O.

8. Demonstrate that at a dielectric-conductor interface the surface density of the dielectric's bound charge $\sigma' = -\sigma(\varepsilon - 1)/\varepsilon$, where ε is the permittivity, σ is the surface density of the charge on the conductor.

9. The gap between the plates of a parallel-plate capacitor is filled with isotropic dielectric whose permittivity ε varies linearly from ε_1 to ε_2 ($\varepsilon_2 > \varepsilon_1$) in the direction perpendicular to the plates. The area of each plate equals S, the separation between the plates is equal to d. Find:

   (a) the capacitance of the capacitor;

   (b) the space density of the bound charges as a function of ε if the charge of the capacitor is q and the field $\vec{E}$ in it is directed toward the growing ε values.

10. There is a double-layer cylindrical capacitor whose parameters are shown in Fig. 2. The breakdown field strength values for these dielectrics are equal to E_1 and E_2 respectively. What is the breakdown voltage of this capacitor if $\varepsilon_1 R_1 E_1 < \varepsilon_2 R_2 E_2$?

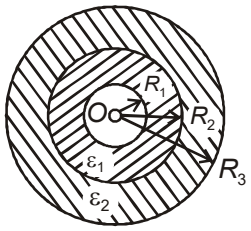


Fig. 2

11. Find the capacitance of a system of two identical metal balls of radius a if the distance between their centres is equal to b, with $b >> a$. The system is located in a uniform dielectric with permittivity ε.

12. What charges will flow after the shorting of the switch Sw in the circuit illustrated in Fig. 3 through sections 1 and 2 in the directions indicated by the arrows?

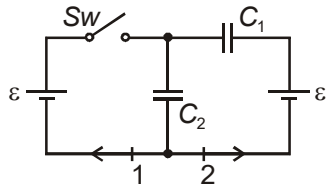


Fig. 3

13. There is an infinite straight chain of alternating charges q and $-q$. The distance between the neighbouring charges is equal to a. Find the interaction energy of each charge with all the others.

    Instruction: Make use of the expansion of $\ln(1 + \alpha)$ in a power series in α.

14. What amount of heat will be generated in the circuit shown in Fig. 4 after the switch Sw is shifted from position 1 to position 2?

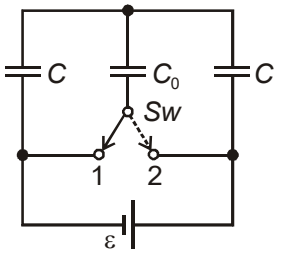


Fig. 4

15. A point charge q is located at the centre O of a spherical uncharged conducting layer provided with a small orifice (Fig. 5). The inside and outside radii of the layer are equal to a and b respectively. What amount of work has to be performed to slowly transfer the charge q from the point O through the orifice and into infinity?

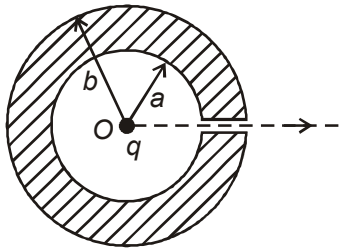


Fig. 5

16. A capacitor consists of two stationary plates shaped as a semi-circle of radius R and a movable plate made of dielectric with permittivity ε and capable of rotating about an axis O between the stationary plates (Fig. 7). The thickness of the movable plate is equal to d which is practically the separation between the stationary plates. A potential difference V is applied to the capacitor. Find the magnitude of the moment of forces relative to the axis O acting on the movable plate in the position shown in the figure.

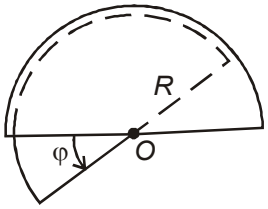


Fig. 7

17. Find the resistance of a wire frame shaped as a cube (Fig. 8) when measured between points

    (a) 1–7; (b) 1–2; (c) 1–3.

    The resistance of each edge of the frame is R

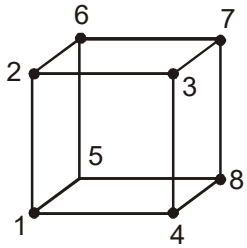


Fig. 8

18. There is an infinite wire grid with square cells (Fig. 9). The resistance of each wire between neighbouring joint connections is equal to R_0. Find the resistance R of the whole grid between points A and B.

Instruction: Make use of principles of symmetry and superposition.

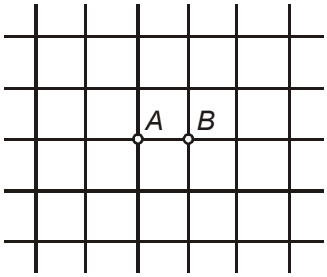


Fig. 9

19. The space between two conducting concentric spheres of radii a and b ($a < b$) is filled up with homogeneous poorly conducting medium. The capacitance of such a system equals C. Find the resistivity of the medium if the potential difference between the spheres, when they are disconnected from an external voltage, decreases η-fold during the time interval Δt.

20. Demonstrate that the law of refraction of direct current lines at the boundary between two conducting media has the form $\frac{\tan\alpha_2}{\tan\alpha_1} = \frac{\sigma_2}{\sigma_1}$, where σ_1 and σ_2 are the conductivities of the media, α_2 and α_1 are the angles between the current lines and the normal of the boundary surface.

21. Two sources of current of equal emf are connected in series and have different internal resistances R_1 and R_2. ($R_2 > R_1$). Find the external resistance R at which the potential difference across the terminals of one of the sources (which one in particular?) becomes equal to zero.

22. Find the magnitude and direction of the current flowing through the resistance R in the circuit shown in Fig. 10 if the emf's of the sources are equal to ε_1 = 1.5 V and ε_2 = 3.7 V and the resistances are equal to R_1 = 10 Ω, R_2 = 20 Ω, R = 5.0 Ω. The internal resistances of the sources are negligible.

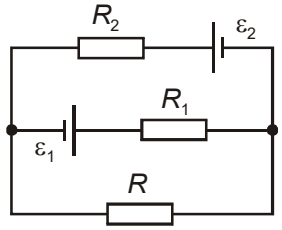


Fig. 10

23. Find the resistance between points A and B of the circuit shown in Fig. 11.

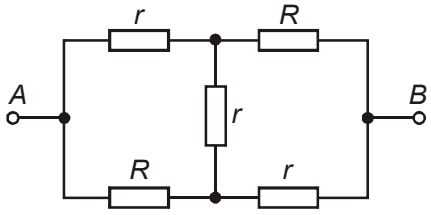


Fig. 11

24. A conductor has a temperature-independent resistance R and a total heat capacity C. At the moment t = 0 it is connected to a dc voltage V. Find the time dependence of a conductor's temperature T assuming the thermal power dissipated into surrounding space to vary as $q = k(T - T_0)$, where k is a constant, T_0 is the environmental temperature (equal to the conductor's temperature at the initial moment).

25. The radii of spherical capacitor electrodes are equal to a and b, with $a < b$. The interelectrode space is filled with homogeneous substance of permittivity ε and resistivity ρ. Initially the capacitor is not charged. At the

moment $t = 0$ the internal electrode gets a charge q_0. Find:

(a) the time variation of the charge on the internal electrode;

(b) the amount of heat generated during the spreading of the charge.

26. Two large parallel plates are located in vacuum. One of them serves as a cathode, a source of electrons whose initial velocity is negligible. An electron flow directed toward the opposite plate produces a space charge causing the potential in the gap between the plates to vary as $\varphi = ax^{4/3}$, where a is a positive constant, and x is the distance from the cathode. Find:

(a) the volume density of the space charge as a function of x;

(b) the current density.

27. A current I flows along a thin wire shaped as a regular polygon with n sides which can be inscribed into a circle of radius R. Find the magnetic induction at the centre of the polygon. Analyse the obtained expression at $n \to \infty$.

28. A direct current I flows along a lengthy straight wire. From the point O (Fig. 12) the current spreads radially all over an infinite conducting plane perpendicular to the wire. Find the magnetic induction at all points of space.

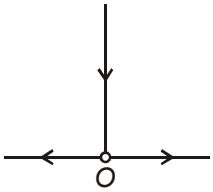


Fig. 12

29. Find the magnitude and direction of a force vector acting on a unit length of a thin wire, carrying a current $I = 8.0$ A, at a point O, if the wire is bent as shown in

(a) Fig. 13*a*, with curvature radius $R = 10$ cm;

(b) Fig. 13*b*, the distance between the long parallel segments of the wire being equal to $\ell = 20$ cm.

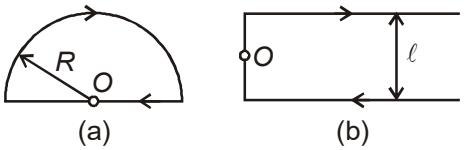


Fig. 13 *a* and *b*

30. A small coil C with $N = 200$ turns is mounted on one end of a balance beam and introduced between the poles of an electromagnet as shown in Fig. 14. The cross-sectional area of the coil is $S = 1.0$ cm^2, the length of the arm OA of the balance beam is $\ell = 30$ cm. When there is no current in the coil the balance is in equilibrium. On passing a current $I = 22$ mA through the coil the equilibrium is restored by putting the additional counterweight of mass $\Delta m = 60$ mg on the balance pan. Find the magnetic induction at the spot where the coil is located.

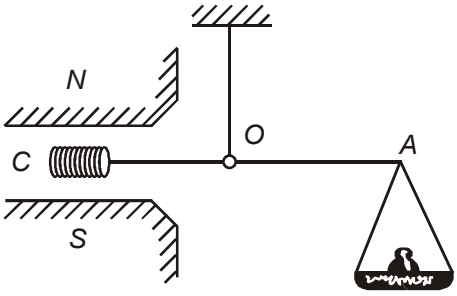


Fig. 14

31. Two long thin parallel conductors of the shape shown in Fig. 15 carry direct currents I_1 and I_2. The separation between the conductors is a, the width of the right-hand conductor is equal to b. With both conductors lying in one plane, find the magnetic interaction force between them reduced to a unit of their length.

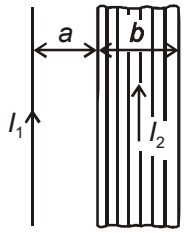


Fig. 15

32. In an electromagnetic pump designed for transferring molten metals a pipe section with metal is located in a uniform magnetic field of induction B (Fig. 16). A current I is made to flow across this pipe section in the direction perpendicular both to the vector B and to the axis of the pipe. Find the gauge pressure produced by the pump if B = 0.10 T, I = 100 A, and a = 2.0 cm.

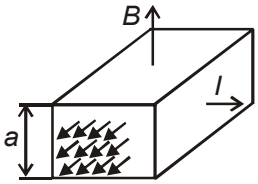


Fig. 16

33. A small current-carrying coil having a magnetic moment p_m is located at the axis of a round loop of radius R with current I flowing through it. Find the magnitude of the vector force applied to the coil if its distance from the centre of the loop is equal to x and the vector p_m, coincides in direction with the axis of the loop.

34. A metal rod of mass m can rotate about a horizontal axis O, sliding along a circular conductor of radius a (Fig. 17). The arrangement is located in a uniform magnetic field of induction B directed perpendicular to the ring plane. The axis and the ring are connected to an emf source to form a circuit of resistance R. Neglecting the friction, circuit inductance, and ring resistance, find the law according to which the source emf must vary to make the rod rotate with a constant angular velocity ω.

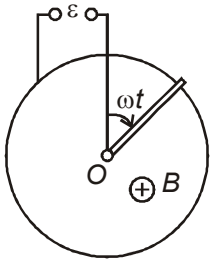


Fig. 17

35. A square wire frame with side a and a straight conductor carrying a constant current I are located in the same plane (Fig. 18).

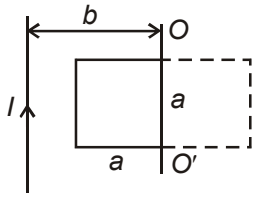


Fig. 18

The inductance and the resistance of the frame are equal to L and R respectively. The frame was turned through 180° about the axis OO' separated from the current-carrying conductor by a distance b. Find the electric charge having flown through the frame.

36. A conducting rod AB of mass m slides without friction over two long conducting rails separated by a distance ℓ (Fig. 19). At the left end the rails are interconnected by a resistance R. The system is located in a uniform magnetic field perpendicular to the plane of the loop. At the moment $t = 0$ the rod AB starts moving to the right with an initial velocity v_0. Neglecting the resistances of the rails and the rod AB, as well as the self-inductance, find:

    (a) the distance covered by the rod until it comes to a standstill;

    (b) the amount of heat generated in the resistance R during this process.

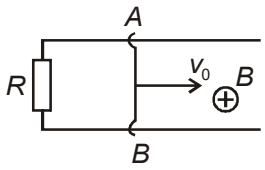


Fig. 19

37. Calculate the inductance of a doughnut solenoid whose inside radius is equal to b and cross-section has the form of a square with side a. The solenoid winding consists of N turns. The space inside the solenoid is filled up with uniform paramagnetic having permeability μ.

38. A ring of radius $a = 50$ mm made of thin wire of radius $b = 1.0$ mm was located in a uniform magnetic field with induction $B = 0.50$ mT so that the ring plane was perpendicular to the vector $\vec{B}$. Then the ring was cooled down to a superconducting state, and the magnetic field was switched off. Find the ring current after that. Note that the inductance of a thin ring along which the surface current flows is equal to $L = \mu_0 a\left(\ln\frac{8a}{b} - 2\right)$.

39. Demonstrate that the magnetic energy of interaction of two current-carrying loops located in vacuum can be represented as $W_{ia} = \left(\frac{1}{\mu_0}\right)\int \vec{B}_1 \cdot \vec{B}_2 dV$, where $\vec{B}_1$ and $\vec{B}_2$ are the magnetic inductions within a volume element dV, produced individually by the currents of the first and the second loop respectively.

40. A non-relativistic charged particle flies through the electric field of a cylindrical capacitor and gets into a uniform transverse magnetic field with induction B (Fig. 20). In the capacitor the particle moves along the arc of a circle, in the magnetic field, along a semi-circle of radius r. The potential difference applied to the capacitor is equal to V, the radii of the electrodes are equal to a and b, with $a < b$. Find the velocity of the particle and its specific charge q/m.

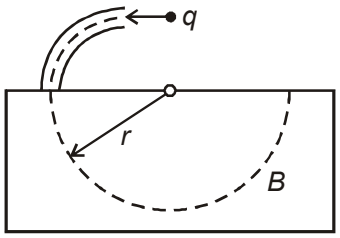


Fig. 20

41. A particle with specific charge q/m is located inside a round solenoid at a distance r from its axis. With the current switched into the winding, the magnetic induction of the field generated by the solenoid amounts to

B. Find the velocity of the particle and the curvature radius of its trajectory, assuming that during the increase of current flowing in the solenoid the particle shifts by a negligible distance.

OSCILLATIONS AND WAVES

1. A point moves along the *x* axis according to the law $x = a \sin^2\left(\omega t - \frac{\pi}{4}\right)$. Find:

   (a) the amplitude and period of oscillations; draw the plot $x(t)$;

   (b) the velocity projection v_x as a function of the coordinate *x*; draw the plot $v_x(x)$.

2. A particle performs harmonic oscillations along the *x* axis according to the law $x = a \cos \omega t$. Assuming the probability *P* of the particle to fall within an interval from $-a$ to $+a$ to be equal to unity, find how the probability density $\frac{dP}{dx}$ depends on *x*. Here *dP* denotes the probability of the particle falling within an interval from *x* to $x + dx$. Plot $\frac{dP}{dx}$ as a function of *x*.

3. Determine the period of small oscillations of a mathematical pendulum, that is a ball suspended by a thread ℓ = 20 cm in length, if it is located in a liquid whose density is η = 3.0 times less than that of the ball. The resistance of the liquid is to be neglected.

4. Determine the period of oscillations of mercury of mass *m* = 200 g poured into a bent tube (Fig. 1) whose right arm forms an angle θ = 30° with the vertical. The cross-sectional area of the tube is *S* = 0.50 cm^2. The viscosity of mercury is to be neglected.

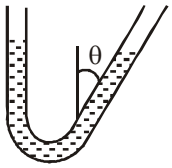


Fig. 1

5. Find the period of small oscillations of a mathematical pendulum of length ℓ if its point of suspension *O* moves relative to the Earth's surface in an arbitrary direction with a constant acceleration *w* (Fig. 2). Calculate that period if $\ell = 21\,\text{cm}$, $w = \frac{g}{2}$, and the angle between the vectors $\vec{w}$ and $\vec{g}$ equals β = 120°.

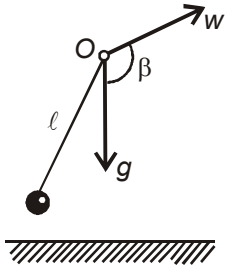


Fig. 2

6. A plank with a body of mass *m* placed on it starts moving straight up according to the law $y = a(1 - \cos \omega t)$, where *y* is the displacement from the initial position, ω = 11 s^{-1}. Find:

   (a) the time dependence of the force that the body exerts on the plank if *a* = 4.0 cm; plot this dependence;

   (b) the minimum amplitude of oscillation of the plank at which the body starts falling behind the plank;

   (c) the amplitude of oscillation of the plank at which the body springs up to a height *h* = 50 cm relative to the initial position (at the moment *t* = 0).

7. A particle of mass *m* moves in the plane *xy* due to the force varying with velocity as $\vec{F} = a\left(\dot{y}\hat{i} - \dot{x}\hat{j}\right)$, where

a is a positive constant, $\hat{i}$ and $\hat{j}$ are the unit vectors of the *x* and *y* axes. At the initial moment *t* = 0 the particle was located at the point *x* = *y* = 0 and possessed a velocity v_0 directed along the unit vector $\hat{j}$. Find the law of motion *x* (*t*) , *y* (*t*) of the particle, and also the equation of its trajectory.

8. A uniform rod of mass *m* = 1.5 kg suspended by two identical threads $\ell = 90$ cm in length (Fig. 3) was turned through a small angle about the vertical axis passing through its middle point *C*. The threads deviated in the process through an angle α = 5.0°. Then the rod was released to start performing small oscillations. Find:

   (a) the oscillation period;

   (b) the rod's oscillation energy.

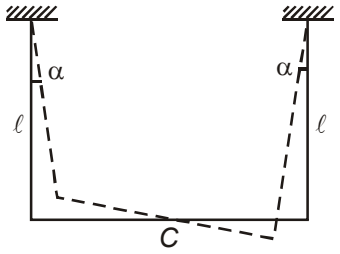


Fig. 3

9. A uniform rod of mass *m* and length ℓ performs small oscillations about the horizontal axis passing through its upper end. Find the mean kinetic energy of the rod averaged over one oscillation period if at the initial moment it was deflected from the vertical by an angle θ_0 and then imparted an angular velocity $\dot{\theta}_0$.

10. A uniform cylindrical pulley of mass *M* and radius *R* can freely rotate about the horizontal axis *O* (Fig. 4). The free end of a thread tightly wound on the pulley carries a deadweight *A*. At a certain angle α it counterbalances a point mass *m* fixed at the rim of the pulley. Find the frequency of small oscillations of the arrangement.

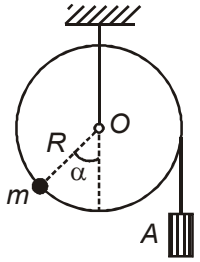


Fig. 4

11. A mock-up of a CO_2 molecule consists of three balls interconnected by identical light springs and placed along a straight line in the state of equilibrium. Such a system can freely perform oscillations of two types, as shown by the arrows in Fig. 5. Knowing the masses of the atoms, find the ratio of frequencies of these oscillations.

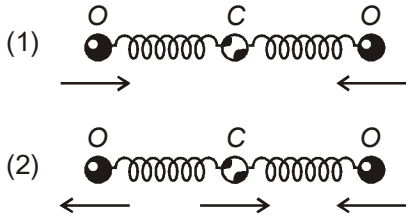


Fig. 5

12. A coil of inductance *L* connects the upper ends of two vertical copper bars separated by a distance ℓ. A horizontal conducting connector of mass *m* starts falling with zero initial velocity along the bars without losing contact with them. The whole system is located in a uniform magnetic field with induction *B* perpendicular to the plane of the bars. Find the law of motion *x* (*t*) of the connector.

13. A body performs torsional oscillations according to the law $\varphi = \varphi_0 e^{-\beta t} \cos \omega t$. Find :
    (a) the angular velocity $\dot{\varphi}$ and the angular acceleration $\ddot{\varphi}$ of the body at the moment $t = 0$;
    (b) the moments of time at which the angular velocity becomes maximum.
14. An oscillating circuit consists of a capacitor with capacitance C, a coil of inductance L with negligible resistance, and a switch. With the switch disconnected, the capacitor was charged to a voltage V_m and then at the moment $t = 0$ the switch was closed. Find:
    (a) the current $I(t)$ in the circuit as a function of time;
    (b) the emf of self-inductance in the coil at the moments when the electric energy of the capacitor is equal to that of the current in the coil.
15. The free damped oscillations are maintained in a circuit, such that the voltage across the capacitor varies as $V = V_m e^{-\beta t} \cos \omega t$. Find the moments of time when the modulus of the voltage across the capacitor reaches
    (a) peak values;
    (b) maximum (extremum) values.
16. A capacitor with capacitance C whose interelectrode space is filled up with poorly conducting medium with active resistance R is connected to a source of alternating voltage $V = V_m \cos \omega t$. Find the time dependence of the steady-state current flowing in lead wires. The resistance of the wires is to be neglected.
17. A series circuit consisting of an inductance-free resistance $R = 0.16\ \text{k}\Omega$ and a coil with active resistance is connected to the mains with effective voltage $V = 220$ V. Find the heat power generated in the coil if the effective voltage values across the resistance R and the coil are equal to $V_1 = 80$ V and $V_2 = 180$ V respectively.
18. A ring of thin wire with active resistance R and inductance L rotates with constant angular velocity ω in the external uniform magnetic field perpendicular to the rotation axis. In the process, the flux of magnetic induction of external field across the ring varies with time as $\Phi = \Phi_0 \cos \omega t$. Demonstrate that
    (a) the inductive current in the ring varies with time as $I = I_m \sin(\omega t - \varphi)$, where $I_m = \dfrac{\omega \Phi_0}{\sqrt{R^2 + \omega^2 L^2}}$ with $\tan\varphi = \dfrac{\omega L}{R}$;
    (b) the mean mechanical power developed by external forces to maintain rotation is defined by the formula
    $$P = \frac{\frac{1}{2}\omega^2 \Phi_0^2 R}{\left(R^2 + \omega^2 L^2\right)}.$$
19. Demonstrate that any differentiable function $f(t + \alpha x)$, where α is a constant, provides a solution of wave equation. What is the physical meaning of the constant α?
20. A plane wave $\xi = a \cos(\omega t - kx)$ propagates in a homogeneous elastic medium. For the moment $t = 0$ draw
    (a) the plots of ξ, $\dfrac{\partial \xi}{\partial t}$, and $\dfrac{\partial \xi}{\partial x}$ vs x;
    (b) the velocity direction of the particles of the medium at the points where $\xi = 0$, for the cases of longitudinal and transverse waves;
    (c) the approximate plot of density distribution $\rho(x)$ of the medium for the case of longitudinal waves.
21. The equation of a plane standing wave in a homogeneous elastic medium has the form $\xi = a \cos kx \cdot \cos \omega t$. Plot:

(a) ξ and $\frac{\partial \xi}{\partial x}$ as functions of x at the moments $t = 0$ and $t = \frac{T}{2}$, where T is the oscillation period;

(b) the distribution of density $\rho(x)$ of the medium at the moments $t = 0$ and $t = \frac{T}{2}$ in the case of longitudinal oscillations;

(c) the velocity distribution of particles of the medium at the moment $t = \frac{T}{4}$; indicate the directions of velocities at the antinodes, both for longitudinal and transverse oscillations.

OPTICS

1. Write the law of reflection of a light beam from a mirror in vector form, using the directing unit vectors e and e' of the incident and reflected beams and the unit vector n of the outside normal to the mirror surface.

2. Determine the focal length of a concave mirror if:

   (a) with the distance between an object and its image being equal to $\ell = 15$ cm, the transverse magnification $\beta = -2.0$;

   (b) in a certain position of the object the transverse magnification is $\beta_1 = -0.50$ and in another position displaced with respect to the former by a distance $\ell = 5.0$ cm the transverse magnification $\beta_2 = -0.25$.

3. A parallel beam of light falls from vacuum on a surface enclosing a medium with refractive index n (Fig. 1). Find the shape of that surface, x (r), if the beam is brought into focus at the point F at a distance f from the crest O. What is the maximum radius of a beam that can still be focussed?

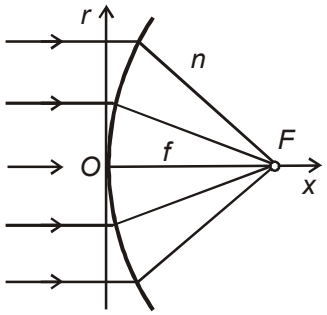


Fig. 1

4. There are two thin symmetrical lenses: one is converging, with refractive index $n_1 = 1.70$, and the other is diverging with refractive index $n_2 = 1.51$. Both lenses have the same curvature radius of their surfaces equal to $R = 10$ cm. The lenses were put close together and submerged into water. What is the focal length of this system in water?

5. On passing through a telescope a flux of light increases its intensity $\eta = 4.0 \times 10^4$ times. Find the angular dimension of a distant object if its image formed by that telescope has an angular dimension $\psi' = 2.0°$.

6. A system illustrated in Fig. 2 consists of two coherent point sources 1 and 2 located in a certain plane so that their dipole moments are oriented at right angles to that plane. The sources are separated by a distance d, the radiation wavelength is equal to λ. Taking into account that the oscillations of source 2 lag in phase behind the oscillations of source 1 by φ ($\varphi < \pi$), find:

   (a) the angles θ at which the radiation intensity is maximum;

   (b) the conditions under which the radiation intensity in the direction $\theta = \pi$ is maximum and in the opposite direction, minimum.

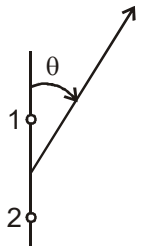


Fig. 2

7. In Lloyd's mirror experiment (Fig. 3) a light wave emitted directly by the source *S* (narrow slit) interferes with the wave reflected from a mirror *M*. As a result, an interference fringe pattern is formed on the screen *Sc*. The source and the mirror are separated by a distance ℓ = 100 cm. At a certain position of the source the fringe width on the screen was equal to Δx = 0.25 mm, and after the source was moved away from the mirror plane by Δh = 0.60 mm, the fringe width decreased η = 1.5 times. Find the wavelength of light.

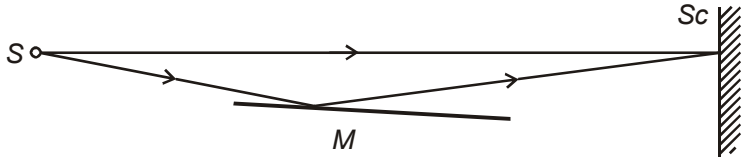


Fig. 3

8. A plane light wave with wavelength λ = 0.70 μm falls normally on the base of a biprism made of glass (n = 1.520) with refracting angle θ = 5.0°. Behind the biprism (Fig. 5) there is a plane-parallel plate, with the space between them filled up with benzene (n' = 1.500). Find the width of a fringe on the screen *Sc* placed behind this system.

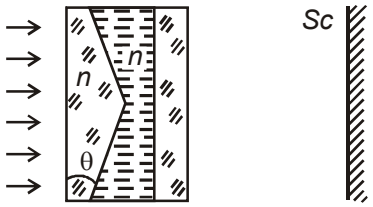


Fig. 5

9. Figure 6 illustrates an interferometer used in measurements of refractive indices of transparent substances. Here *S* is a narrow slit illuminated by monochromatic light with wavelength λ = 589 nm, 1 and 2 are identical tubes with air of length ℓ = 10.0 cm each, *D* is a diaphragm with two slits. After the air in tube 1 was replaced with ammonia gas, the interference pattern on the screen *Sc* was displaced upward by N = 17 fringes. The refractive index of air is equal to n = 1.000277. Determine the refractive index of ammonia gas.

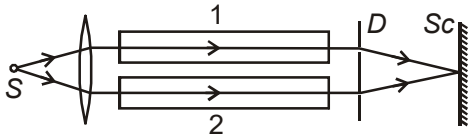


Fig. 6

10. Two plano-convex thin glass lenses are brought into contact with their spherical surfaces. Find the optical power of such a system if in reflected light with wavelength λ = 0.60 μm the diameter of the fifth bright ring is d = 1.50 mm.

11. The spherical surface of a plano-convex lens comes into contact with a glass plate. The space between the lens and the plate is filled up with carbon dioxide. The refractive indices of the lens, carbon dioxide, and the plate are equal to n_1 = 1.50, n_2 = 1.63, and n_3 = 1.70 respectively. The curvature radius of the spherical surface of the lens is equal to R = 100 cm. Determine the radius of the fifth dark Newton's ring in reflected light with wavelength λ = 0.50 μm.

12. A plane monochromatic light wave falls normally on a round aperture. At a distance b = 9.0 m from it there is a screen showing a certain diffraction pattern. The aperture diameter was decreased η = 3.0 times. Find the new distance b' at which the screen should be positioned to obtain the diffraction pattern similar to the previous one but diminished η times.

ATOMIC AND NUCLEAR PHYSICS

1. To what minimum distance will an alpha particle with kinetic energy T = 0.40 MeV approach in the case of a head-on collision to

   (a) a stationary Pb nucleus;

   (b) a stationary free Li^7 nucleus?

2. An alpha particle with kinetic energy T = 0.50 MeV is deflected through an angle of $\theta = 90°$ by the Coulomb field of a stationary Hg nucleus. Find:

   (a) the least curvature radius of its trajectory;

   (b) the minimum approach distance between the particle and the nucleus.

3. A particle with kinetic energy T is deflected by a spherical potential well of radius R and depth U_0, i.e. by the field in which the potential energy of the particle takes the form

$$U = \begin{cases} 0 & \text{for } r > R, \\ -U_0 & \text{for } r < R, \end{cases}$$

   where r is the distance from the centre of the well. Find the relationship between the aiming parameter b of the particle and the angle θ through which it deflects from the initial motion direction.

4. A stationary ball of radius R is irradiated by a parallel stream of particles whose radius is r. Assuming the collision of a particle and the ball to be elastic, find:

   (a) the deflection angle θ of a particle as a function of its aiming parameter b;

   (b) the fraction of particles which after a collision with the ball are scattered into the angular interval between θ and $\theta + d\theta$;

   (c) the probability of a particle to be deflected, after a collision with the ball, into the front hemisphere $\left(\theta < \frac{\pi}{2}\right)$.

5. In accordance with classical electrodynamics an electron moving with acceleration $\vec{w}$ loses its energy due to radiation as

$$\frac{dE}{dt} = -\frac{2e^2}{3c^3} w^2,$$

   where e is the electron charge, c is the velocity of light. Estimate the time during which the energy of an electron performing almost harmonic oscillations with frequency $\omega = 5 \times 10^{15}$ s^{-1} will decrease η = 10 times.

6. Making use of the formula of the foregoing problem, estimate the time during which an electron moving in a hydrogen atom along a circular orbit of radius r = 50 pm would have fallen onto the nucleus. For the sake of simplicity assume the vector $\vec{w}$ to be permanently directed toward the centre of the atom.

7. A particle of mass m moves along a circular orbit in a centro-symmetrical potential field $U(r) = \frac{kr^2}{2}$. Using the Bohr quantization condition, find the permissible orbital radii and energy levels of that particle.

8. For hydrogen-like systems find the magnetic moment μ_n corresponding to the motion of an electron along the nth orbit and the ratio of the magnetic and mechanical moments μ_n/M_n. Calculate the magnetic moment of an electron occupying the first Bohr orbit.

9. Find the quantum number n corresponding to the excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm.

10. A stationary He^+ ion emitted a photon corresponding to the first line of the Lyman series. That photon liberated a photoelectron from a stationary hydrogen atom in the ground state. Find the velocity of the photoelectron.

11. Calculate the separation between the particles of a system in the ground state, the corresponding binding energy, and the wavelength of the first line of the Lyman series, if such a system is

    (a) a mesonic hydrogen atom whose nucleus is a proton (in a mesonic atom an electron is replaced by a meson whose charge is the same and mass is 207 that of an electron);

    (b) a positronium consisting of an electron and a positron revolving around their common centre of masses.

12. Two identical non-relativistic particles move at right angles to each other, possessing de Broglie wavelengths λ_1 and λ_2. Find the de Broglie wavelength of each particle in the frame of their centre of inertia.

13. At what value of kinetic energy is the de Broglie wavelength of an electron equal to its Compton wavelength?

14. A small amount of solution containing Na^{24} radionuclide with activity $A = 2.0 \times 10^3$ disintegrations per second was injected in the bloodstream of a man. The activity of 1 cm^3 of blood sample taken $t = 5.0$ hours later turned out to be $A' = 16$ disintegrations per minute per cm^3. The half-life of the radionuclide is $T = 15$ hours. Find the volume of the man's blood.

15. A P^{32} radionuclide with half-life $T = 14.3$ days is produced in a reactor at a constant rate $q = 2.7 \times 10^9$ nuclei per second. How soon after the beginning of production of that radionuclide will its activity be equal to $A = 1.0 \times 10^9$ dis/s?

16. A radionuclide A_1 with decay constant λ_1 transforms into a radionuclide A_2 with decay constant λ_2. Assuming that at the initial moment the preparation contained only the radionuclide A_1, find:

    (a) the equation describing accumulation of the radionuclide A_2 with time;

    (b) the time interval after which the activity of radionuclide A_2 reaches the maximum value.

17. Solve the foregoing problem if $\lambda_1 = \lambda_2 = \lambda$.

18. A neutron collides elastically with an initially stationary deuteron. Find the fraction of the kinetic energy lost by the neutron

    (a) in a head-on collision;

    (b) in scattering at right angles.

Book Crunch- 2

Chemistry

Chemistry

1. It is known that atom contain protons, neutrons and electrons. If the mass of neutron is assumed to be half of its original value whereas that of proton is assumed to be twice of its original value then the atomic mass of $^{14}_{6}C$ will be

   (a) Same
   (b) 14.28% less
   (c) 14.28% more
   (d) 28.56% less

2. Dieldrin, an insecticide, contains C, H, Cl and O. Combustion of 29.72 mg of Dieldrin gave 41.21 mg CO_2 and 5.63 mg of H_2O. In a separate analysis 25.31 mg of Dieldrin was converted into 67.16 mg AgCl. What is the empirical formula of Dieldrin?

   (a) $C_6H_4Cl_3O$
   (b) C_8H_8ClO
   (c) $C_{12}H_8Cl_6O$
   (d) $C_6H_4Cl_3O_2$

3. Suppose two elements X and Y combine to form two compounds XY_2 and X_2Y_3 . 0.05 mole of XY_2 weighs 5 g while 3.011×10^{23} molecules of X_2Y_3 weigh 85 g. The atomic masses of X and Y are respectively

   (a) 20, 30
   (b) 30, 40
   (c) 40, 30
   (d) 80, 60

4. Some older emergency oxygen masks containing potassium superoxide, KO_2 which reacts with CO_2 and water in exhaled air to produce oxygen according to the given equation. If a person exhales 0.667 g of CO_2 per minute, how many grams of KO_2 are consumed in 5.0 minutes?

   $4KO_2 + 2H_2O + 4CO_2 \rightarrow 4KHCO_3 + 3O_2$

   (a) 10.7
   (b) 0.0757
   (c) 1.07
   (d) 5.38

5. 9 moles of "D" and 14 moles of E are allowed to react in a closed vessel according to given reactions. Calculate number of moles of B formed in the end of reaction, if 4 moles of G are present in reaction vessel. (Percentage yield of reaction is mentioned in the reaction)

   Step-I $3D + 4E \xrightarrow{80\%} 5C + A$

   Step-2 $3C + 5G \xrightarrow{50\%} 6B + F$

   (a) 2.4
   (b) 30
   (c) 4.8
   (d) 1

6. The chief ore of Zn is the sulphide, ZnS. The ore is concentrated by froth floatation process and then heated in air to convert ZnS to ZnO.

$2ZnS + 3O_2 \xrightarrow{80\%} 2ZnO + 2SO_2$

$ZnO + H_2SO_4 \xrightarrow{100\%} ZnSO_4 + H_2O$

$2ZnSO_4 + 2H_2O \xrightarrow{80\%} 2Zn + 2H_2SO_4 + O_2$

The number of moles of ZnS required for producing 2 moles of Zn will be

(a) 3.125 (b) 2

(c) 2.125 (d) 4

7. One gram of the silver salt of an organic dibasic acid yields, on strong heating, 0.5934 g of silver. If the weight percentage of carbon in it is 8 times the weight percentage of hydrogen and one-half the weight percentage of oxygen, determine the molecular formula of the acid.

(a) $C_4H_6O_4$ (b) $C_4H_6O_6$

(c) $C_2H_6O_2$ (d) $C_5H_{10}O_5$

8. 2 mole of N_2H_4 loses 16 mole of electron on being converted to a new compound X. Assuming that all of the N appears in the new compound. What is the oxidation state of 'N' in X?

(a) –1 (b) –2

(c) +2 (d) +4

9. H_2O_2 is used as bleaching reagent because on dissociation it gives oxygen $\left(H_2O_2 \longrightarrow H_2O + \frac{1}{2}O_2\right)$.

If "Chachi 420" used H_2O_2 solution to bleach her hair and she required 2.24 L O_2 gas at 1 atm and 273 K. She has a H_2O_2 solution labelled as "5.6 V" then what volume of such solution must she required to bleach her hair?

(a) 200 mL (b) 300 mL

(c) 400 mL (d) 500 mL

10. 0.80 g of impure $(NH_4)_2SO_4$ was boiled with 100 mL of a 0.2 N NaOH solution till all the $NH_3(g)$ evolved. The remaining solution was diluted to 250 mL. 25 mL of this solution was neutralized using 5 mL of a 0.2 N H_2SO_4 solution. The percentage purity of the $(NH_4)_2SO_4$ sample is

(a) 82.5 (b) 72.5

(c) 62.5 (d) 17.5

11. Stannous sulphate ($SnSO_4$) and potassium permanganate are used as oxidising agents in acidic medium for oxidation of ferrous ammonium sulphate to ferric sulphate. The ratio of number of moles of stannous sulphate required per mole of ferrous ammonium sulphate to the number of moles of $KMnO_4$ required per mole of ferrous ammonium sulphate, is

(a) 2.5 (b) 0.2

(c) 0.4 (d) 2.0

12. In an iodometric estimation, the following reactions occur

$2Cu^{2+} + 4I^- \longrightarrow Cu_2I_2 + I_2$; $I_2 + 2Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$

0.12 mole of $CuSO_4$ was added to excess of KI solution and the liberated iodine required 120 mL of hypo. The molarity of hypo solution was

(a) 2 (b) 0.20

(c) 0.1 (d) 1.0

13. A mixture of NaOH and Na_2CO_3 required 25 mL of 0.1 M HCl using phenolphthalein as the indicator. However, the same amount of the mixture required 30 mL of 0.1 M HCl when methyl orange was used as the indicator. The molar ratio of NaOH and Na_2CO_3 in the mixture was

(a) 2 : 1 (b) 1 : 2

(c) 4 : 1 (d) 1 : 4

14. 6.2 g of a sample containing Na_2CO_3, $NaHCO_3$ and non-volatile inert impurity on gentle heating loses 5% of its weight due to reaction $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$. Residue is dissolved in water and formed 100 mL solution and its 10 mL portion requires 7.5 mL of 0.2 M aqueous solution of $BaCl_2$ for complete precipitation of carbonates. Determine weight (in gram) of Na_2CO_3 in the original sample.

(a) 1.59 (b) 1.06

(c) 0.53 (d) None of these

15. A 1.0 g sample of a pure organic compound containing chlorine is fused with Na_2O_2 to convert chlorine to NaCl. The sample is then dissolved in water, and the chloride precipitated with $AgNO_3$, giving 1.96 g of AgCI. If the molecular weight of organic compound is 147, how many chlorine atoms does each molecule contain?

(a) 1 (b) 2

(c) 3 (d) 4

16. 11.6 g of an organic compound having formula C_nH_{2n+2} is burnt in excess of $O_2(g)$ initially taken in a 22.41 litre steel vessel. Before reaction the gaseous mixture was at 273 K with pressure reading 2 atm. After complete combustion and loss of considerable amount of heat, the mixture of product and excess of O_2 had a temperature of 546 K and 4.6 atm pressure. The formula of organic compound is

(a) C_2H_6 (b) C_3H_8

(c) C_5H_{12} (d) C_4H_{10}

17. $H_2O_2 + 2KI \xrightarrow{40\% \text{ yield}} I_2 + 2KOH$

$H_2O_2 + 2KMnO_4 + 3H_2SO_4 \xrightarrow{50\% \text{ yield}} K_2SO_4 + 2MnSO_4 + 3O_2 + 4H_2O$

150 mL of H_2O_2 sample was divided into two parts. First part was treated with KI and formed KOH required 200 mL of M/2 H_2SO_4 for neutralisation. Other part was treated with $KMnO_4$ yielding 6.74 litre of O_2 at 1 atm and 273 K. Using % yield indicated find volume strength of H_2O_2 sample used.

(a) 5.04 (b) 10.08

(c) 3.36 (d) 33.6

18. 0.10 g of a sample containing $CuCO_3$ and some inert impurity was dissolved in dilute sulphuric acid and volume made up to 50 mL. This solution was added into 50 mL of 0.04 M KI solution where copper precipitates as CuI and I^- is oxidized into I_3^-. A 10 mL protion of this solution is taken for analysis, filtered and made free from I_3^- and then treated with excess of acidic permanganate solution. Liberated iodine required 20 mL of 2.5 mM sodium thiosulphate solution to reach the end point. Determine weight percentage of $CuCO_3$ in the original sample.

(a) 7.41

(b) 74.1

(c) 61.75

(d) None of these

19. When an equimolar mixture of Cu_2S and CuS is titrated with $Ba(MnO_4)_2$ in acidic medium, the final product's contains Cu^{2+}, SO_2 and Mn^{2+}. If the mol. wt. of Cu_2S, CuS and $Ba(MnO_4)_2$, are M_1, M_2 and M_3 respectively then

(a) Eq. wt. of Cu_2S is $\frac{M_1}{8}$

(b) Eq. wt. of CuS is $\frac{M_2}{6}$

(c) Eq. wt. of $Ba(MnO_4)_2$ is $\frac{M_3}{5}$

(d) Cu_2S and CuS both have same equivalents in mixture

20. Match Column-I with Column-II.

Column-I			**Column-II**
[Atomic masses (M)]			**(% composition of heavier isotope)**
Isotope-I	Isotope-II	Average	
(A) (z – 1)	(z + 3)	z	(P) 25% by moles
(B) (z + 1)	(z + 3)	(z + 2)	(Q) 50% by moles
(C) z	3z	2z	(R) % by mass dependent on *z*
(D) (z – 1)	(z + 1)	z	(S) 75% by mass

21. A sample of raw material contain $NaNO_3$ also contains $NaIO_3$. The $NaIO_3$ can be used as a source of iodine, produced in the following reactions:

$$IO_3^- + HSO_3^- \longrightarrow I^- + SO_4^{2-} \quad \text{... (1)}$$

$$I^- + IO_3^- \longrightarrow I_2 + H_2O \quad \text{... (2)}$$

One litre of sample solution containing 396 g of $NaIO_3$ is treated with stoichiometric quantity of $NaHSO_3$. Now a substantial amount of same solution is added to reaction mixture to bring about the reaction (2).

Column-I	**Column-II**
(A) n-factor of IO_3^- in reaction (2)	(P) 6
(B) Number of moles of HSO_3^- used in reaction (1)	(Q) 1.2
(C) Moles of I_2 produced	(R) 2
(D) Equivalents of IO_3^- used in reaction (2)	(S) 5

22. 10 g mixture of $K_2Cr_2O_7$ and $KMnO_4$ was treated with excess of KI in acidic medium. Iodine liberated required 100 cm^3 of 2.2 N sodium thiosulphate solution for titration. If the mass percent of $KMnO_4$ in the mixture is Z, then what is the value of 2Z/5?

23. Electronic transition in He^+ ion takes from n_2 to n_1 shell such that :

$2n_2 + 3n_1 = 18$

$2n_2 - 3n_1 = 6$

What will be the total number of photons emitted when electrons transit to n_1 shell ?

(a) 21 (b) 15

(c) 20 (d) 10

24. A light source of wavelength λ illuminates a metal and ejects photoelectrons with $(K.E.)_{max}$ = 1 eV.

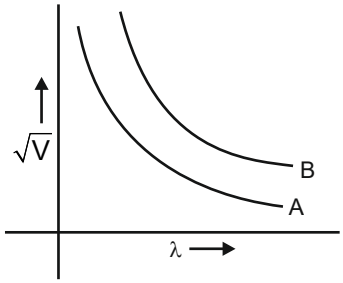
Another light source of wavelength $\frac{\lambda}{3}$, ejects photoelectrons from same metal with $(K.E.)_{max}$ = 4 eV.

Find the value of work function.

(a) 1 eV (b) 2 eV

(c) 0.5 eV (d) None of these

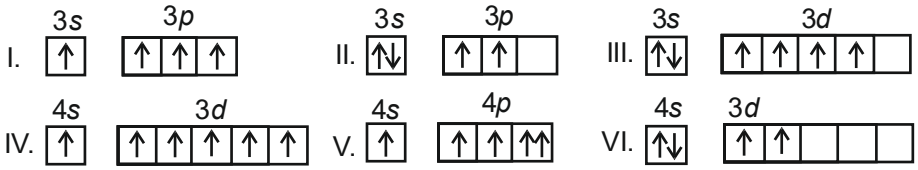
25. $\sqrt{V}$ on two particles A and B are plotted against de-Broglie wavelengths. Where V is potential difference applied on the particles. Which of the following relation is correct about the mass of particles? (Assume equal charge on the two particles)



(a) $m_A = m_B$ (b) $m_A > m_B$

(c) $m_A < m_B$ (d) $m_A \leq m_B$

26. Consider the following six electronic configurations (remaining inner orbitals are completely filled) and mark the incorrect option.



(a) Stability order : II > I > IV > III

(b) Order of spin multiplicity : IV > III = I > II

(c) V does not violate all the three rules of electronic configuration

(d) If VI represents A then A^+ when kept near a magnet, acts as diamagnetic substance.

27. The Schrodinger wave equation for hydrogen atom is

$$\psi\,(\text{radial}) = \frac{1}{16\sqrt{4}}\left(\frac{Z}{a_0}\right)^{3/2}\left[(\sigma-1)\,(\sigma^2-8\sigma+12)\right]e^{-\sigma/2}$$

where a_0 and Z are the constant in which answer can be expressed and $\sigma = \frac{2Zr}{a_0}$.

Minimum and maximum position of radial nodes from nucleus are ..... respectively.

(a) $\frac{a_0}{z}, \frac{3a_0}{z}$ (b) $\frac{a_0}{2z}, \frac{a_0}{z}$

(c) $\frac{a_0}{2z}, \frac{3a_0}{z}$ (d) $\frac{a_0}{2z}, \frac{4a_0}{z}$

28. For a hypothetical hydrogen like atom, the potential energy of the system is given by $U(r) = \frac{-Ke^2}{r^3}$, where r is the distance between the two particles. If Bohr's model of quantization of angular momentum is applicable then velocity of particle is given by

(a) $v = \frac{n^2h^3}{Ke^2 8\pi^3 m^2}$ (b) $v = \frac{n^3h^3}{8Ke^2\pi^3 m^2}$

(c) $v = \frac{n^3h^3}{24Ke^2\pi^3 m^2}$ (d) $v = \frac{n^2h^3}{24Ke^2\pi^3 m^2}$

29. A small particle of mass m moves in such a way that $P.E. = -\frac{1}{2}mkr^2$, where k is a constant and r is the distance of the particle from origin. Assuming Bohr's model of quantization of angular momentum is applicable then r is directly proportional to

(a) n^2 (b) n

(c) $\sqrt{n}$ (d) None of these

30. An element undergoes a reaction as shown :

$X + 2e^- \rightarrow X^{2-}$, energy released = 30.87 eV/atom. If the energy released, is used to dissociate 4 gms of H_2 molecules, equally into H^+ and H*, where H* is excited state of H atoms where the electron travel in orbit whose circumference equal to four times its de Broglie's wavelength. Determine the least moles of X that would be required :

Given : I.E. of H = 13.6 eV/atom, bond energy of H_2 = 4.526 eV/molecule.

(a) 1 (b) 2

(c) 3 (d) 4

31. In a measurement of quantum efficiency of photosynthesis in green plants, it was found that 10 quanta of red light of wavelength 6850 Å were needed to release one molecule of O_2. The average energy storage in this process is 112 kcal/mol O_2 evolved.

What is the energy conversion efficiency in this experiment?

Given: 1 cal = 4.18 J; $N_A = 6 \times 10^{23}$; $h = 6.63 \times 10^{-34}$ Js

(a) 23.5 (b) 26.9

(c) 66.34 (d) 73.1

32. The radial distribution functions [P(r)] is used to determine the most probable radius, which is used to find the electron in a given orbital $\left(\frac{dP(r)}{dr}\right)$ for 1*s*-orbital of hydrogen like atom having atomic number Z, is $\frac{dP}{dr} = \frac{4Z^3}{a_0^3}\left(2r - \frac{2Zr^2}{a_0}\right)e^{-2Zr/a_0}$.

Then which of the following statements is/are correct?

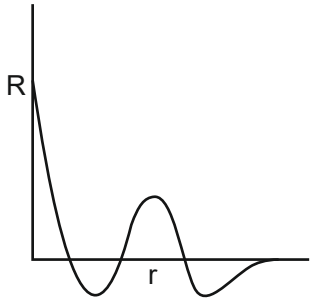
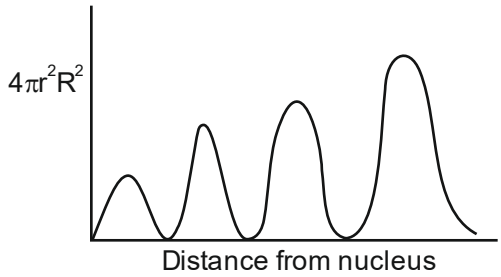
(a) At the point of maximum value of radial distribution function $\frac{dP(r)}{dr} = 0$

(b) Most probable radius of Li^{2+} is $\frac{a_0}{3}$pm

(c) Most probable radius of He^+ is $\frac{a_0}{2}$pm

(d) Most probable radius of hydrogen atom is a_0 pm

33. Match Column-I with Column-II

Column-I	Column-II
(A) 	(P) 4*s*
(B) 	(Q) $5p_y$
(C) Angular probability is dependent of θ and ϕ	(R) 3*s*
(D) At least one angular node is present	(S) $6d_{xy}$

34. For 1*s* orbital of Hydrogen atom radial wave function is given as:

$$R(r) = \frac{1}{\sqrt{\pi}}\left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}$$ (where a_0 = 0.529 Å)

The ratio of radial probability density of finding electron at r = a_0 to the radial probability density of finding electron at the nucleus is given as ($x.e^{-y}$). Calculate the value of (x + y).

35. Air entering the lungs ends up in tiny sacs called alveoli. It is from the alveoli that oxygen diffuses into the blood. The average radius of the alveoli is 0.0050 mm and the air inside contains 14 per cent oxygen. Assuming that the pressure in the alveoli is 1.0 atm and the temperature is 37°C, calculate the number of oxygen molecules in one of the alveoli.

(a) 6×10^{13} (b) 10^{24}

(c) 1.7×10^{22} (d) 1.7×10^{9}

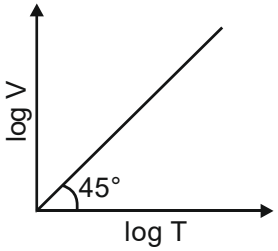
36. The density of liquid gallium at 30°C is 6.095 g/mL. Because of its wide liquid range (30 to 2400°C), gallium could be used as a barometer fluid at high temperature. What height (in cm) of gallium will be supported on a day when the mercury barometer reads 740 torr?

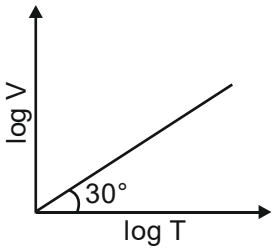
(The density of mercury is 13.6 g/mL.).

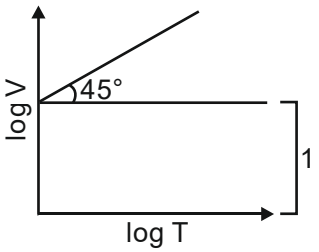
(a) 322 (b) 285

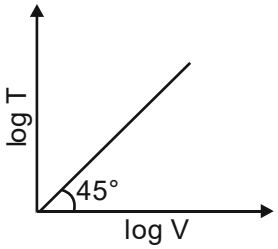
(c) 165 (d) 210

37. For a closed (not rigid) container containing n = 10 moles of an ideal gas fitted with movable, frictionless, weightless piston operating such that pressure of gas remains constant at 0.821 atm, which graph represents correct variation of log V vs log T, where V is in litre and T in kelvin?

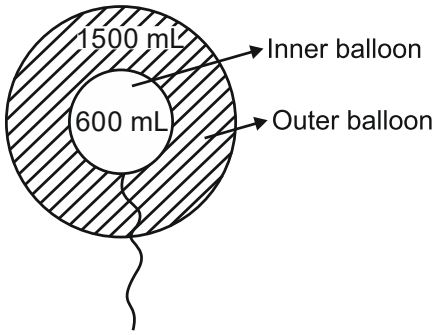
(a) 

(b) 

(c) 

(d) 

38. Two inflated balloons I and II (thin skin) having volume 600 mL and 1500 mL at 300 K are taken as shown in diagram. If maximum volume of inner and outer balloons are 800 mL and 1800 mL respectively then find the balloon which will burst first on gradual heating.



(a) Inner balloon
(b) Outer balloon
(c) Both simultaneously
(d) Unpredictable

39. Consider the following statements. If the van der Waal's parameters of two gases are given as

	a/dm^6 bar mol^{-2}	b/dm^3 mol^{-1}
Gas A	6.5	0.055
Gas B	2	0.01

then:

1. Critical volume of A < Critical volume of B.
2. Critical pressure A > Critical pressure of B.
3. Critical temperature of A > Critical temperature of B.

Which of the above statements is incorrect?

(a) 1 alone
(b) 1 and 2
(c) 1, 2 and 3
(d) 2 and 3

40. The van der Waals' equation for one mole may be expressed as

$$V_m^{3.} - \left(b + \frac{RT}{p}\right)V_m^2 + \frac{aV_m}{p} - \frac{ab}{p} = 0$$

where V_m is the molar volume of the gas. Which of the following is incorrect?

(a) For a temperature less than T_c, V has three real roots

(b) For a temperature less than T_c, V has three imaginary roots

(c) For a temperature equal to T_c, all three roots of V are real and identical

(d) On increasing the temperature ($T < T_c$), the three roots become closer to one another

41. For a gas obeying the van der Waals' equation, at the critical temperature

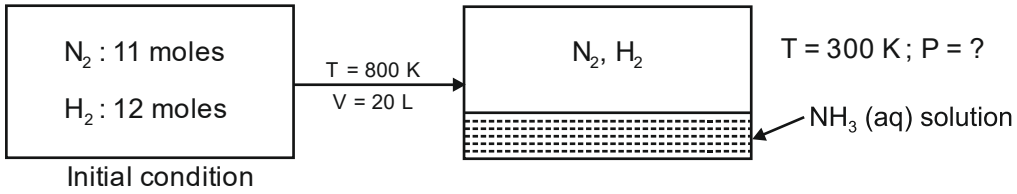
(a) Both $\left(\frac{\partial P}{\partial V}\right)_T$ and $\left(\frac{\partial^2 P}{\partial V^2}\right)_T$ are zero

(b) Only $\left(\frac{\partial^2 P}{\partial V^2}\right)_T$ is not zero

(c) $\left(\frac{\partial P}{\partial V}\right)_T$ is zero but $\left(\frac{\partial^2 P}{\partial V^2}\right)_T$ is non-zero

(d) $\left(\frac{\partial P}{\partial V}\right)_T$ is non-zero but $\left(\frac{\partial^2 P}{\partial V^2}\right)_T$ is zero

42. 11 moles N_2 and 12 moles of H_2 mixture reacted in 20 litre vessel at 800 K. After equilibrium was reached, 6 mole of H_2 was present. 3.58 litre of liquid water is injected in equilibrium mixture and resultant gaseous mixture suddenly cooled to 300 K. What is the final pressure of gaseous mixture? Neglect vapour pressure of liquid solution. Assume (i) all NH_3 dissolved in water (ii) no change in volume of liquid (iii) no reaction of N_2 and H_2 at 300 K



(a) 18.47 atm (b) 60 atm

(c) 22.5 atm (d) 45 atm

43. A mixture of $NH_3(g)$ and $N_2H_4(g)$ is placed in a sealed container at 300 K. The total pressure is 0.5 atm. The container is heated to 1200 K at which time both substances decompose completely according to the equations

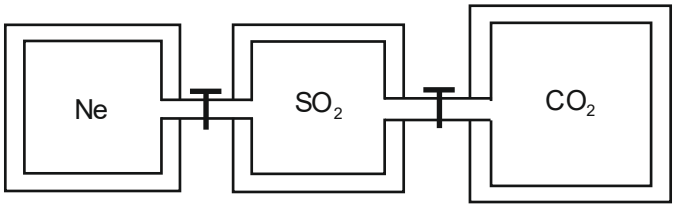
$2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ and $N_2H_4(g) \rightarrow N_2(g) + 2H_2(g)$

After decomposition is complete, the total pressure at 1200 K is found to be 4.5 atm. Find the mole% of N_2H_4 in the original mixture.

(a) 20% (b) 25%

(c) 50% (d) 75%

44. Three closed rigid vessels, A, B and C without energy transfer with surroundings, which initially contain three different gases at different temperatures are connected by tube of negligible volume. The vessel A contain 2 mole Ne gas, at 300 K, vessel 'B' contain 2 mole SO_2 gas at 400 K and vessel 'C' contain 3 mole CO_2 gas at temperature 500 K. What is the final pressure (in atm) attained by gases when all valves of connecting three vessels are opened and additional 15.6 kcal heat supplied to vessel through valve. The volume of A, B and C vessel is 2, 2 and 3 litre respectively

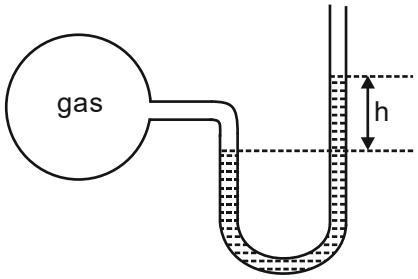


Given: R = 2 calorie/mol-K; C_v (Ne) = 3/2R, C_v (CO) = 5/2 R and C_v (SO_2) = 3R

(a) 73.89 atm (b) 67.31 atm

(c) 80 atm (d) None of these

45. A bulb of constant volume is attached to a manometer tube open at other end as shown in figure. The manometer is filled with a liquid of density $(1/3)^{rd}$ that of mercury. Initially *h* was 228 cm.



Through a small hole in the bulb gas leaked assuming pressure decreases as $\frac{dp}{dt} = -kP$.

If value of h is 114 cm after 14 minutes. What is the value of k (in $hour^{-1}$)?

[Use : ln (4/3) = 0.28 and density of Hg = 13.6 g/mL]

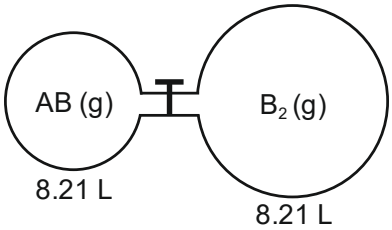
(a) 0.6 (b) 1.2

(c) 2.4 (d) None of these

46. At room temperature following reaction goes to completion

$2AB(g) + B_2(g) \rightarrow 2AB_2(s)$

AB_2 is solid with negligible vapour pressure below 0°C. At 300 K, the AB in the smaller flask exerts a pressure of 3 atm and the larger flask exerts a pressure of 1 atm at 400 K when they are separated out by a close valve. The gases are mixed by opening the stop cock and after the end of the reaction the flask are cooled to 250 K.



The final pressure is

(a) 0.156 atm (b) 0.3125 atm

(c) 0.625 atm (d) 3.2 atm

47. van der Waal's gas equation can be reduced to virial equation (in terms of volume) as $Z = A + \frac{B}{V_m} + \frac{C}{V_m^2} + ...$ where A = first virial coefficient, B = second virial coefficient, C = third virial coefficient. The third virial coefficient of Hg(g) is 625 $(cm^3/mol)^2$. What volume is available for movement of 10 moles He(g) atoms present in 50 L vessel?

(a) 49.75 L (b) 49.25 L

(c) 25 L (d) 50 L

Paragraph for Q. Nos. 48 to 51

van der Waals' equation for calculating the pressure of a non ideal gas is

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

van der Waals' suggested that the pressure exerted by an ideal gas, P_{ideal}, is related to the experimentally measured pressure, P_{real} by the equation

$$P_{ideal} = P_{real} + \frac{an^2}{V^2}$$

P_{real}: observed pressure; $\frac{an^2}{V^2}$: correction term

Constant a is measure of intermolecular interaction between gaseous molecules that gives rise to non ideal behavior and depends on how frequently any two molecules approach each other closely. Another correction concerns the volume occupied by the gas molecules. In the ideal gas equation, V represents the volume of the container. However, each molecule does occupy a finite, although small, intrinsic volume, so the effective volume of the gas becomes (V –nb), where n is the number of moles of the gas and b is a constant. The term nb represents the volume occupied by n moles of the gas.

Having taken into account the corrections for pressure and volume, we can rewrite the ideal gas equation as follows:

$$\left(P + \frac{an^2}{V^2}\right)(v - nb) = nRT$$

corrected pressure; corrected volume

48. At relatively high pressures, the van der Waals' equation of state reduces to

(a) $PV = RT - a/V$ (b) $PV = aRT/V^2$

(c) $P = RT - a/V^2$ (d) $PV = RT + Pb$

49. For non-zero value of force of attraction between gas molecules for large volume, gas equation will be

(a) $PV = nRT - \frac{n^2a}{V}$ (b) PV = nRT + nbP

(c) $P = \frac{nRT}{V - b}$ (d) PV = nRT

50. The van der Waals' constant 'a' for CO_2 gas is greater than that of H_2 gas. Its mean that the

(a) Strength of van der Waals' force of CO_2 gas is less than that of H_2 gas

(b) Strength of van der Waals' force of CO_2 gas is equal to that of H_2 gas

(c) CO_2 gas can be more easily liquified

(d) H_2 gas can be more easily liquified

51. Using van der Waals' equation, find the constant 'a' (in atm L^2 mol^{-2}) when two moles of a gas confined in 4 L flask exerts a pressure of 11.0 atmospheres at a temperature of 300 K. The value of b is 0.05 L mol^{-1}. (R = 0.082 atm. L/K mol)

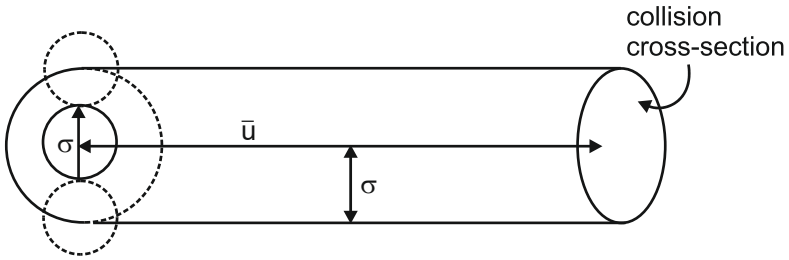
(a) 2.62 (b) 2.64

(c) 6.24 (d) 6.46

Paragraph for Q. Nos. 52 to 54

Collision cross-section is an area of an imaginary sphere of radius σ around the molecule within which the centre of another molecule cannot penetrate.

The volume swept by a single molecule in unit time is



$V = (\pi\sigma^2)\bar{u}$ where $\bar{u}$ is the average speed

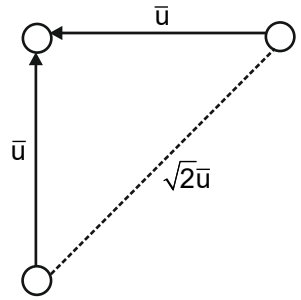
If N* is the number of molecules per unit volume, then the number of molecules within the volume V is

$$N = VN^* = (\pi\sigma^2\bar{u})N^*$$

Hence, the number of collision made by a single molecule in unit time will be

$$Z = N = (\pi\sigma^2\bar{u})N^*$$

In order to account for the movements of all molecules, we must consider the average velocity along the line of centres of two colliding molecules instead of the average velocity of a single molecule. If it is assumed that, on an average, molecules collide while approaching each other perpendicularly, then the average velocity along their centres is $\sqrt{2}\,\bar{u}$ as shown below. Number of collision made by a single molecule with other molecules per unit time is given by



$$Z_1 = \pi\sigma^2(\bar{u}_{rel})N^* = \sqrt{2}\,\pi\sigma^2\,\bar{u}N^*$$

The total number of bimolecular collisions Z_{11} per unit volume per unit time is given by

$$Z_{11} = \frac{1}{2}(Z_1N^*) \text{ or } Z_{11} = \frac{1}{2}(\sqrt{2}\,\pi\sigma^2\bar{u}N^*)N^* = \frac{1}{\sqrt{2}}\pi\sigma^2\bar{u}N^{*2}$$

If the collision involve two unlike molecules then the number of collisions Z_{12} per unit volume per unit time is given as

$$Z_{12} = \pi\sigma_{12}^2\left(\sqrt{\frac{8kT}{\pi\mu}}\right)N_1N_2$$

where N_1 and N_2 are the number of molecules per unit volume of the two types of molecules, σ_{12} is the average diameter of the two molecules and μ is the reduced mass.

The mean free path is the average distance travelled by a molecule between two successive collisions. We can express it as follows :

$$\lambda = \frac{\text{Average distance travelled per unit time}}{\text{No. of collisions made by a single molecule per unit time}} = \frac{\bar{u}}{Z_1}$$

or $$\lambda = \frac{\bar{u}}{\sqrt{2}\,\pi\sigma^2\bar{u}N^*} \Rightarrow \frac{1}{\sqrt{2}\,\pi\sigma^2 N^*}$$

52. For a given gas the mean free path at a particular pressure is

(a) Independent of temperature (b) Decreases with rise in temperature

(c) Increases with rise in temperature (d) Directly proportional to T^2

53-54 Three ideal gas samples in separate equal volume containers are taken and following data is given :

	Pressure	Temperature	Mean free paths	Mol. wt.
Gas A	1 atm	1600 K	0.16 nm	20
Gas B	2 atm	200 K	0.16 nm	40
Gas C	4 atm	400 K	0.04 nm	80

53. Calculate ratio of collision frequencies (Z_{11}) (A : B : C) of following for the three gases.

(a) 1 : 2 : 4 (b) 4 : 2 : 1

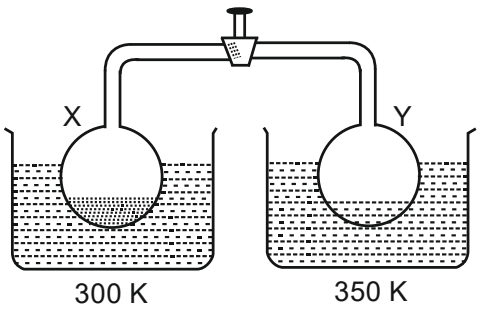
(c) 1 : 4 : 16 (d) 16 : 4 : 1

54. Calculate number of collision by one molecule per sec (Z_1).

(a) 4 : 1 : 4 (b) 1 : 4 : 4

(c) 4 : 3 : 2 (d) 1 : 2 : 4

55. Two container each containing liquid water are connected as shown in diagram.



Given that vapour pressure of $H_2O(l)$ at 300 K and 350 K are 22 torr and 40 torr. Select write statement(s).

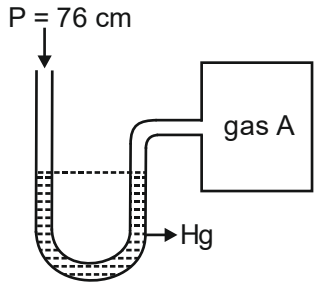
(a) The final pressure in each container if valve is opened while keeping the containers at the given temperature is 22 torr

(b) The final pressure in each container if valve is opened while keeping the containers at the given temperature is 40 torr

(c) Mass of $H_2O(l)$ is decreased in vessel X

(d) Mass of $H_2O(l)$ is decreased in vessel Y

56. An open ended mercury manometer is used to measure the pressure exerted by a trapped gas as shown in the figure. Initially manometer shows no difference in mercury level in both columns as shown in diagram.



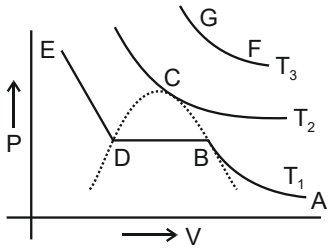
After sparking 'A' dissociates according to following reaction

$A(g) \rightarrow B(g) + 3C(g)$

If pressure of Gas "A" decreases to 0.9 atrn. Then (Assume temperature to be constant and is 300 K)

(a) Total pressure increased to 1.3 atm
(b) Total pressure increased by 0.3 atm
(c) Total pressure increased by 22.3 cm of Hg
(d) Difference in mercury level is 228 mm

57. Match the correct column from Column-I to Column-II on the basis of following Andrews isotherm of real gas.



Column-I	Column-II
(A) Substance exist in both liquid and gas state	(P) At AB part
(B) Only liquid state exist	(Q) At BD part
(C) Substance exist in gas state only	(R) At DE part
(D) Real gas is called super critical fluid	(S) At point C
	(T) At GF curve

58. Determine which of the following reactions taking place at constant pressure represent surrounding that do work on the system

I. $4NH_3(g) + 7O_2(g) \rightarrow 4NO_2(g) + 6H_2O(g)$

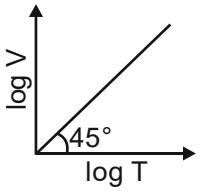
II. $CO(g) + 2H_2(g) \rightarrow CH_3OH(l)$

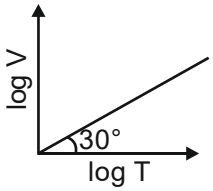
III. $C(s, \text{graphite}) + H_2O(g) \rightarrow CO(g) + H_2(g)$

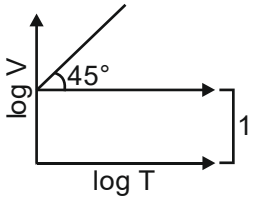
IV. $H_2O(s) \rightarrow H_2O(l)$

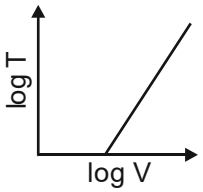
(a) III, IV
(b) II and III
(c) II, IV
(d) I, II and IV

59. For a closed container containing 100 mole of an ideal gas fitted with movable, frictionless, weightless piston operating such that pressure of gas remains constant at 8.21 atm, which graph represents correct variation of log V versus log T where V is in litre and T in kelvin.

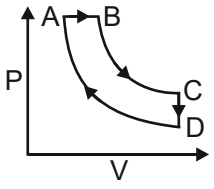
(a) 

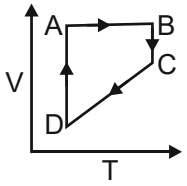
(b) 

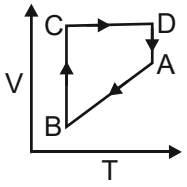
(c) 

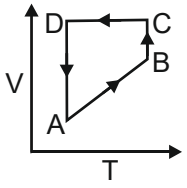
(d) 

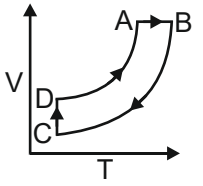
60. A cyclic process ABCD is shown in P–V diagram for an ideal gas. Which of the following diagram represents the same process?



(a) 

(b) 

(c) 

(d) 

61. One mole of a non-ideal gas undergoes a change of state from (1.0 atm, 3.0 L, 200 K) to (4.0 atm, 5.0 L, 250 K) with a change in internal energy (ΔU) = 40 L atm. The change in enthalpy of the process in 'L atm' is

(a) 43 (b) 57

(c) 42 (d) None of these

62. At 5×10^5 bar pressure density of diamond and graphite are 3 g/cc and 2 g/cc respectively, at certain temperature 'T'. Find the value of $\Delta U - \Delta H$ for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T'.

(a) 100 kJ/mol (b) 50 kJ/mol

(c) –100 kJ/mol (d) None of these

63. Stearic acid [$CH_3(CH_2)_{16}CO_2H$] is a fatty acid, the part of fat that stores most of the energy. 1.0 g of stearic acid was burned in a bomb calorimeter. The bomb had a heat capacity of 652 J/°C. If the temperature of 500 g water (c = 4.18J/g°C) rose from 25.0 to 39.3°C, how much heat was released when the stearic acid was burned? [Given C_P (H_2O) = 4.18 J/g°C]

(a) 39.21 kJ (b) 29.91 kJ

(c) 108 kJ (d) 9.32 kJ

64. The enthalpies of neutralization of a weak base AOH and a strong base BOH by HCl are –12250 cal/mol and –13000cal/mol respectively. When one mole of HCl is added to a solution containing 1 mole of AOH and 1 mole of BOH, the enthalpy change was –12500 cal/mol. In what ratio is the acid distribution between AOH and BOH?

(a) 2 : 1 (b) 2 : 3

(c) 1 : 2 (d) None of these

65. Boron can undergo the following reactions with the given enthalpy changes :

$$2B(s)+\frac{3}{2}O_2(g)\rightarrow B_2O_3(s);\ \Delta H=-1260\ kJ$$

$$2B(s)+3H_2(g)\rightarrow B_2H_6(g);\ \Delta H=30\ kJ$$

Assume no other reactions are occurring.

If in a container (operating at constant pressure) which is isolated from the surrounding, mixture of H_2 (gas) and O_2 (gas) are passed over excess of B(s), then calculate the molar ratio (O_2 : H_2) so that temperature of the container do not change.

(a) 15 : 3 (b) 42 : 1

(c) 1 : 42 (d) 1 : 84

66. A rigid and insulated tank of 3 m^3 volume is divided into two compartments. One compartment of volume of 2 m^3 contains an ideal gas at 0.8314 MPa and 400 K and while the second compartment of volume 1 m^3 contains the same gas at 8.314 MPa and 500 K. If the partition between the two compartments is ruptured, the final temperature of the gas is

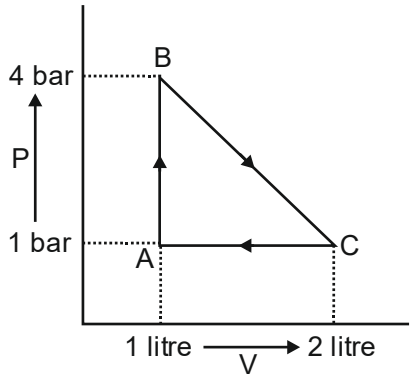
(a) 420 K

(b) 450 K

(c) 480 K

(d) None of these

67. One mole of an ideal gas is carried through the reversible cyclic process as shown in figure. The max. temperature attained by the gas during the cycle



(a) $\frac{7}{6R}$ (b) $\frac{12}{49R}$

(c) $\frac{49}{12R}$ (d) None of these

68. Which of the following statement(s) is/are correct?

Statement (i) : The entropy in isolated system with P-V work only, is always maximized at equilibrium.

Statement (ii) : It is possible for the entropy of a close system to decrease substantially in an irreversible process.

Statement (iii) : Entropy can be created but not destroyed.

Statement (iv) : ΔS_{system} is zero for reversible process in an isolated system.

(a) Statement i, ii, iii
(b) Statement ii, iv
(c) Statement i, ii, iv
(d) All

69. For the hypothetical reaction

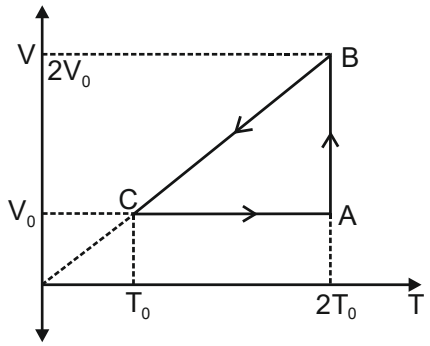
$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

$\Delta_r G°$ and $\Delta_r S°$ are 20 kJ/mol and –20 JK^{-1} mol^{-1} respectively at 200 K

If $\Delta_r C_p$ is 20 JK^{-1} mol^{-1} then $\Delta_r H°$ at 400 K is

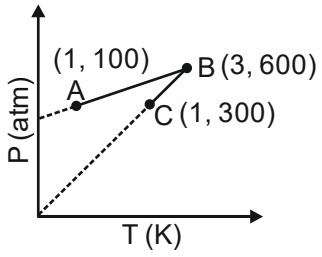
(a) 20 kJ/mol
(b) 7.98 kJ/mol
(c) 28 kJ/mol
(d) None of these

70. The efficiency of an ideal gas with adiabatic exponent 'γ' for the shown cyclic process would be



(a) $\frac{(\gamma-1)(2\ln 2-1)}{1+(\gamma-1)2\ln 2}$
(b) $\frac{(\gamma-1)(1-2\ln 2)}{(\gamma-1)2\ln 2-1}$
(c) $\frac{(2\ln 2+1)(\gamma-1)}{(\gamma-1)2\ln 2+1}$
(d) $\frac{(2\ln 2-1)}{\gamma/(\gamma-1)}$

71. One mole of an ideal gas is subjected to a two step reversible process (A–B and B–C). The pressure at A and C is same. Mark the correct statement(s).



(a) Work involved in the path AB is zero
(b) In the path AB work will be done on the gas by the surrounding
(c) Volume of gas at C = 3 × volume of gas at A
(d) Volume of gas at B is 16.42 litres

72. Assume ideal gas behaviour for all the gases considered and neglect vibrational degrees of freedom. Separate equimolar samples of Ne, O_2, CO_2 and SO_2 were subjected to a two process as mentioned. Initially all are at same state of temperature and pressure.

Step I $\rightarrow$ All undergo reversible adiabatic expansion to attain same final volume, which is double the original volume thereby causing the decrease in their temperature.

Step II $\rightarrow$ After step I, all are given appropriate amount of heat isochorically to restore the original temperature.

Mark the correct option(s).

(a) Due to step I only, the decrease in temperature will be maximum for Ne

(b) During step II, heat given will be minimum for SO_2

(c) There will be no change in internal energy for any of the gas after both the steps of process are competed

(d) The P–V graph of O_2 and CO_2 will be same

73. Match Column-I with Column-II

Column-I **(Partial derivative)**	**Column-II** **(Thermodynamic variable)**
(a) $\left(\frac{\partial U}{\partial T}\right)_V$	(p) C_P
(b) $\left(\frac{\partial H}{\partial T}\right)_P$	(q) C_V
(c) $\left(\frac{\partial G}{\partial T}\right)_P$	(r) $-S$
(d) $\left(\frac{\partial G}{\partial P}\right)_T$	(s) V

74. A diatomic ideal gas is expanded according to PV^3 = constant, under very high temperature (Assume vibration mode active). Calculate the molar heat capacity of gas (in cal/mol K) in this process.

75. For a reversible gaseous reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ at equilibrium, if some moles of H_2 are replaced by same number of moles of T_2 (T is tritium, isotope of H and assume isotopes do not have different chemical properties) without affecting other parameters, then

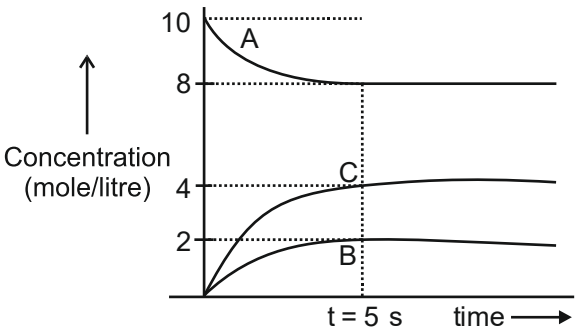
(a) The sample of ammonia obtained after sometime will be radioactive.

(b) Moles of N_2 after the change will be different as compared to moles of N_2 present before the change

(c) The value of K_p or K_c will change

(d) The average molecular mass of new equilibrium will be same as that of old equilibrium

76. Attainment of the equilibrium $A(g) \rightleftharpoons 2C(g) + B(g)$ gave the following graph. Find the correct option. (% dissociation = fraction dissociated × 100)



(a) At t = 5 s equilibrium has been reached and K_c = 40 (mol/litre)^2

(b) At t = 5 s equilibrium has been reached and % dissociation of A is 20%

(c) At t = 5 s equilibrium has been reached and % dissociation of A is 30%

(d) None of these

77. $CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4 \cdot 3H_2O(s) + 2H_2O(g); K_p = 4 \times 10^{-4}$ atm^2 . If the vapour pressure of water is 38 torr then the percentage relative humidity is (Assume all data at constant temperature)

(a) 4 (b) 10

(c) 40 (d) None of these

78. Ammonia under a pressure of 15 atm at 27°C is heated to 347°C in a closed vessel in the presence of a catalyst. Under the conditions, NH_3 is partially decomposed according to the equation,

$2NH_3 \rightleftharpoons N_2 + 3H_2$ the vessel is such that the volume remains effectively constant whereas pressure increases to 50 atm. Calculate the percentage of NH_3 actually decomposed.

(a) 65% (b) 61.3%

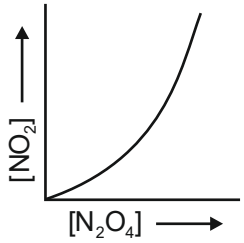
(c) 62.5% (d) 64%

79. The graph which represents all the equilibrium concentrations for the reaction

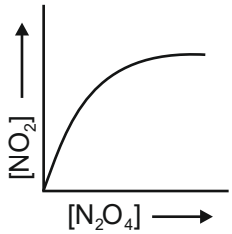
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

Then select the correct graph for concentrations of NO_2 against concentrations of NO_4.

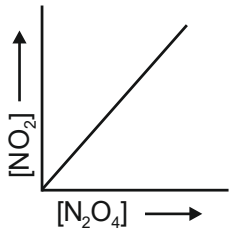
(a)



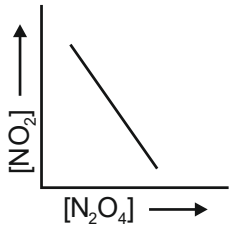
(b)



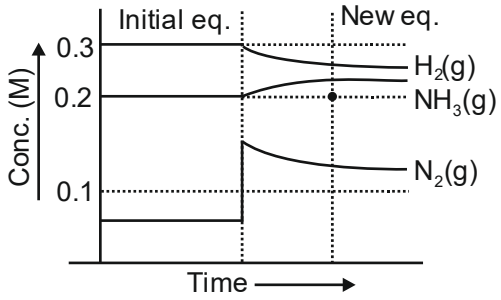
(c)

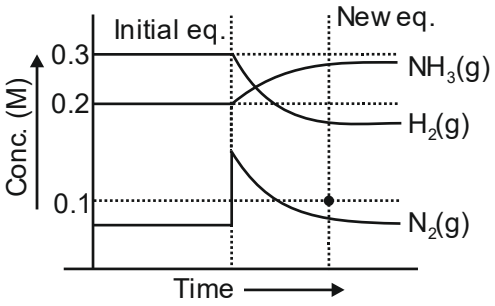


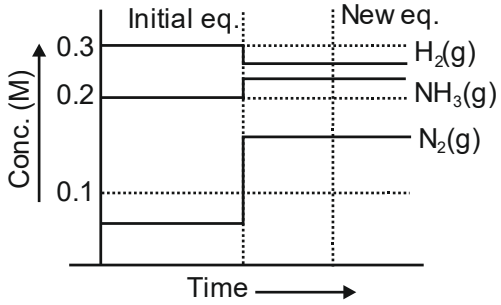
(d)

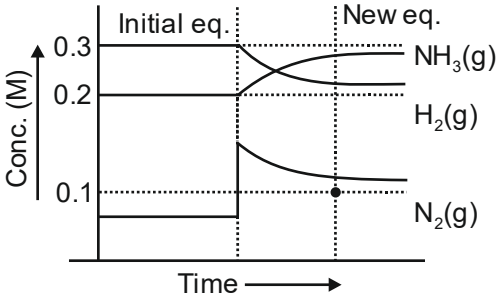


80. An equilibrium mixture at 700 K of 0.05 M $N_2(g)$, 0.3 M $H_2(g)$ and 0.2 M $NH_3(g)$ is present in a container. Now if this equilibrium is disturbed by adding $N_2(g)$ so that its concentration becomes 0.15 M just after addition then which of the following graphs represents the above situation more appropriately?

(a)


(b)


(c)


(d)


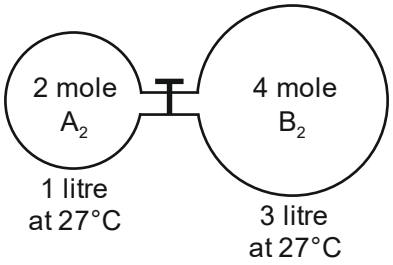
81. Solid $Ca(HCO_3)_2$ decomposes as

$$Ca(HCO_3)_2(s) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(g)$$

If the total pressure is 0.2 bar at 420 K, what is the standard free energy change for the given reaction ($\Delta_r G°$)?

(a) 840 kJ/mol
(b) 3.86 kJ/mol
(c) 6.98 kJ/mol
(d) 16.083 kJ/mol

82. The gas A_2 in the left flask allowed to react with gas B_2 present in right flask as $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$; $K_c = 4$ at 27°C. What is the concentration of AB when equilibrium is established?



(a) 1.33 M
(b) 2.66 M
(c) 0.66 M
(d) 0.33 M

83. A vessel of 250 litre was filled with 0.01 mole of Sb_2S_3 and 0.01 mole of H_2 to attain the equilibrium at 440°C as

$$Sb_2S_3(s) + 3H_2(g) \rightleftharpoons 2Sb(s) + 3H_2S(g)$$

After equilibrium, the H_2S formed was analysed by dissolving it in water and treating with excess of Pb^{2+} to give 1.19 g of PbS as precipitate. What is the value of K, at 440°C?

(a) 1
(b) 2
(c) 4
(d) 8

84. $N_2(g)$ and $H_2(g)$ are allowed to react in a closed vessel at given temperature and pressure for the formation of $NH_3(g)$ $[N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 22.4 \text{ kcal}]$ if He(g) is added at equilibrium at constant pressure then which is/are correct?

(a) Concentration of $N_2(g)$, $H_2(g)$ and $NH_3(g)$ decrease

(b) Moles of $NH_3(g)$ decreases

(c) The extent of cooling depends on amount of He(g) added

(d) Concentration of N_2 and H_2 increases and concentration of NH_3 decreases

85. Given : Enthalpy of ionization of two acids :

$\Delta H°(HCN) = 45.2 \text{ kJ mol}^{-1}$

$\Delta H°(CH_3COOH) = 2.1 \text{ kJ mol}^{-1}$

which relationship for the two acids is true?

(a) $pK_a(HCN) = pK_a(CH_3COOH)$

(b) $pK_a(HCN) > pK_a(CH_3COOH)$

(c) $pK_a(HCN) < pK_a(CH_3COOH)$

(d) $pK_a(HCN) = \frac{45.2}{2.1} pK_a(CH_3COOH)$

86. At 25°C dissociation constants of acid HA and base BOH in aqueous solution are same. The pH of 0.01 M solution of HA is 5. The pOH of 10^{-4} M solution of BOH at the same temperature is

(a) 3.5

(b) 4

(c) 6

(d) None of these

87. The enthalpy of neutralisation of four acids HA, HB, HC and HD with NaOH are –13, –12, –11, –10 kcal/mol. Which salt has maximum degree of hydrolysis?

(a) 1 M NaA

(b) 1 M NaB

(c) 1 M NaC

(d) 1 M NaD

88. 1.0 L solution is prepared by mixing 61 gm benzoic acid (pKa = 4.2) with 72 gm of sodium benzoate and then 300 mL 1.0 M HBr solution was added. The pH of final solution is

(a) 3.6

(b) 3.8

(c) 4.2

(d) 4.8

89. An acid-base indicator which is a weak acid has a pK_a value = 5.45. At what concentration ratio of sodium acetate to acetic acid would the indicator show a colour half-way between those of its acid and conjugate base forms ? [pK_a of acetic acid = 4.75]

(a) 4 : 1

(b) 6 : 1

(c) 5 : 1

(d) 3 : 1

90. A 1 litre solution containing NH_4Cl and NH_4OH has hydroxide ion concentration of 10^{-6} mol/litre. Which of the following hydroxides could be precipitated when the solution is added to 1 litre solution of 0.1 M metal ions?

(I) $Ba(OH)_2$ ($K_{sp} = 5 \times 10^{-3}$)

(II) $Ni(OH)_2$ ($K_{sp} = 1.6 \times 10^{-16}$)

(III) $Mn(OH)_2$ ($K_{sp} = 2 \times 10^{-13}$)

(IV) $Fe(OH)_2$ ($K_{sp} = 8 \times 10^{-16}$)

(a) I, II, IV

(b) IV

(c) II and IV

(d) II, III, IV

91. A 1.025 g sample containing a weak acid HX (mol. wt. = 82) is dissolved in 60 mL water and titrated with 0.25 M NaOH. When half of the acid was neutralised the pH was found to be 5.0 and at the equivalence point the pH is 9.0. Calculate weight percentage of HX in sample.

(a) 50% (b) 75%

(c) 80% (d) None of these

92. A solution of weak acid HA was titrated with base NaOH. The equivalent point was reached when 40 mL of 0.1 M NaOH has been added. Now 20 mL of 0.1 M HCl were added to titrated solution, the pH was found to be 5.0. What will be the pH of the solution obtained by mixing 20 mL of 0.2 M NaOH and 20 mL of 0.2 M HA?

(a) 7 (b) 9

(c) 11 (d) None of these

93. A buffer solution having 0.04 M in Na_2HPO_4 and 0.02 M in Na_3PO_4 is prepared. The electrolytic oxidation of 1.0 milli-mole of the organic compound RNHOH is carried out in 100 mL of the buffer. The reaction is

$RNHOH + H_2O \rightarrow RNO_2 + 4H^+ + 4e^-$

The approximate pH of solution after the complete oxidation is

[Given : for H_3PO_4, $pK_{a_1} = 2.2$; $pK_{a_2} = 7.20$; $pK_{a_3} = 12$]

(a) 6.90 (b) 7.20

(c) 7.5 (d) None of these

94. A solution is $0.1\,M$ in Cl^- and 10^{-4} M in CrO_4^{-2}. If solid $AgNO_3$ is gradually added to this solution, what will be the concentration of Cl^- when Ag_2CrO_4 begins to precipitate?

[K_{sp} (AgCl) = 10^{-10} M^2; K_{sp} (Ag_2CrO_4) = 10^{-12} M^3

(a) 10^{-6} M (b) 10^{-4} M

(c) 10^{-5} M (d) 10^{-9} M

95. If 500 mL of 0.4 M $AgNO_3$ is mixed with 500 mL of 2 M NH_3 solution then what is the concentration of $Ag(NH_3)^+$ in solution?

Given : $K_{f_1}[Ag(NH_3)^+] = 10^3$; $K_{f_2}[Ag(NH_3)_2^+] = 10^4$

(a) 3.33×10^{-7} M (b) 3.33×10^{-5} M

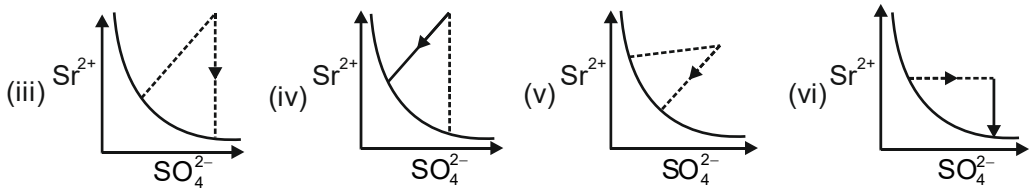
(c) 3×10^{-4} M (d) 10^{-7} M

96. The simultaneous solubility of AgCN ($K_{sp} = 2.5 \times 10^{-16}$) and AgCl ($K_{sp} = 6 \times 10^{-10}$) in 1.0 M NH_3 (aq.) are respectively : [Given : $K_f[Ag(NH_3)_2^+] = 10^7$]

(a) 0.037, 5.78×10^{-8} (b) 5.78×10^{-8}, 0.037

(c) 0.04, 6.25×10^{-8} (d) 1.58×10^{-3}, 1.26×10^{-5}

97. There exist an equilibrium between solid $SrSO_4$ and Sr^{2+} and SO_4^{2-} ion in aqueous medium. The possible equilibrium states are shown in figure as thick line. Now, if equilibrium is disturbed by addition of (I) $Sr(NO_3)_2$ and (II) K_2SO_4 and dotted line represent approach of system towards equilibrium. Match the columns given below :



(I) Addition of $Sr(NO_3)_2$ (II) Addition of K_2SO_4

(a) (I) – (iii), (II) – (iv) (b) (I) – (iv), (II) – (v)

(c) (I) – (vi), (II) – (v) (d) (I) – (iv), (II) – (vi)

98. The salt $Al(OH)_3$ is involved in the following two equilibria,

$$Al(OH)_3(s) \rightleftharpoons Al^{3+}(aq.) + 3OH^-(aq.); K_{sp}$$

$$Al(OH)_3(s) + OH^-(aq.) \rightleftharpoons Al(OH)_4^-(aq.); K_c$$

Which of the following relationship is correct at which solubility is minimum?

(a) $[OH^-] = \left(\frac{K_{sp}}{K_c}\right)^{\frac{1}{3}}$

(b) $[OH^-] = \left(\frac{K_c}{K_{sp}}\right)^{\frac{1}{4}}$

(c) $[OH^-] = \sqrt{\left(\frac{K_{sp}}{K_c}\right)^{\frac{1}{4}}}$

(d) $\left[OH^-\right] = \left(\frac{3K_{sp}}{K_c}\right)^{\frac{1}{4}}$

99. If K_{a_1}, K_{a_2} and K_{a_3} be the first, second and third ionization constant of H_3PO_4 and $K_{a_1} >> K_{a_2} >> K_{a_3}$ which is/are correct?

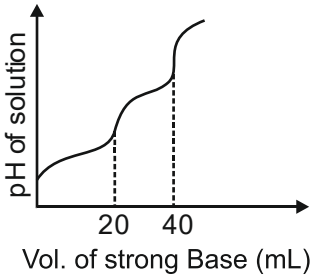
(a) $[H^+] \simeq \sqrt{K_{a_1}[H_3PO_4]}$

(b) $[H^+] \simeq [HPO_4^{2-}]$

(c) $K_{a_2} \simeq [HPO_4^{2-}]$

(d) $[HPO_4^{2-}] = [PO_4^{3-}]$

100. 10 mL of H_2A (weak diprotic acid) solution is titrated against 0.1 M NaOH. pH of the solution is plotted against volume of strong base added and following observation is made.



If pH of the solution at first equivalence point is pH_1 and at second equivalence point is pH_2. Calculate the value of $(pH_2 - pH_1)$ at 25°C.

Given : For H_2A, $pK_{a_1} = 4.6$ and $pK_{a_2} = 8$, $\log 25 - 1.4$

101. The rate of reaction is expressed in different ways as follows:

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is

(a) $4A + B \rightarrow 2C + 3D$

(b) $B + 3D \rightarrow 4A + 2C$

(c) $A + B \rightarrow C + D$

(d) $B + D \rightarrow A + C$

102. The decomposition of azo methane, at certain temperature according to the equation

$(CH_3)_2N_2 \rightarrow C_2H_6 + N_2$ is a first order reaction.

After 40 minutes from the start, the total pressure developed is found to be 350 mm Hg in place of initial pressure 200 mm Hg of azo methane. The value of rate constant k is :

(a) $2.88 \times 10^{-4}\ s^{-1}$ (b) $1.25 \times 10^{-4}\ s^{-1}$

(c) $5.77 \times 10^{-4}\ s^{-1}$ (d) None of these

103. A $\rightarrow$ B first order reaction, A is optically active and B is optically inactive. A series of experiment were conducted on a solution of A

Time	0	60 min	∞
Optical rotation	82°	77°	2°

Assume some impurity is present, calculate the optical rotation after 5 hours.

(Given, In 1.066 = 0.064 $e^{0.16}$ = 1.17)

(a) 60 (b) 30

(c) 20 (d) 120

104. A decomposes as $A(g) \xrightarrow[k_2 = 1 \times 10^{-3}\ s^{-1}]{k_1 = 2 \times 10^{-3}\ s^{-1}} \begin{matrix} 2B(g) \\ C(g) \end{matrix}$, the rate of appearance of B, taking 2 M concentration of A, is equal to

(a) $2 \times 10^{-3}\ M\ s^{-1}$ (b) $4 \times 10^{-3}\ M\ s^{-1}$

(c) $8 \times 10^{-3}\ M\ s^{-1}$ (d) None of these

105. For a complex reaction $A \xrightarrow{k}$ products

$E_{a_1} = 180\ kJ/mol; E_{a_2} = 80\ kJ/mol; E_{a_3} = 50\ kJ/mol$

Overall rate constant k is related to individual rate constant by the equation $k = \left(\frac{k_1 \cdot k_2}{k_3}\right)^{\frac{2}{3}}$ (Activation energy (kJ/mol) for the overall reaction is

(a) 100 (b) 43.44

(c) 150 (d) 140

106. A radioactive sample had an initial activity of 56 dpm. After 69.3 minutes, it was found to have an activity of 28 dpm. Find the number of atoms in a sample having an activity of 100 dpm.

(a) 693 (b) 100

(c) 1000 (d) 10,000

107. In a sample of wood, the reading of a counter is 32 dpm and in a fresh sample of tree it is 122 dpm. Due to error counter gives the reading 2 dpm in absence of ^{14}C. Half life of ^{14}C is 5770 years. The approximate age (in years) of wood sample is

(a) 7997.2 (b) 57570

(c) 11,540 (d) 15140

108. $_{84}Po^{218}$ ($t_{1/2}$ = 183 s) decay to $_{82}Pb$ ($t_{1/2}$ = 161 s) by α-emission, while Pb^{214} is a β-emitter. In an experiment starting with 1 mole of pure Po^{218}, how much time would be required for the number of nuclei of $_{82}Pb^{214}$ to reach maximum?

(a) 147.5 (b) 247.5

(c) 182 (d) 304

109. The reaction A(g) $\rightarrow$ B(g) + 2C(g) is a first order reaction with rate constant 2.772 × 10^{-3} s^{-1}. Starting with 0.1 mole of A in 2 litre vessel, find the concentration of A after 250 s when the reaction is allowed to take place at constant pressure at 300 K.

(a) 0.0125 M (b) 0.025 M

(c) 0.05 M (d) None of these

110. A(aq) $\rightarrow$ B(aq) + C(aq) is a first order reaction.

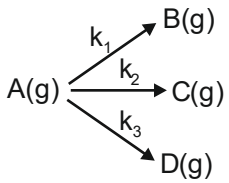
Time	t	∞
moles of reagent	n_1	n_2

Reaction progress is measured with the help of titration of reagent 'R'. If all A, B and C react with reagent and have 'n' factors [n factor; $\text{eq. wt.} = \frac{\text{mol. wt.}}{n}$] in the ratio of 1 : 2 : 3 with the reagent. The k in terms of t, n_1 and n_2 is

(a) $k = \frac{1}{t}\ln\left(\frac{n_2}{n_2 - n_1}\right)$ (b) $k = \frac{1}{t}\ln\left(\frac{2n_2}{n_2 - n_1}\right)$

(c) $k = \frac{1}{t}\ln\left(\frac{4n_2}{n_2 - n_1}\right)$ (d) $k = \frac{1}{t}\ln\left(\frac{4n_2}{5(n_2 - n_1)}\right)$

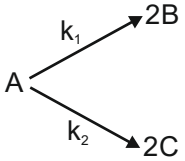
111. A gaseous compound A reacts by three independent first order processes (as shown in figure) with rate constant 2 × 10^{-3}, 3 × 10^{-3} and 1.93 × 10^{-3} s^{-1} for products B, C and D respectively. If initially pure A was taken in a closed container with P = 8 atm, then the partial pressure of B (in atm) after 100 s from starting of experiment is



(a) 0.288 (b) 0.577

(c) 1.154 (d) None of these

112. For given hypothetical elementary parallel reaction,



where $\frac{k_1}{k_2} = \frac{1}{2}$

Initially only 2 moles of A are present. The total no. of moles of A, B and C at the end of 75% reaction are

(a) 2 (b) 3

(c) 4 (d) 3.5

113. The reaction cis-X $\underset{k_b}{\overset{k_f}{\rightleftharpoons}}$ trans-X is first order in both directions. At 25°C, the equilibrium constant is 0.10 and the rate constant $k_f = 3 \times 10^{-4}\ s^{-1}$. In an experiment starting with the pure cis-form, how long would it take for half of the equilibrium amount of the trans-isomer to be formed?

(a) 150 s

(b) 200 s

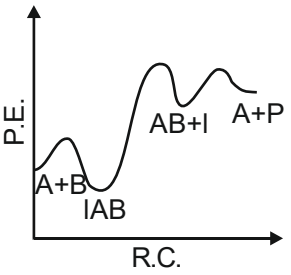
(c) 240 s

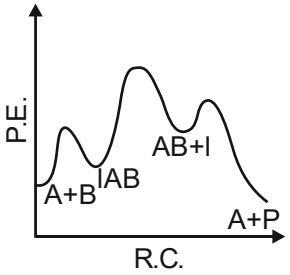
(d) 210 s

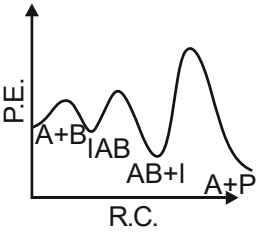
114. The following mechanism has been proposed for the exothermic catalyzed complex reaction

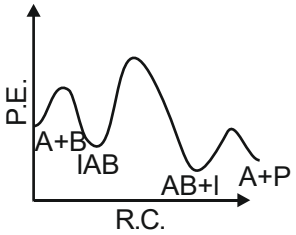
$$A + B \underset{\text{fast}}{\overset{\text{slow}}{\rightleftharpoons}} IAB \xrightarrow{k_1} AB + I \xrightarrow{k_2} P + A$$

If k_1 is much smaller than k_2, the most suitable qualitative plot of potential energy (P. E.) versus reaction co-ordinate (R.C.) for the above reaction is

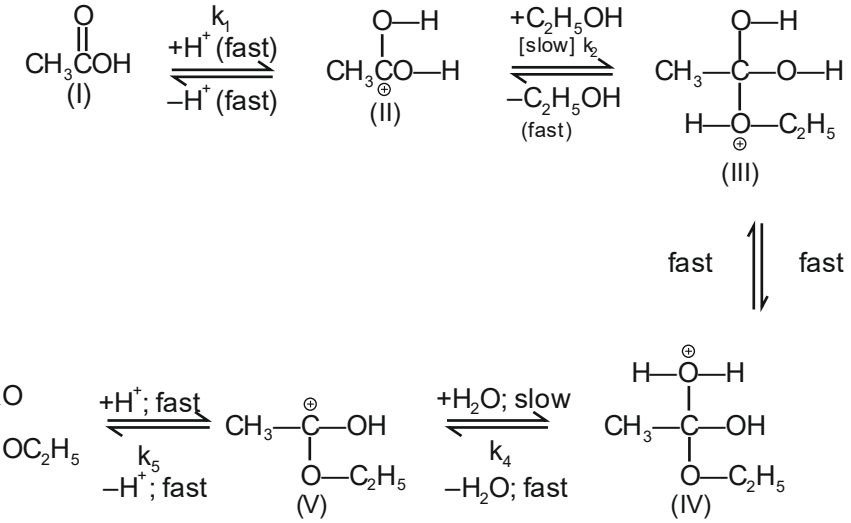
(a) 

(b) 

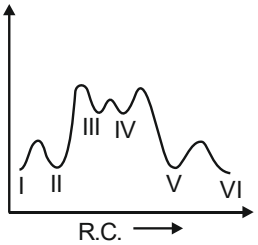
(c) 

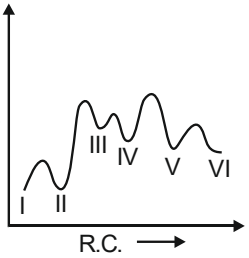
(d) 

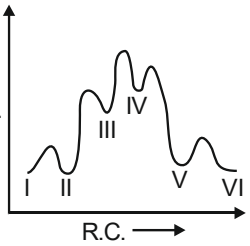
115. The mechanism of esterification in presence of acid catalyst (H_2SO_4) is proposed as follows :

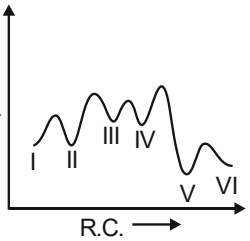


Which of the following potential energy versus reaction co-ordinate diagram is consistent with given mechanism?

(a) 

(b) 

(c) 

(d) 

116. A fresh radioactive mixture containing short lived species A and B. Both emitting α-particles initially of 8000 α-particles per minute. 20 minutes later, they emit at the rate of 3500 α-particles per minute. If the half-lives of the species A and B are 10 minutes and 500 hours respectively, then the ratio of activities of A : B in the initial mixture was

(a) 4 : 6

(b) 6 : 4

(c) 3 : 4

(d) 3 :1

Paragraph for Q. Nos. 117 to 119

Two consecutive irreversible first order reactions can be represented by

$$A \xrightarrow{k_1} B \xrightarrow{k_2} C$$

The rate equation for A is readily integrated to obtain

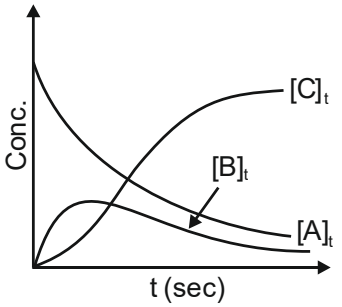
$$[A]_t = [A]_0 \cdot e^{-k_1 t}; \text{ and } [B]_t = \frac{k_1[A]_0}{k_2 - k_1}[e^{-k_1 t} - e^{-k_2 t}]$$

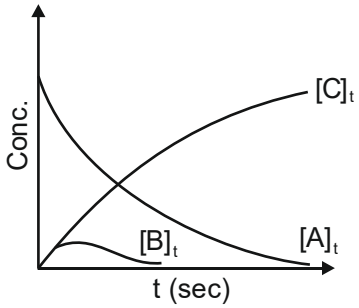
117. At what time will B be present in greatest concentration?

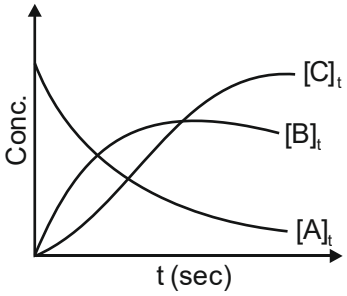
(a) $\frac{k_1}{k_2 - k_1}$ (b) $\frac{1}{k_1 - k_2} \ln \frac{k_1}{k_2}$

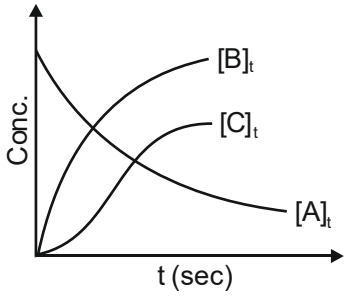
(c) $\frac{1}{k_2 - k_1} \ln \frac{k_1}{k_2}$ (d) None of these

118. When $k_1 = 1\ s^{-1}$ and $k_2 = 50\ s^{-1}$; select most appropriate graph.

(a)


(b)


(c)


(d)


119. If k_1 and k_2 both are almost same then which graph is most suitable?

(a) Graph A (b) Graph B

(c) Graph C (d) Graph D

120. Select the correct statement(s).

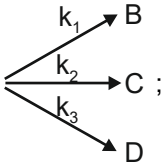
(a) The rate law of the elementary reaction; $2A \rightarrow B + C$, must be $r = k\ (A)^2$

(b) The rate law for the complex reaction

$A + B \rightarrow C$, might not be $r = k[A][B]$

(c) If the partial orders differ from the coefficients in the balanced reaction, the reaction must be complex

(d) If the partial orders are equal to corresponding coefficients in the balanced reaction, the reaction must be elementary

121. 

; at time t = 0 (initial mole) of A is 1.

Overall half life of the reaction is 15 days. The number of moles of C after 45 days is n, the value of 100n is if the ratio of $k_1 : k_2 : k_3$ is 4 : 2 : 1

122. In the given radioactive disintegration series

$$^{232}_{90}Th \rightarrow ^{208}_{82}Pb$$

Calculate value of (n + 2).

Where value of n is number of isobars formed in this series, suppose there is successive emission of β-particles.

123. An aqueous solution containing 1 M each of Au^{3+}, Cu^{2+}, Ag^{+}, Li^{+} is being electrolysed by using inert electrodes. The value of standard reduction potentials are

$$E^{o}_{Ag^{+}/Ag} = 0.80 \text{ V}, E^{o}_{Cu^{+}/Cu} = 0.34 \text{ V}, \text{and } E^{o}_{Au^{3+}/Au} = 1.50 \text{ V}, E^{o}_{Li^{+}/Li} = -3.03 \text{ V},$$

With increasing voltage, the sequence of deposition of metals on the cathode will be

(a) Li, Cu, Ag, Au

(b) Cu, Ag, Au

(c) Au, Ag, Cu

(d) Au, Ag, Cu, Li

124. Ammonium perchlorate, NH_4ClO_4, used in the solid fuel in the booster rockets on the space shuttle, is prepared from sodium perchlorate, $NaClO_4$, which is produced commercially by the electrolysis of a hot, stirred solution of sodium chloride. How many faradays are required to produce 1.0 kg of sodium perchlorate?

$$NaCl + 4H_2O \rightarrow NaClO_4 + 4H_2$$

(a) 40.3 (b) 18.3

(c) 31.6 (d) 65.3

125. Consider the following half-cell reactions and associated standard half-cell potentials and determine the maximum voltage that can be obtained by combination resulting in spontaneous processes :

$$AuBr_4^{-}(aq) + 3e^{-} \rightarrow Au(s) + 4Br^{-}(aq); E^{\circ} = -0.86 \text{ V}$$

$$Eu^{3+}(aq) + e^{-} \rightarrow Eu^{2+}(aq); E^{\circ} = -0.43 \text{ V}$$

$$Sn^{2+}(aq) + 2e^{-} \rightarrow Sn(s); E^{\circ} = -0.14 \text{ V}$$

$$IO^{-}(aq) + H_2O(l) + 2e^{-} \rightarrow I^{-}(aq) + 2OH^{-}(aq); E^{\circ} = +0.49 \text{ V}$$

(a) +0.72 (b) +1.54

(c) +1.00 (d) +1.35

126. Consider the following equation for an electrochemical cell reaction. Which of the following changes in condition will increase the cell voltage?

$$H_2(g) + PbCl_2(s) \rightarrow Pb(s) + 2HCl(aq)$$

(I) Dissolve concentrated $HClO_4$ in the cell solution

(II) Increase the pressure of $H_2(g)$

(III) Increase the amount of Pb(s)

(a) III (b) I and II

(c) II and III (d) II

127. The temperature coefficient of a cell whose operation is based on the reaction

$Pb(s) + HgCl_2(aq) \rightarrow PbCl_2(aq) + Hg(l)$ is :

$\left(\frac{dE}{dT}\right)_p = 1.5 \times 10^{-4}\ VK^{-1}$ at 298 K

The change in entropy (in J/K mol) during the operation is

(a) 8627 (b) 57.9

(c) 28.95 (d) 14.475

128. The specific conductance of a saturated solution of silver bromide is κ S cm^{-1}. The limiting ionic conductivity of Ag^+ and Br^- ions are x and y, respectively. The solubility of silver bromide in gL^{-1} is (molar mass of AgBr = 188)

(a) $\frac{\kappa \times 1000}{x - y}$ (b) $\frac{\kappa}{x + y} \times 188$

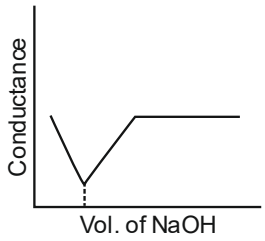
(c) $\frac{\kappa \times 1000 \times 188}{x + y}$ (d) $\frac{x + y}{\kappa} \times \frac{1000}{188}$

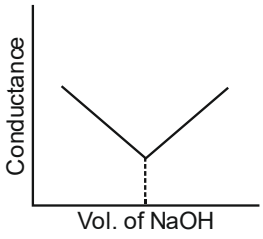
129. The conductance of a salt solution (AB) measured by two parallel electrodes of area 100 cm^2 separated by 10 cm was found to be 0.0001 Ω^{-1}. If volume enclosed between two electrode contain 0.1 mole of salt, what is the molar conductivity (S cm^2 mol^{-1}) of salt at same concentration?

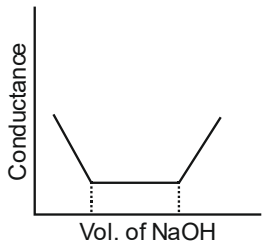
(a) 10 (b) 0.1

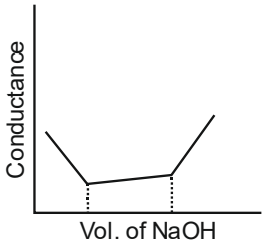
(c) 1 (d) None of these

130. Conductometric titration curve of a equimolar mixture of a HCl and HCN with NaOH(aq) is

(a)


(b)


(c)


(d)


131. In the Hall's process, aluminum is produced by the electrolysis of molten Al_2O_3. How many second would it take to produce enough aluminum by the Hall's process to make a case of 24 cans of aluminum soft-drink, if each can uses 5.0 g of Al, a current of 9650 A is employed, and the current efficiency of the cell is 90.0%?

(a) 203.2 (b) 148.14

(c) 333 (d) 6.17

132. 100 mL of 0.05 M $CuSO_4$(aq) solution was electrolyzed by using inert electrodes by passing current till the pH of the resulting solution was 2. The solution after electrolysis was neutralized and then treated with excess KI and formed I_2 titrated with 0.04 M $Na_2S_2O_3$. Calculate the required volume (in mL) of $Na_2S_2O_3$.

(a) 112.5 mL (b) 100 mL

(c) 125 mL (d) None of these

133. Copper reduces NO_3^- into NO and NO_2 depending upon concentration of HNO_3 in solution. Assuming $[Cu^{2+}] = 0.1\,M$, and $P_{NO} = P_{NO_2} = 10^{-3}$ bar. At which concentration of HNO_3, thermodynamic tendency for reduction of NO_3^- into NO and NO_2 by copper is same?

$\left[\text{Given}: E^o_{Cu^{2+}|Cu} = +0.34 \text{ volt}, E^o_{NO_3^-|NO} = +0.96 \text{ volt}, E^o_{NO_3^-|NO_2} = +0.79 \text{ volt}\right]$

(a) $10^{1.23}$ M (b) $10^{0.56}$ M

(c) $10^{0.66}$ M (d) $10^{0.12}$ M

134. Given the following molar conductivities at 25°C; HCl, 426 Ω^{-1} cm^2mol^{-1}; NaCl, 126 Ω^{-1} cm^2 mol^{-1}; NaC (sodium crotonate), 83 Ω^{-1} cm^2 mol^{-1}. What is the ionization constant of crotonic acid, if the conductivity of a 0.001 M crotonic acid solution is 3.83×10^{-5} Ω^{-1} cm^{-1} ?

(a) 10^{-5} (b) 1.11×10^{-5}

(c) 1.11×10^{-4} (d) 0.01

135. The conductivity of 0.001 M Na_2SO_4 solution is 2.6×10^{-4} S crn^{-1} and increase to 7.0×10^{-4} S cm^{-1}, when the solution is saturated with $CaSO_4$. The molar conductivities of Na^+ and Ca^{2+} are 50 and 120 S cm^2 mol^{-1}, respectively. Neglect conductivity of used water.

What is the solubility product of $CaSO_4$?

(a) 4×10^{-6} (b) 1.57×10^{-3}

(c) 4×10^{-4} (d) 2.46×10^{-6}

136. Three electrolytic cells X, Y, Z containing solution of NaCl, $AgNO_3$ and $CuSO_4$ respectively are connected in series combination. During electrolysis 21.6 gm of silver deposite at cathode in cell Y. Which is incorrect statement?

(a) 6.35 gm copper deposite at cathode in cell Z

(b) 2.24 litre Cl_2 liberated at 1 atm and 273 K at cathode in cell X

(c) 2.24 litre O_2 liberated at 1 atm and 273 K at anode in cell Y

(d) 2.24 litre H_2 liberated at 1 atm and 273 K at cathode in cell X

137. Standard electrode potential of two half-reactions are given below :

$Fe^{2+} \rightleftharpoons Fe \qquad E^\circ = -0.44$ V

$Fe^{3+} \rightleftharpoons Fe^{2+} \qquad E^\circ = +0.77$ V

If Fe^{2+}, Fe^{3+} and Fe are kept together

(a) The concentration of Fe^{3+} increases

(b) The concentration of Fe^{3+} decreases

(c) The mass of Fe increases

(d) The concentration of Fe^{2+} decreases

138. A dilute solution of KCl was placed between two Pt electrode 10 cm apart across which a potential difference of 10 volt was applied. Which is/are correct statement? (Given : molar conductivity of K^+ at infinite dilution is 96.5 S cm^2 mol^{-1}).

(a) Ionic mobility of K^+ is 10^{-3} cm^2 s^{-1} $volt^{-1}$

(b) The speed of K^+ is 10^{-3} cm s^{-1}

(c) Distance travelled by K^+ is 5×10^3 s is 5 cm

(d) The potential gradient is 1.0 volt cm^{-1}

139. Given : $Pt(s)\,|\,\underset{1\,bar}{H_2(g)}\,|\,0.1\,M\ NH_4OH\ (aq)\,||\,0.1\,M\ CH_3COOH\ (aq)\,|\,\underset{1\,bar}{H_2(g)}\,|\,Pt(s)$

$pK_b(NH_4OH) = 5;\ pK_a(CH_3COOH) = 5;\ \dfrac{2.303RT}{F} = 0.06$

Volume of 0.1 M NH_4OH in anode half cell = 100 mL,

Volume of 0.1 M CH_3COOH in cathode half cell = 100 mL

Which is/are correct statement?

(a) The emf of given cell is 0.48 V

(b) The emf of given cell is 0.36 V when 50 mL, 0.1 M NaOH added to cathode compartment

(c) The emf of given cell is 0.36 V when 50 mL 0.1 M HCl added to anode compartment

(d) The emf of given cell is 0.192 V when 100 mL 0.1 M NaOH added to anode compartment

140. What is the potential of an electrode which originally contained 0.1 M NO_3^- and 0.4 M H^+ and which has been treated by 80% of the cadmium necessary to reduce all the NO_3^- to NO(g) at 1 bar?

Given : $NO_3^- + 4H^+ + 3e^- \rightarrow NO + 2H_2O;\ E^\circ = 0.96\ V;\ \log 2 = 0.3$

(a) 0.84 V (b) 1.08 V

(c) 1.23 V (d) 1.36 V

141. Given : $Pt(s)\,|\,\underset{P_1\,atm}{Cl_2(g)}\,|\,Cl^-(C_1)\,||\,Cl^-(C_2)\,|\,\underset{P_2\,atm}{Cl_2(g)}\,|\,Pt(s)$

Identify in which of following condition working of cell takes place.

(a) $C_1 > C_2$ and $P_1 = P_2$

(b) $P_2 > P_1$ and $C_1 = C_2$

(c) $C_1 < C_2$ and $P_1 = P_2$

(d) $P_2 < P_1$ and $C_1 = C_2$

142. A sample of liquid H_2O of mass 18.0 g is injected into an evacuated 7.6 L flask maintained at 27.0°C. If vapour pressure of H_2O at 27°C is 24.63 mm Hg. What weight percentage of the water will be vapourised when the system comes to equilibrium? Assume water vapours behaves as an ideal gas. The volume occupied by the liquid water is negligible compared to the volume of the container.

(a) 1% (b) 10%

(c) 18% (d) 20%

143. The solubility of a specific non-volatile salt is 4 g in 100 g of water at 25°C. If 2.0 g, 4.0 g and 6.0 g of the salt added to 100 g of water at 25°C, in system X, Y and Z. The vapour pressure would be in the order:

(a) X < Y < Z (b) X > Y > Z

(c) Z > X = Y (d) X > Y = Z

144. Equal weight of a solute are dissolved in equal weight of two solvents A and B and formed very dilute solution. The relative lowering of vapour pressure for the solution B has twice the relative lowering of vapour pressure for the solution A. If M_A and M_B are the molecular weights of solvents A and B respectively, then

(a) $M_A = M_B$ (b) $M_B = 2 \times M_A$

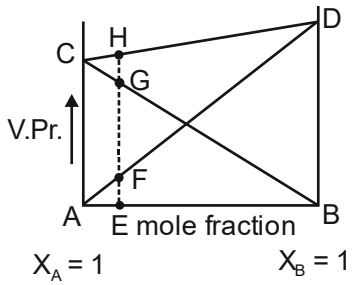
(c) $M_A = 4M_B$ (d) $M_A = 2M_B$

145. Two liquids A and B from ideal solutions. At 300 K, the vapour pressure of solution containing 1 mole of A and 3 mole of B is 550 mm Hg. At the same temperature, if one more mole of B is added to this solution, the vapour pressure of the solution increases by 10 mm Hg. Determine the vapour pressure of A and B in their pure states (in mm Hg)

(a) 400, 600 (b) 500, 500

(c) 600, 400 (d) None of these

146. Based on the given diagram, which of the following statements regarding the homogeneous solutions of two volatile liquids are correct?



(1) Plots AD and BC show that Raoult's law is obeyed for the solution in which B is a solvent and A is the solute and as well as for that in which A is solvent and B is solute.

(2) Plot CD shows that Dalton's law of partial pressures is obeyed by the binary solution of components A and B.

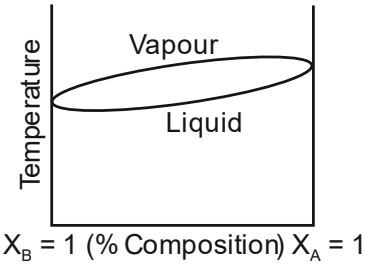
(3) EF + EG = EH; and AC and BD correspond to the vapour pressures of the pure solvents A and B respectively.

Select the correct answer using the options given below

(a) Only 1 (b) 2 and 3

(c) 1 and 3 (d) All

147. Boiling point composition diagram of the liquid-vapour equilibrium for A and B is shown in the figure. If a binary liquid mixture of A and B is distilled fractionally, which of the following would be correct observation?



(a) Composition of the still (residue) will approach pure liquid B

(b) Composition of the distillate will approach pure A

(c) Composition of distillate and residue will approach pure B and A respectively

(d) Neither of the component can be obtained in pure state

148. When a liquid that is immiscible with water was steam distilled at 95.2°C at a total pressure of 748 torr, the distillate contained 1.25 g of the liquid per gram of water. The vapour pressure of water is 648 torr at 95.2°C, what is the molar mass of liquid?

(a) 7.975 g/mol (b) 166 g/mol

(c) 145.8 g/mol (d) None of these

149. A very dilute saturated solution of a sparingly soluble salt X_3Y_4 has a vapour pressure of 20 mm Hg at temperature T, while pure water exerts a pressure of 20.0126 mm Hg at the same temperature. Calculate molality (m) at temperature T.

(a) 6.3×10^{-4} (b) 3.5×10^{-2}

(c) 5×10^{-3} (d) None of these

150. A complex is represented as $CoCl_3xNH_3$. Its 0.1 molal solution in water shows $\Delta T_f = 0.558$ K. K_f for H_2O is 1.86 K molality^{-1}. Assuming 100% ionisation of complex and co-ordination number of Co is six, calculate formula of complex.

(a) $[Co(NH_3)_6]Cl_3$ (b) $[Co(NH_3)_5Cl]Cl_2$

(c) $[Co(NH_3)_4Cl_2]Cl$ (d) None of these

151. If 1 m solution of benzoic acid in benzene has a freezing point depression of 3.84°C. ($K_f = 5.12$°C mol^{-1} kg) and boiling point elevation of 2.53°C (K_b, = 2.53°C mol^{-1} kg), then select the correct statement(s)

Statement I : There is dimer formation when undergoing freezing

Statement II : There is no change when undergoing boiling

Statement III : Reverse of I and II

Statement IV : Dimer formation in freezing and boiling state

(a) I, II (b) II, III

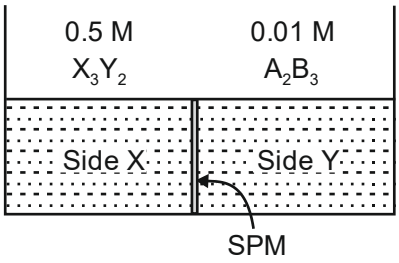
(c) III, I (d) Only I

152. Insulin is dissolved in a suitable solvent and the osmotic pressure (π) of solutions of various concentrations C(g/cm^3) is measured at 20°C. The slope of a plot of π against C is found to be 4.65×10^{-3}. The molecular weight of the insulin (g/mol) is

(a) 3×10^5 (b) 9×10^5

(c) 4.5×10^5 (d) 5.16×10^6

153. X_3Y_2 (i = 5) when reacted with A_2B_3 (i = 5) in aqueous solution gives brown colour. These are separated by a semipermeable membrane AB as shown. Due to osmosis there is



(a) Brown colour formation in side X

(b) Brown colour formation in side Y

(c) Formation in both of the sides X and Y

(d) No brown colour formation

154. According to Henry's law, the partial pressure of gas (P_g) is directly proportional to mole fraction of gas in liquid solution, $P_{gas} = K_H \cdot X_{gas}$ where K_H is Henry's constant. Which is incorrect?

(a) K_H is characteristic constant for a given gas-solvent system

(b) Higher is the value of K_H, lower is solubility of gas for a given partial pressure of gas

(c) K_H has temperature dependence

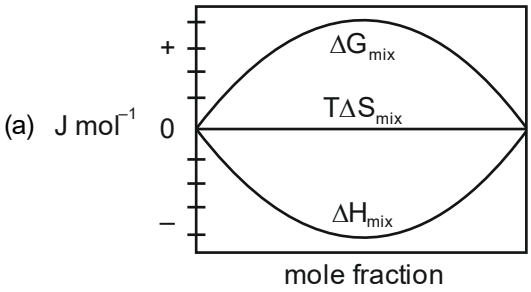
(d) K_H decreases with increase of temperature

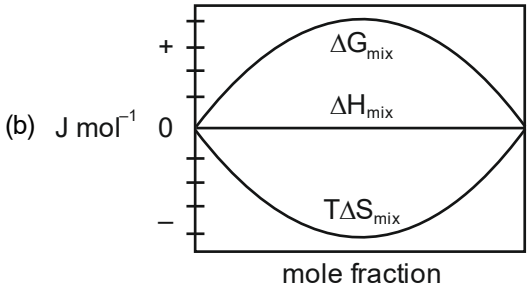
155. The vapour pressure of two pure liquids A and B, that form an ideal solution are 100 and 900 torr respectively at temperature T. This liquid solution of A and B is composed of 1 mole of A and 1 mole of B. What will be the pressure, when 1 mole of mixture has been vaporized?

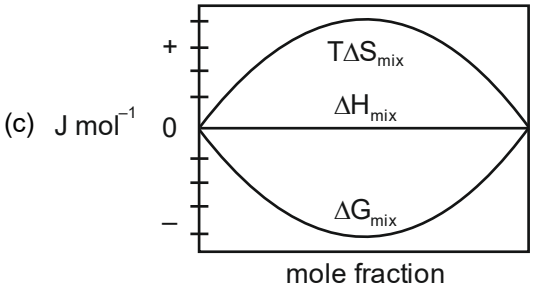
(a) 800 torr (b) 500 torr

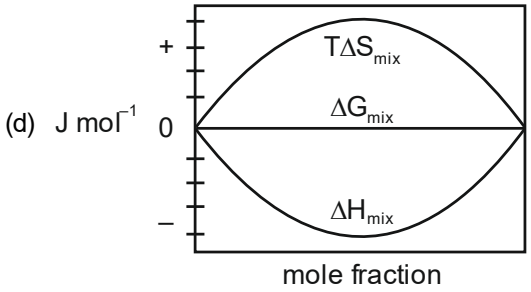
(c) 300 torr (d) None of these

156. Which of the following represents correctly the changes in thermodynamics properties during the formation of 1 mole of an ideal binary solution?

(a) 

(b) 

(c) 

(d) 

157. Ratio of $\frac{\Delta T_b}{K_b}$ of 10 g AB_2 and 14 g A_2B per 100 g of solvent in their respective, solution (AB_2 and A_2B both are non-electrolytes) is 1 mol/kg in both cases. Hence, atomic wt. of A and B are respectively

(a) 100, 40

(b) 60, 20

(c) 20, 60

(d) None of these

158. Each edge of a cubic unit cell is 400 pm long. If atomic weight of the element is 120 and its density is 6.25 g/cm^3. The crystal lattice is (use $N_A = 6 \times 10^{23}$)

(a) Primitive (b) Body centered

(c) Face centered (d) End centered

159. By X-ray diffraction it is found that nickel (atomic mass = 59 g mol^{-1}), crystallizes with ccp. The edge length of the unit cell is 3.5 Å. If density of Ni crystal is 9.0 g/cm^3. Then value of Avogadro's number from the data is

(a) 6.05×10^{23} (b) 6.11×10^{23}

(c) 6.02×10^{23} (d) 6.023×10^{23}

160. Which one of the following schemes of ordering closed packed sheets of equal sized spheres do not generate closest packed lattice?

(a) ABCABC

(b) ABACABAC

(c) ABBAABBA

(d) ABCBCABCBC

161. An alloy of copper, silver and gold is found to have copper constituting the ccp lattice. If silver atoms occupy the edge centres and gold is present at body centre, the alloy will have the formula

(a) Cu_4Ag_2Au (b) Cu_4Ag_4Au

(c) Cu_4Ag_3Au (d) CuAgAu

162. A certain sample of cuprous sulphide is found to have formula $Cu_{1.8}S$, because of incorporation of Cu^{2+} ions in the lattice. What is the mole % of Cu^{2+} in total copper content in this crystal?

(a) 99.8% (b) 11.11%

(c) 88.88% (d) None of these

163. When heated above 916°C, iron changes its bcc crystalline form to fcc without the change in the radius of atom. The ratio of density of the crystal before heating and after heating is:

(a) 1.069 (b) 0.918

(c) 0.725 (d) 1.231

164. A body centered cubic lattice is made up of hollow spheres of B. Spheres of solid A are present in hollow spheres of B. Radius of A is half of radius of B. What is the ratio of total volume of spheres of B unoccupied by A in a unit cell and volume of unit cell?

(a) $\frac{7\sqrt{3\pi}}{64}$ (b) $\frac{7\sqrt{3}}{128}$

(c) $\frac{7 \cdot \pi}{24}$ (d) None of these

165. The density of a pure substance 'A' whose atoms are in cubic close pack arrangement is 1 g/cc. If all the tetrahedral voids are occupied by 'B' atom, what is the density of resulting solid in g/cc? [Atomic mass (A) = 30 g/mol and atomic mass (B) = 50 g/mol]

(a) 3.33 (b) 4.33

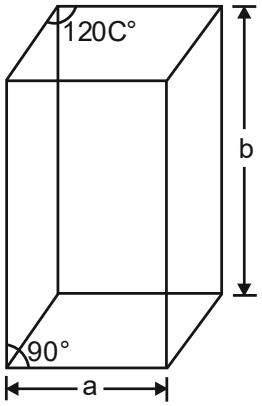
(c) 2.33 (d) 5.33

166. An element X (atomic weight 24 gm/mol) forms a face centered cubic lattice. If the edge length of the lattice is 4×10^{-8} cm and the observed density is 2.40×10^3 kg/m^3, then the percentage occupancy of lattice points by element X is (Use $N_A = 6 \times 10^{23}$)

(a) 96 (b) 98

(c) 99.9 (d) None of these

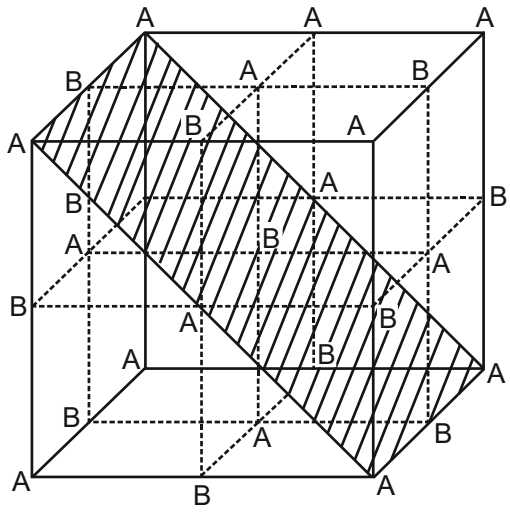
167. A_2B molecules (molar mass = 259.8 g/mol) crystallises in a hexagonal lattice as shown in figure. The lattice constants were a = 5 Å and b = 8 Å. If density of crystal is 5 g/cm^3 then how many molecules are contained in given unit cell? (use $N_A = 6 \times 10^{23}$)



(a) 6 (b) 4

(c) 3 (d) 2

168. A crystal is made of particles A and B. A forms fcc packing and B occupies all the octahedral voids. If all the particles along the plane as shown in figure are removed, then, the formula of the crystal would be:



(a) AB (b) A_5B_7

(c) A_7B_5 (d) None of these

Paragraph for Q. Nos. 169 to 172

A spinel is an important class of oxides consisting of two types of metal ions with the oxide ions arranged in ccp layers. The normal spinel has one-eighth of the tetrahedral holes occupied by one type of metal ion and one-half of the octahedral holes occupied by another type of metal ion. Such a spinel is formed by Mg^{2+}, Al^{3+} and O^{2-}. The neutrality of the crystal is being maintained.

169. The formula of the spinel is

(a) Mg_2AlO_4 (b) $MgAl_2O_4$

(c) $Mg_3Al_2O_6$ (d) None of these

170. Type of hole occupied by Al^{3+} ions is

(a) Tetrahedral (b) Octahedral

(c) Both (a) and (b) (d) None of these

171. Type of hole occupied by Mg^{2+} ions is

(a) Tetrahedral (b) Octahedral

(c) Both (a) and (b) (d) None of these

172. If oxide ion is replaced by $X^{-8/3}$, the number of anionic vacancy per unit cell is

(a) 1 (b) 2

(c) 3 (d) $\frac{3}{4}$

173. Select the correct statement(s).

(a) A cubic system possesses a total of 23 elements of symmetry

(b) A cubic system contains centre of symmetry, planes of symmetry as well as axes of symmetry

(c) For triclinic system $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma \neq 90°$

(d) The total no. of Bravais space lattice belonging to all the seven crystals are 14

174. Given: Radius of A^{2+} = 100 pm; Radius of C^{+} = 240 pm; Radius of B^{2-} = 300 pm; Radius of D^{-} = 480 pm. Which is/are correct statement?

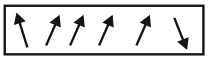
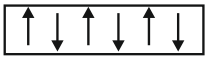
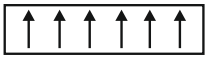
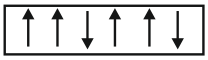
(a) Coordination number of A^{2+} in compound AB is 4

(b) Coordination number of A^{2+} in compound AB is 6.

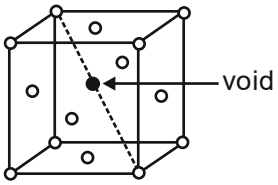
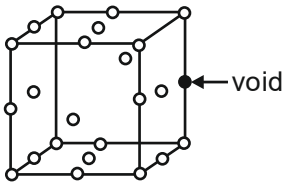
(c) Coordination number of C^{+} in compound CD is 6

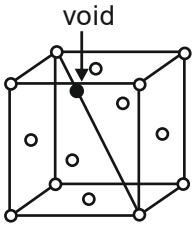
(d) Coordination number of C^{+} in compound CD is 8.

175. Match Column-I with Column-II.

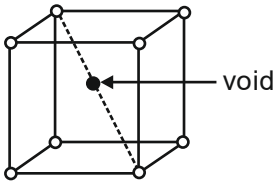
Column-I	**Column-II**
(a) Magnetic moment in a paramagnetic substance	(p) 
(b) Magnetic moment in a ferromagnetic	(q) 
(c) Magnetic moment in a antiferromagnetic	(r) 
(d) Magnetic moment in a ferrimagnetic	(s) 

176. Match Column-I with Column-II.

Column-I (Void representation)	**Column-II (Name of void)**
(a) 	(p) Tetrahedral void
(b) 	(q) Octahedral void

(c) 

(r) Cubic void

(d) 

(s) Triangular void

177. Ionic solid Na^+A^- crystallise in rock salt type structure. 2.592 gm of ionic solid salt NaA dissolved in water to make 2 litre solution. The pH of this solution is 8. If distance between cation and anion is 300 pm. Calculate density of ionic solid (in gm/cm^3).

(Given: $pK_w = 13$, $pK_a(HA) = 5$, $N_A = 6 \times 10^{23}$)

178. Calculate the value of $\frac{Z}{10}$. Where

Z = Co-ordination number of 2D-square close packing

+

Co-ordination number of 2D-hcp

+

Co-ordination number of 3D-square close packing

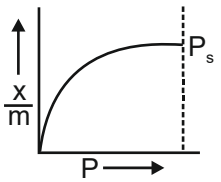
+

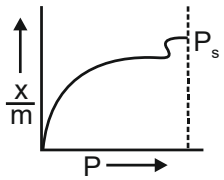
Co-ordination number of 3D, ABCABC ..... packing

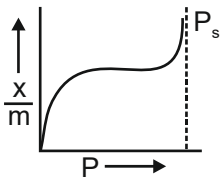
+

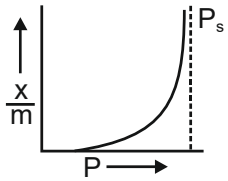
Co-ordination number of 3D, ABAB ..... packing

179. Which of the following adsorption isotherms represents the adsorption of a gas by a solid involving monolayer formation? (P_s = saturation pressure)

(a) 

(b) 

(c) 

(d) 

180. On adding $AgNO_3$ solution into KI solution, a negatively charged colloidal sol is obtained when they are in

(a) 50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.1 M KI (b) 50 mL of 0.1 M $AgNO_3$ + 50 mL of 0.2 M KI

(c) 50 mL of 0.2 M $AgNO_3$ + 50 mL of 0.1 M KI (d) None of these

181. The gold numbers of protective colloids A, B, C and D are 0.04, 0.004, 10 and 40 respectively. The protective powers of A, B, C and D are in the order

(a) A > B > C > D (b) B > A > C > D

(c) D > C > A > B (d) D > C > B > A

182. Under the influence of an electric field, the particles in a sol migrate towards cathode. The coagulation of the same sol is studied using NaCl, Na_2SO_4 and Na_3PO_4 solutions. Their coagulating values will be maximum for

(a) NaCl (b) Na_2SO_4

(c) Na_3PO_4 (d) Same for all

183. 100 mL of 0.6 M acetic acid is shaken with 2 g activated carbon. The final concentration of the solution after adsorption is 0.5 M. What is the amount of acetic acid adsorbed per gram of carbon?

(a) 0.6 g (b) 0.3 g

(c) 1.2 g (d) None of these

184. A detergent ($C_{12}H_{25}SO_4^-Na^+$) solution becomes a colloidal sol at a concentration of 10^{-3} M. On an average 10^{13} colloidal particles are present in 1 mm^3. What is the average number of ions are contain in one colloidal particle (micelle)? [Given: $N_A = 6 \times 10^{23}$]

(a) 6×10^7 (b) 10

(c) 60 (d) None of these

185. At 1 atm and 273 K the volume of nitrogen gas required to cover a sample of silica gel, assuming Langmuir monolayer adsorption, is found to be 1.30 $cm^3\ g^{-1}$ of the gel. The area occupied by a nitrogen molecule is 0.16 nm^2. What is the surface area per gram of silica gel?

[Given : $N_A = 6 \times 10^{23}$]

(a) 5.568 $m^2\ g^{-1}$ (b) 3.48 $m^2\ g^{-1}$

(c) 1.6 $m^2\ g^{-1}$ (d) None of these

186. 10% sites of catalyst bed have been adsorbed by H_2. On heating H_2 gas is evolved from sites and collected at 0.03 atm and 300 K in a small vessel of 2.46 cm^3. No. of sites available is 5.4×10^{16} per cm^2 and surface area is 1000 cm^2. Find out the no. of surface sites occupied per molecule of H_2. (Given $N_A = 6 \times 10^{23}$)

(a) 1 (b) 2

(c) 3 (d) None of these

187. A sample of 16 g charcoal was brought into contact with CH_4 gas contained in a vessel of 1 litre at 27°C. The pressure of gas was found to fall from 760 to 608 torr. The density of charcoal sample is 1.6 g/cm^3. What is the volume of the CH_4 gas adsorbed per gram of the adsorbent at 608 torr and 27°C?

(a) 125 mL/g (b) 16.25 mL/g

(c) 26 mL/g (d) None of these

❑ ❑ ❑

Book Crunch- 2

Mathematics

MATHEMATICS

1. Let $f : D \to R$ be defined as $f(x) = \dfrac{x^2 + 2x + a}{x^2 + 4x + 3a}$ where D and R denote the domain of f and the set of all real numbers respectively. If f is surjective mapping then the range of a is

   (1) $0 \leq a \leq 1$ (2) $0 < a \leq 1$

   (3) $0 \leq a < 1$ (4) $0 < a < 1$

2. If $f(x) = \pi\left(\dfrac{\sqrt{x+7} - 4}{x - 9}\right)$, then the range of function $y = \sin(2f(x))$ is

   (1) $[0, 1]$ (2) $\left(0, \dfrac{1}{\sqrt{2}}\right]$

   (3) $\left(0, \dfrac{1}{\sqrt{2}}\right) \cup \left(\dfrac{1}{\sqrt{2}}, 1\right]$ (4) $(0, 1]$

3. If the range of function $f(x) = \dfrac{x^2 + x + c}{x^2 + 2x + c}$, $x \in R$ is $\left[\dfrac{5}{6}, \dfrac{3}{2}\right]$, then c is equal to

   (1) –4 (2) 3

   (3) 4 (4) 5

4. If a polynomial function f satisfies the relation

   $$\log_2[f(x)] = \log_2\left(2 + \frac{2}{3} + \frac{2}{9} + \ldots.. + \infty\right) \cdot \log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right)$$ and $f(10) = 1001$, then the value of $f(20)$ is

   (1) 2002 (2) 7999

   (3) 8001 (4) 16001

5. The sum of all positive integral values of a, $a \in [1, 500]$ for which the equation $[x]^3 + x - a = 0$ has solution is
   (Note : [ ] denotes the greatest integer function]

   (1) 462 (2) 512

   (3) 784 (4) 812

6. Let $A = \{1, 2, 3, 4\}$. Number of function $f : A \to A$ are there satisfying $fof(x) = x \ \forall \ x \in A$, is

   (1) 10 (2) 11

   (3) 12 (4) 13

Paragraph for Q. Nos. 7 and 8

Let f be an even function satisfying

$$f(x-2)=f\left(x+\left[\frac{6x^2+13}{x^2+2}\right]\right)\forall x\in R \text{ and } f(x)=\begin{cases}3x, & 0\le x<1\\ 4-x, & 1\le x\le 4\end{cases}$$

(Note : [y] denotes greatest integer function of y]

7. The area bounded by the graph of $f(x)$ and the x-axis from $x = -1$ to $x = 9$ is

(1) $\frac{31}{2}$ (2) 15

(3) 12 (4) $\frac{15}{2}$

8. The value of $f(-89) - f(-67) + f(46)$ is equal to

(1) 4 (2) 5

(3) 6 (4) 7

9. Let f, g and h be three functions defined as follows :

$f(x)=\frac{32}{4+x^2+x^4}$, $g(x) = 9 + x^2$ and $h(x) = -x^2 - 3x + k$

Identify which of the following statement(s) is(are) correct.

(1) Number of integers in the range of $f(x)$ is 8

(2) Number of integral values of k for which $h(f(x)) > 0$ and $h(g(x)) < 0\ \forall\ x \in R$ is 20

(3) Number of integral values of k for which $h(f(x)) > 0$ and $h(g(x)) < 0\ \forall\ x \in R$ is 19

(4) Maximum value of $g(f(x))$ is 73

10. Consider, $P=\frac{x^2-2x}{x^2+x+1}$, $Q=\frac{y-1}{y^2+y+1}$ and $R=\frac{2}{z^2+z+1}$ where $x, y, z \in R$.

If $k = [P + Q + R] - ([P] + [Q] + [R])$ then the possible value(s) of k is(are) (where $[\cdot]$ denotes greatest integer less than or equals to x)

(1) 0 (2) 1

(3) 2 (4) 3

11. Let the function $f:(-\infty, \infty)\to\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ be given by $f(x)=\sin^{-1}\left(\log_3\left(\frac{x^2-x+1}{x^2+x+1}\right)\right)$, then

(1) $f\left(\frac{1}{x}\right)=-f(-x)$ (2) $f(x)$ is a strictly increasing function in $(-\infty, \infty)$

(3) $f(x)$ is a surjective function (4) $f(x)$ is a injective function

12. Let $f : R \to [1, \infty)$ be a quadratic surjective function such that $f(2 + x) = f(2 - x)$ and $f(1) = 2$. If $g : (-\infty, \ln 2] \to [1, 5)$ is given by $g(\ln x) = f(x)$ then which of the following is(are) correct?

(1) The value of $f(3)$ is equal to 2

(2) $g^{-1}(x)=\ln(2-\sqrt{x-1})$

(3) $g^{-1}(x)=\ln(2+\sqrt{x-1})$

(4) The sum of values of x satisfying the equation $f(x) = 5$ is 4

13. Let $f(x)=\begin{cases}x+3 & \text{if} \quad x\in[-4,\ -2)\\ 1 & \text{if} \quad x\in[-2,\ 2)\\ 3-x & \text{if} \quad x\in[2,\ 4]\end{cases}$, $g(x)=\begin{bmatrix}x+6 & x<0\\ 2x+6 & x\geq 0\end{bmatrix}$ then

(1) $gof\ (x) = k$ will have one at least solution if $k \in [5, 8]$

(2) Range of $fog\ (x)$ is $[-1, 1]$

(3) $\lim\limits_{x\to -2} fog(x) = -1$

(4) $gof\ (x)$ is an even function

14. Let f be a function defined in $[-2, 3]$ given as

$$f(x)=\begin{cases}3(x+1)^{1/3}, & -2\leq x<0\\ -(x-1)^2, & 0\leq x<1\\ 2(x-1)^2, & 1\leq x<2\\ -x^2+4x-3, & 2\leq x\leq 3\end{cases}$$

	Column-I	**Column-II**
(A)	The number of integers in the range of $f(x)$ is	(p) 2
(B)	The number of integral values of x which are in the domain of $f(1 - \lvert x\rvert)$, is	(q) 4
(C)	The number of integers in the range of $\lvert f(-\lvert x\rvert)\rvert$, is	(r) 6
(D)	The number of integral values of k for which the equation $f(\lvert x\rvert) = k$ has exactly four distinct solutions is	(s) 7

15. Let $f(x) = |\ x^2 - 9\ | - |\ x - a\ |$. Find the number of integers in the range of a so that $f(x) = 0$ has 4 distinct real root.

16. The set of real values of x satisfying the equality $\left[\frac{3}{x}\right]+\left[\frac{4}{x}\right]=5$ (where [ ] denotes the greatest integer function) belongs to the interval $\left(a,\ \frac{b}{c}\right]$ where $a, b, c \in N$ and $\frac{b}{c}$ is in its lowest form. Find the value of $a + b + c + abc$.

17. Let f be an odd periodic function from R to R. If the period of f is 2 and it is given that

$f\left((3.65+\log_{\frac{9}{4}}\left(\frac{1}{2\sqrt{3}}\sqrt{6-\frac{1}{2\sqrt{3}}\sqrt{6-\frac{1}{2\sqrt{3}}.....\infty}}\right)\right)=1$, then find the absolute value of $f(0.85)$.

18. Let $f: R\to\left(0,\ \frac{\pi}{6}\right]$ be defined as $f(x)=\sin^{-1}\left(\frac{4}{4x^2-12x+17}\right)$ then $f(x)$ is

(1) Injective as well as surjective (2) Surjective but not injective

(3) Injective but not surjective (4) Neither injective nor surjective

19. Let $g : R \to \left[\frac{\pi}{6}, \frac{\pi}{2}\right)$ is defined by $g(x) = \sin^{-1}\left(\frac{x^2 - c}{1 + x^2}\right)$. Then the possible values of c for which g is surjective function, is

(1) $\left\{\frac{1}{2}\right\}$ (2) $\left(-1, -\frac{1}{2}\right]$

(3) $\left\{-\frac{1}{2}\right\}$ (4) $\left[-\frac{1}{2}, 1\right)$

20. The sum of series $\cot^{-1}\left(\frac{9}{2}\right) + \cot^{-1}\left(\frac{33}{4}\right) + \cot^{-1}\left(\frac{129}{8}\right) + ...\infty$ is equal to

(1) $\cot^{-1}$ (2) (2) $\cot^{-1}$ (3)

(3) $\cot^{-1}$ (–1) (4) $\cot^{-1}$ (1)

21. If 'α' is the only real root of the equation $x^3 + bx^2 + cx + 1 = 0$ ($b < c$), then the value of $\tan^{-1}\alpha + \tan^{-1}(\alpha^{-1})$ is equal to

(1) $\frac{-\pi}{2}$ (2) $\frac{\pi}{2}$

(3) 0 (4) Can't be determined

22. If the sum of the series $\sum_{n=1}^{\infty}\left(\frac{\pi - \sec^{-1}\sqrt{|x|+1} - \operatorname{cosec}^{-1}\sqrt{|x|+1}}{\pi a}\right)^n$ is finite, where $x \in R$ and $a > 0$ then a lies in the interval

(1) $(1, \infty)$ (2) $\left(0, \frac{1}{2}\right)$

(3) $\left(\frac{1}{2}, \infty\right)$ (4) $\left[\frac{1}{2}, \infty\right)$

Paragraph for Q. Nos. 23 and 24

Consider, $f(x) = \tan^{-1}\left(\frac{\sqrt{1-4x^2} - 2\sqrt{3}x}{\sqrt{3-12x^2} + 2x}\right)$

23. If $x \in \left(\frac{-\sqrt{3}}{4}, \frac{1}{2}\right)$ then range of $f(x)$ is

(1) $\left[0, \frac{\pi}{6}\right]$ (2) $\left[\frac{-\pi}{3}, \frac{\pi}{6}\right)$

(3) $\left[\frac{-\pi}{3}, 0\right)$ (4) $\left[\frac{-\pi}{3}, \frac{\pi}{2}\right)$

24. The value of $f\left(\frac{\sqrt{3}+1}{4\sqrt{2}}\right) + f\left(\frac{1}{2\sqrt{2}}\right)$ is equal to

(1) $\frac{-5\pi}{12}$ (2) $\frac{-\pi}{12}$

(3) $\frac{-\pi}{3}$ (4) $\frac{5\pi}{12}$

25. If $f : R \to \left[-\frac{\pi}{4}, \frac{\pi}{2}\right)$ is a function defined by $f(x) = \tan^{-1}\left(x^4 - x^2 - \frac{7}{4} + \tan^{-1}\alpha\right)$ and f is surjective then

(1) $\cos^{-1}\left(\frac{1-\alpha^2}{1+\alpha^2}\right) = 2$

(2) $\alpha + \frac{1}{\alpha} = 2 \text{ cosec } 2$

(3) $\sin^{-1}\left(\frac{2\alpha}{\alpha^2+1}\right) = \pi - 2$

(4) $\tan^{-1}\left(\frac{2\alpha}{\alpha^2-1}\right) = 2 - \pi$

26. If $f(n) = \cot^{-1}(n+3) - 2\cot^{-1}(n+1) + \cot^{-1}(n-1)$, $n \in N$, then $\sum_{n=1}^{\infty} f(n)$ is equal to

(1) $\sum_{k=1}^{3} \tan^{-1}\left(\frac{1}{k}\right)$

(2) $\lim_{x\to\pi} \frac{-\sin(x)}{\pi - x}$

(3) $\lim_{x\to 0^-} \cot^{-1}\left(\frac{1}{x}\right)$

(4) $\lim_{x\to\pi} \frac{\pi(x-\pi)^2}{4(\cos x + 1)}$

27. Let $\alpha = 3\cos^{-1}\left(\frac{5}{\sqrt{28}}\right) + 3\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$ and $\beta = 4\sin^{-1}\left(\frac{7\sqrt{2}}{10}\right) - 4\tan^{-1}\left(\frac{3}{4}\right)$, then which of the following does not hold(s) good?

(1) $\alpha < \pi$ but $\beta > \pi$

(2) $\alpha > \pi$ but $\beta < \pi$

(3) Both α and β are equal

(4) $\cos(\alpha + \beta) = 0$

28. If $\sin^{-1}\left(\frac{\sqrt{x}}{2}\right) + \sin^{-1}\left(\sqrt{1-\frac{x}{4}}\right) + \tan^{-1} y = \frac{2\pi}{3}$, then

(1) Maximum value of $x^2 + y^2$ is $\frac{49}{3}$

(2) Maximum value of $x^2 + y^2$ is 4

(3) Minimum value of $x^2 + y^2$ is $\frac{1}{3}$

(4) Minimum value of $x^2 + y^2$ is 3

29. Match column-I with column-II

Column-I	Column-II
(A) $2\cot(\cot^{-1}(3)+\cot^{-1}(7)+\cot^{-1}(13)+\cot^{-1}(21))$ has the value equal to	(p) 3
(B) If $\tan\left(\tan^{-1}\left(\frac{1}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)+\tan^{-1}\left(\frac{1}{13}\right)+.....+\tan^{-1}\left(\frac{1}{381}\right)\right)=\frac{m}{n}$, where $m, n \in N$, then the least value of $(m+n)$ is divisible by	(q) 4
(C) Number of integral ordered pairs (x, y) satisfying the equation $\text{arc tan}\frac{1}{x}+\text{arc tan}\frac{1}{y}=\text{arc tan}\frac{1}{10}$, is	(r) 5
(D) The smallest positive integral value of n for which $(n-2)x^2+8x+n+4 > \sin^{-1}(\sin 12)+\cos^{-1}(\cos 12)\ \forall\ x \in R$, is	(s) 8
	(t) 10

30. If $S_n=\sum_{r=1}^{n} r!$ then for $n > 6$ $\left(\text{given, } \sum_{r=1}^{6} r! = 873\right)$

Column-I	Column-II
(A) $\sin^{-1}\left(\sin\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	(p) $5-2\pi$
(B) $\cos^{-1}\left(\cos\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	(q) $2\pi-5$
(C) $\tan^{-1}\left(\tan\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	(r) $6-2\pi$
(D) $\cot^{-1}\left(\cot\left(S_n-7\left[\frac{S_n}{7}\right]\right)\right)$	(s) $5-\pi$
	(t) $\pi-4$

(where [ ] denotes greatest integer function)

31. Consider $f(x)=\tan^{-1}\left(\frac{(\sqrt{12}-2)x^2}{x^4+2x^2+3}\right)$ and m and M are respectively minimum and maximum values of $f(x)$ and $x = a$ ($a > 0$) is the point in the domain of $f(x)$ where $f(x)$ attains its maximum value.

Column-I	Column-II
(A) If $\sin^{-1}2\sqrt{x}=3\tan^{-1}(\tan(m+M))$ then $8x$ equals	(p) 0
(B) If $\cos^{-1}x+\cos^{-1}y=3\left\{\tan^{-1}\left(\tan\frac{7M}{2}\right)+\tan^{-1}\left(m+\tan\frac{3\pi}{8}\right)\right\}$, then $(x + y)$ equals	(q) 2
(C) The value of $\tan\left(\sec^{-1}\left(\frac{2}{a^2}\right)+M\right)$ equals	(r) –2
(D) If α and β are roots of the equation $x^2 - (\tan(3\sin^{-1}(\sin M)))\,x + a^4 = 0$, then $\alpha\beta - (\alpha + \beta)$ equals	(s) 1
	(t) –1

32. If $x\in\left(0,\ \frac{\pi}{2}\right)$ satisfies the inequality $|\tan x-\sqrt{3}|+|4\sin^2 x-3|+\left|\tan(\tan^{-1}x)-\frac{\pi}{3}\right|\le 0$, then find the value of $\left[\tan\left(\cot^{-1}\left(\frac{\sqrt{2}}{30x}\cos\left(\frac{3x}{4}\right)\right)\right)\right]$.

[Note : [.] denotes greatest integer function.]

33. If the $\lim_{n\to\infty}\sum_{k=2}^{n}\cos^{-1}\left(\frac{1+\sqrt{(k-1)k(k+1)(k+2)}}{k(k+1)}\right)=\frac{120\pi}{k}$ then find the value of k.

34. If $\alpha=\sum_{n=1}^{\infty}\tan^{-1}\left\{\frac{16(2n+1)}{(2n+1)^4-4(2n+1)^2+16}\right\}$ and $\sin\alpha=\frac{a}{b}$ where $a, b \in N$, then find the least value of $a^2 + b^2$.

35. Let $f(x)=\frac{1}{\pi}(\sin^{-1}x+\cos^{-1}x+\tan^{-1}x)+\frac{x+1}{x^2+2x+10}$. If the absolute maximum value of $f(x)$ is M, find $52M$.

36. Consider, $\alpha=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$, $x\in[-1,\ 1]$, $\beta=\cos^{-1}\left(\frac{3\cos y-4\sin y}{10}\right)$, $y\in[0,\ 2\pi]$ and $\gamma=2\tan^{-1}(z^2-4z+5)$, $z\in R$. If α, β and γ are interior angles of a triangle such that $(\beta + \gamma)$ is minimum then $x+\tan y+z=\frac{a-\sqrt{b}}{c}$, where, $a, b, c \in N$. Find the least value of $(a + b + c)$.

37. Let $f : R \to R$ be a continuous function such that $f(x) = 19\ \forall\ x \in Q$ and $g(x) = \sqrt{39 + x^2}$, then the value of $g(f(x))$, is

(1) 18 (2) 20

(3) 25 (4) 17

38. Let $P(x) = a_1x + a_2x^2 + a_3x^3 + ... + a_{100}x^{100}$, where $a_1 = 1$ and $a_i \in R\ \forall\ i = 2, 3, 4, ..., 100$, then $\lim\limits_{x\to 0} \dfrac{\sqrt[100]{1+P(x)}-1}{x}$ has the value equal to

(1) 100 (2) $\dfrac{1}{100}$

(3) 1 (4) 5050

39. Let $f(x) = \max.\{|x^2 - 2|x||, |x|\}$ and $g(x) = \min.\{|x^2 - 2|x||, |x|\}$ then

(1) Both $f(x)$ and $g(x)$ are non-differentiable at 5 points

(2) $f(x)$ is not differentiable at 5 points where $g(x)$ is non-differentiable at 7 points

(3) Number of points of non-differentiability for $f(x)$ and $g(x)$ are 7 and 5 respectively

(4) Both $f(x)$ and $g(x)$ are non-differentiable at 3 and 5 points respectively

40. If $\alpha_1, \alpha_1, \alpha_2, \alpha_3, ..., \alpha_n$ are the roots of equation $x^{n+1} - 5x^2 + 6x - 3 = 0$ and $(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_1 - \alpha_4)$ ..... $(\alpha_1 - \alpha_n) = k$, then $n(n+1)\alpha_1^{n-1} - 2k$ equals

(1) 5 (2) 10

(3) 12 (4) 5

41. If $\lim\limits_{n\to\infty} \dfrac{4^n}{(4-\sqrt{3}+2\sin\theta)^{n+2}}$ exists and is equal to p $(p \neq 0)$ where $\theta \in \left(0, \dfrac{\pi}{2}\right)$, then the value of $\left(\dfrac{p+\cos\theta}{p}\right)$ is equal to

(1) 7 (2) 8

(3) 9 (4) 10

42. Number of values of $x \in [0, \pi]$ where $f(x) = [4 \sin x - 7]$ is non-derivable is

(Note : $[k]$ denotes the greatest integer less than or equal to k]

(1) 7 (2) 8

(3) 9 (4) 10

43. Let $f : R \to (0, \infty)$ be such that $f(x) + \dfrac{e^{x+x^2}}{f(x)} \leq e^x + e^{x^2} \forall\ x > 0$, then $\lim\limits_{x\to 1} f(x)$ is

(1) 1 (2) $\dfrac{1}{e}$

(3) e (4) $2e$

44. Let R be the set of real numbers and $f : R \to R$, be a differentiable function such that $|f(x) - f(y)| \le |x - y|^3 \ \forall\ x, y \in R$. If $f(10) = 100$, the value of $f(20)$ is equal to

(1) 0 (2) 20

(3) 100 (4) 10

45. If $f(x)$ is defined $\forall\ x \in R$ and is discontinuous only at $x = 0$ such that $f^3(x) - 6f^2(x) + 11f(x) - 3 = 3\ \forall\ x \in R$, then the number of such functions is equal to

(1) 4 (2) 8

(3) 16 (4) 24

46. Let $\frac{\pi}{3} < A < \frac{\pi}{2}$ and a function $f(x)$ is defined as

$$f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{ax^2(\sin A - \sin^3 A) - (5x - b)|\sin A - \sin^3 A|^n}{(\sin A - \sin^3 A) - |\sin A - \sin^3 A|^n}, & x \in Q \\ \lim\limits_{n\to\infty} \dfrac{ax^2(\sin A + \sin^3 A) + (5x - b)|\sin A + \sin^3 A|^n}{(\sin A + \sin^3 A) + |\sin A + \sin^3 A|^n}, & x \notin Q \end{cases}$$

If $f(x)$ is continuous at $x = 2$ and $x = 3$ then the value of $b - a$, is

(1) 3

(2) 4

(3) 5

(4) 6

Paragraph for Q. Nos. 47 to 49

Let f, g, h be real-valued functions defined on R. Consider $f(x) = -1 + 4\sin^2 x$, $g(x) = 2x - 1$ and $h(x) = 2x + 1$.

47. The range of function $y = \frac{h(f(x))}{g(f(x))}$ is equal to

(1) $\left[\frac{1}{3}, \frac{7}{5}\right]$ (2) $(-\infty, 1) \cup \left[\frac{7}{5}, \infty\right)$

(3) $\left[1, \frac{7}{5}\right]$ (4) $\left(-\infty, \frac{1}{3}\right] \cup \left[\frac{7}{5}, \infty\right)$

48. The value of $\lim\limits_{x\to\frac{1}{2}} \dfrac{f(\sin^{-1} x)}{g(x)}$ is equal to

(1) 0 (2) 2

(3) 4 (4) 10

49. If $\lim\limits_{x\to\frac{-1}{2}} \dfrac{f(\sin^{-1} kx) - 3}{h(x)}$ exists and has value equal to l then $|k| + |l|$ is equal to

(1) 6 (2) 8

(3) 10 (4) 12

50. Let $f(x)=\begin{cases}5x+1, & x\le 2\\ \int_0^x (5+|1-t|)dt, & x>2\end{cases}$, then which of the following statement(s) is(are) incorrect?

(1) $f(x)$ is continuous but not differentiable at $x = 2$ (2) $f(x)$ is not continuous at $x = 2$

(3) $f(x)$ is differentiable for all $x \in R$ (4) The right hand derivative of $f(x)$ at $x = 3$ does not exist

51. Consider, $f(x)=\begin{bmatrix}2-|x|, & -1\le x\le 1\\ |x-2|-x, & 1<x\le 3\end{bmatrix}$ and $g(x)=\begin{bmatrix}\sin x-1, & 0\le x<\frac{\pi}{2}\\ [x]-\cos(x-2), & \frac{\pi}{2}\le x\le \pi\end{bmatrix}$, where $[k]$ denotes greatest integer function of k. Identify the correct statement(s).

(1) $\lim\limits_{x\to 1^+} g(f(x))=-1$ (2) $\lim\limits_{x\to \frac{\pi}{2}^-} g(f\{g(x)\})=0$

(3) $\lim\limits_{x\to 2^+}\frac{f(g(x))}{f(x)-2}=\frac{1}{2}$ (4) $\lim\limits_{x\to 0^+}\frac{g(f(x))}{(f(x)-2)^2}=\frac{1}{2}$

52. Let $f(x)=\lim\limits_{n\to\infty}\frac{\ln(2+x^2)-x^{2n}\sin(x^2)}{1+x^{2n}}$. Which of the following statement(s) is(are) correct?

(1) $f(x)$ is discontinuous at two points (2) Minimum value of $f(x)$ equals $-\sin 1$

(3) There exists some $c \in R$ for which $f(c) = 1$ (4) The equation $f(x) = 0$ has at least one real root in $(1, \infty)$

53. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that

(i) $a_n + b_n + c_n = 2n + 1$

(ii) $a_nb_n + b_nc_n + c_na_n = 2n - 1$

(iii) $a_nb_nc_n = -1$ and

(iv) $a_n > b_n > c_n$

then which of the following is/are correct?

(1) $\lim\limits_{n\to\infty}\frac{a_n}{n}=\frac{1}{2}$ (2) $\lim\limits_{n\to\infty}\frac{a_n}{n}=2$

(3) $\lim\limits_{n\to-\infty}\frac{a_n}{n}=0$ (4) $\lim\limits_{n\to-\infty}\frac{a_n}{n}=2$

54. The possible value(s) of k for which $\lim\limits_{x\to\infty}\frac{2x^3-(\tan^{-1}x)^3}{\frac{8}{\pi}x^3\cot^{-1}|kx|+k^2x^6\sin\frac{1}{x^3}-3kx^3}=\frac{1}{2}$, is

(1) 0 (2) –1

(3) 2 (4) 4

55. If $f(x) = \text{sgn}((\sin^2 x - \sin x - 1)(\sin^2 x + \sin x + 1)) = 0$ has exactly 4 points of discontinuity for $x \in (0, n\pi)$, $n \in N$ then the value of n may be equal to

(1) 2 (2) 3

(3) 4 (4) 5

56. In the following, [ ] and { } are greatest integer function and fractional part function respectively. Match the functions in Column-I with the properties in Column-II.

Column-I	**Column-II**
(A) $f_1(x)=\left[\frac{4x}{\pi}\right]\text{sgn}(x^2-x+1)$	(p) Discontinuous at more than 3 points but less than 6 points in [–2, 2]
(B) $f_2(x)=\cos^{-1}\left(\text{sgn}\left(\cos\frac{2x-1}{2}\pi\right)\right)$	(q) Non-derivable at more than 2 points but atmost 5 points in [–2, 2]
(C) $f_3(x)=\max.(\{x+1\},\{5-x\})$	(r) Range contains at least one integer but not more than seven and no irrational value in [–2, 2]
(D) $f_4(x)=\sqrt{x^2}+[x]^2$	(s) Many one but not even function in [–2, 2]
	(t) Neither odd nor periodic in [–2, 2]

57. Match column-I with column-II

Column-I	**Column-II**
(A) Number of integral values of x satisfying $\lvert x^2-9\rvert+\lvert x^2-4\rvert=5$, is	(p) 0
(B) Let $f(x)=\begin{cases}-5, & -\infty<x\le -1\\ x-4, & -1<x\le 6\\ 2, & 6<x<\infty\end{cases}$, then $\lim\limits_{x\to n}\frac{[x]^2-13[x]+42}{(x-7)(x-6)}$ is equal to [**Note :** Where n is the number of integers in the range of $f(\lvert x\rvert)$, $[x]$ denotes greatest integer less than or equal to x]	(q) 2
(C) If $\lim\limits_{x\to 0}\frac{a\sin x+bxe^x+3x^2}{\sin x-2x+\tan x}$ exists and has value equal to L, then the value of $\frac{b-L}{a}$, is equal to	(r) 3
(D) Let $P_n=\prod\limits_{k=2}^{n}\left(1-\frac{1}{{}^{k+1}C_2}\right)$. If $\lim\limits_{n\to\infty}P_n$ can be expressed as lowest rational in the form $\frac{a}{b}$, then the value of $(b-a)$ is equal to	(s) 4
	(t) Non-existent

58. If $S_n = \sum_{r=1}^{n} \tan^{-1}\left[\frac{2(2r-1)}{4+r^2(r^2-2r+1)}\right]$ then find the value of $\lim_{n\to\infty} \sum_{n=2}^{n} (\cot(S_{n-1}) - \cot(S_n))$.

59. If the equation $\left| |x-1| - 6 \lim_{t\to\infty} \left(\frac{\sqrt{2t^2-t-1}-\sqrt{t^2-t+1}}{t\left(\tan\frac{\pi}{8}\right)} \right) \right| = k$ has four distinct solutions then find the number of integral values of k.

60. If $\sum_{r=1}^{n} \sin^{-1}\alpha_r = \frac{n\pi}{2}$ for any $n \in N$ and $p = \prod_{r=1}^{n} (\alpha_r)^r$.

If $f(x) = \begin{cases} \frac{x^{1/3} - (3-2x)^{1/4}}{x^2 - x}, & x \neq p \\ k, & x = p \end{cases}$ is continuous at $x = p$, then find the value of $6(k + p)$.

61. Let $f(x) = x^2 + ax + 3$ and $g(x) = x + b$, where $F(x) = \lim_{n\to\infty} \frac{f(x) + x^{2n} g(x)}{1 + x^{2n}}$. If $F(x)$ is continuous at $x = 1$ and $x = -1$ then find the value of $(a^2 + b^2)$.

62. The sequence $\{a_n\}_{n=1}^{+\infty}$ is defined by $a_1 = 0$ and $a_{n+1} = a_n + 4n + 3$, $n \geq 1$. Find the value of

$$\lim_{n\to+\infty} \frac{\sqrt{a_n} + \sqrt{a_{4n}} + \sqrt{a_{4^2 n}} + \sqrt{a_{4^3 n}} + ... + \sqrt{a_{4^{10} n}}}{\sqrt{a_n} + \sqrt{a_{2n}} + \sqrt{a_{2^2 n}} + \sqrt{a_{2^3 n}} + ... + \sqrt{a_{2^{10} n}}}$$

63. Let θ_k (k = 1, 2, 3, ..., n), $\theta_1 < \theta_2 < ..... < \theta_n$ be the solution of θ such that $\sqrt{3}\sin n\theta + \cos n\theta = 0$. If $\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \cos\frac{\theta_k}{2} = \frac{k}{\pi}$, then find the value of k ($k \in N$).

64. Let $S = \frac{1}{2}\tan\frac{\pi}{4} + \frac{1}{2^2}\tan\frac{\pi}{8} + \frac{1}{2^3}\tan\frac{\pi}{16} + ... + \frac{1}{2^n}\tan\left(\frac{\pi}{2^{n+1}}\right)$. If $\lim_{n\to\infty} S = L$; then find the value of $(100\ \pi)\ L$.

(you may use the fact $\cot x - \tan x = 2\cot 2x$)

65. If for a differentiable function $y = f(x)$, $(f'(x) > 0)$ $\lim_{x\to1} \frac{e^{2f(x)} - 2e^{f(x)+1} + e^2\cos(x-1)}{\sec^2(x-1) - 1} = \frac{7}{2}e^2$ and area of the triangle formed by the tangent drawn to the curve $y = f(x)$ at $[1, f(1)]$ and co-ordinate axes is Δ, then find the value of $\frac{1}{\Delta}$.

66. If $f(x) = (2x - 3\pi)^5 + \frac{4}{3}x + \cos x$ and g is the inverse function of f, then $g'(2\pi)$ is equal to

(1) $\frac{7}{3}$ (2) $\frac{3}{7}$

(3) $\frac{30\pi^4 + 4}{3}$ (4) $\frac{3}{30\pi^4 + 4}$

67. Given that $\prod_{n=1}^{n}\cos\frac{x}{2^n}=\frac{\sin x}{2^n\sin\left(\frac{x}{2^n}\right)}$.

Let $f(x)=\begin{cases}\lim\limits_{n\to\infty}\sum\limits_{n=1}^{n}\frac{1}{2^n}\tan\left(\frac{x}{2^n}\right), & x\in(0,\pi)-\left\{\frac{\pi}{2}\right\}\\ \frac{2}{x}, & x=\frac{\pi}{2}\end{cases}$

then which one of the following alternative is true?

(1) $f(x)$ has non-removable discontinuity of finite type at $x=\frac{\pi}{2}$

(2) $f(x)$ has missing point discontinuity at $x=\frac{\pi}{2}$

(3) $f(x)$ is continuous at $x=\frac{\pi}{2}$

(4) $f(x)$ has non-removable discontinuity of infinite type at $x=\frac{\pi}{2}$

68. Suppose $A=\frac{dy}{dx}$ of $x^2+y^2=0$ at $(\sqrt{2},\sqrt{2})$, $B=\frac{dy}{dx}$ of $\sin y+\sin x=\sin x\cdot\sin y$ at (π,π) and $C=\frac{dy}{dx}$ of $2e^{xy}+e^xe^y-e^x-e^y=e^{xy+1}$ at (1, 1), then $(A+B+C)$ has the value equal to

(1) –1 (2) e

(3) –3 (4) 0

69. Let f and g be twice differentiable function on R and $f''(5)=8$, $g''(5)=2$ then $\lim\limits_{x\to5}\left[\frac{f(x)-f(5)-(x-5)f'(5)}{g(x)-g(5)-(x-5)g'(5)}\right]$ is equal to

(1) 0 (2) 1

(3) 2 (4) 4

70. For the curve $32x^3y^2=(x+y)^5$, the value of $\frac{d^2y}{dx^2}$ at $P(1,1)$ is equal to

(1) 0 (2) 1

(3) –1 (4) $\frac{1}{2}$

71. If a curve is represented parametrically by the equations $x=f(t)$ and $y=g(t)$ then $\left(\frac{d^2y}{dx^2}\right)\Big/\left(\frac{d^2x}{dy^2}\right)$ is equal to (where $f'(t)\neq0,\ g'(t)\neq0$).

(1) 1 (2) $\frac{g'(t)}{f'(t)}$

(3) $-\left[\frac{g'(t)}{f'(t)}\right]^2$ (4) $-\left[\frac{g'(t)}{f'(t)}\right]^3$

Paragraph for Q. Nos. 72 to 74

Let $f, g, f_1, f_2 : R \to R$ be twice differentiable function, and $f(x) \geq 0$, $f'(x) \geq 0$, $g'(x) > 0 \ \forall \ x \in R$.

Also, $\lim\limits_{x\to\infty} f_1(x) = 5$, $\lim\limits_{x\to\infty} f_2(x) = 12$, $\lim\limits_{x\to\infty} f(x) = \infty$, $\lim\limits_{x\to\infty} g(x) = \infty$,

and $\dfrac{f'(x)}{g'(x)} + f_1(x)\dfrac{f(x)}{g(x)} = f_2(x) \ \forall \ x > 0$.

(Also, $f'(x)$ and $g'(x)$ are continuous)

72. $\lim\limits_{x\to\infty} \dfrac{g'(x)}{f'(x)}$ equals

(1) 0 (2) 2

(3) 1 (4) $\dfrac{1}{2}$

73. $\lim\limits_{x\to\infty} \dfrac{f(x)}{g(x)}$ equals

(1) 1 (2) 0

(3) $\dfrac{1}{2}$ (4) 2

74. $\lim\limits_{x\to\infty} \dfrac{\lambda f'(x) + f''(x)}{\lambda g'(x) + g''(x)}$ equals ($\lambda > 0$)

(1) ∞ (2) 0

(3) 2 (4) 1

75. If the independent variable x is changed to y then the differential equation $x\dfrac{d^2y}{dx^2} + \left(\dfrac{dy}{dx}\right)^3 - \dfrac{dy}{dx} = 0$ is changed to $x\dfrac{d^2x}{dy^2} + \left(\dfrac{dx}{dy}\right)^2 = k$ where k is equal to

(1) $\lim\limits_{x\to0}\left[\dfrac{2\tan x}{x}\right]$ (2) $\lim\limits_{x\to0}\left[\dfrac{2x}{\tan x}\right]$

(3) $\lim\limits_{x\to0}\left[\dfrac{\tan x}{x}\right]$ (4) 1

76. Let f be a differentiable function satisfying $f(x+y) = f(x) + f(y) + (e^x - 1)(e^y - 1) \ \forall \ x, \ y \in R$ and $f'(0) = 2$. Identify the correct statement(s).

(1) $\lim\limits_{x\to0} \dfrac{f(f(x))}{f(x) - x} = 4$ (2) $\lim\limits_{x\to0} (f(x) + \cos x)^{\frac{1}{e^x - 1}} = e^2$

(3) Number of solutions of the equation $f(x) = 0$ is 2 (4) Range of the function $y = f(x)$ is $(-\infty, \infty)$

77. Three functions, f_1, f_2 and f_3 are given as $f_1(x) = \sin^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$, $f_2(x) = x - \cos^{-1}\left(\dfrac{2x}{1+x^2}\right)$ and

$f_3(x) = 4\left\{\dfrac{\tan^{-1} x}{2}\right\} - \{x\}$. [**Note :** $\{k\}$ denotes fractional part function of k]

	Column-I	Column-II
(A)	The differential coefficient of $f_1(x)$ with respect to $f_2(x)$ at $x = \tan 1$, is	(p) –1
(B)	$\left.\frac{d[f_3(x)]}{d[f_2(x)]}\right\|_{x=\tan 1}$ is equal to	(q) 2
(C)	$\left.\frac{d}{dx}\left(\int (f_2(x) - f_1(x)dx)\right)\right\|_{x=\tan 1}$ is equal to	(r) 1 + sec 2
(D)	The slope of the normal to the curve $y = f_3(x)$ at $x = \sqrt{3}$ is	(s) tan 1

78. Let $y = f(x)$ be an infinitely differentiable function on R such that $f(0) \neq 0$ and $\frac{d^n y}{dx^n} \neq 0$ at $x = 0$ for $n = 1, 2, 3, 4$.

If $\lim_{x \to 0} \frac{f(4x) + af(3x) + bf(2x) + cf(x) + df(0)}{x^4}$ exists, then find the value of $(25a + 50b + 100c + 500d)$.

79. Let $g(x) = f\left[\frac{x}{f(x)}\right]$ where $f(x)$ is a differentiable positive function on $(0, \infty)$ such that $f(1) = f'(1)$. Determine $g'(1)$.

80. Let $f(x)$ be a cubic polynomial with leading coefficient unity such that $f(0) = 1$ and all the roots of $f'(x) = 0$ are also roots of $f(x) = 0$. If $\int f(x)dx = g(x) + C$, where $g(0) = \frac{1}{4}$ and C is constant of integration, then $g(3) - g(1)$ is equal to

(1) 27 (2) 48

(3) 60 (4) 81

81. $\int [\sin(101x) \cdot \sin^{99} x]dx$ equals

(1) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (2) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$

(3) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ (4) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

82. If $I_n = \int \cot^n x\,dx$, then $I_0 + I_1 + 2(I_2 + I_3 + ... + I_8) + I_9 + I_{10}$ equals (where $u = \cot x$)

(1) $u + \frac{u^2}{2} + ... + \frac{u^9}{9}$ (2) $-\left(u + \frac{u^2}{2} + ... + \frac{u^9}{9}\right)$

(3) $-\left(u + \frac{u^2}{2!} + ... + \frac{u^9}{9!}\right)$ (4) $\frac{u}{2} + \frac{2u^2}{3} + ... + \frac{9u^9}{10}$

83. If $\int u\frac{d^2v}{dx^2}dx = u\frac{dv}{dx} - v\frac{du}{dx} + w$ then w is equal to

(1) $\int v\frac{d^2u}{dx^2}dx$ (2) $\int u\frac{d^2v}{dx^2}dx$

(3) $\int v\left(\frac{du}{dx}\right)^2 dx$ (4) $\int u\left(\frac{dv}{dx}\right)^2 dx$

Paragraph for Q. Nos. 84 and 85

Consider $\phi(a, b, t) = a^4 - 5a^2 + b^2 + 5t^2 - 4bt - 2t + \frac{33}{4}$ where $a, b, t \in R$. Given that $f(t)$ and $g(b)$ are the minimum value of $\phi(a, b, t)$.

84. $\int\frac{dx}{f(x)}$ is

(1) $\tan^{-1}(x-1) + C$ (2) $\frac{1}{2}\tan^{-1}\left(\frac{x-1}{2}\right) + C$

(3) $\ln[x - 1 + \sqrt{x^2 - 2x + 2}) + C$ (4) $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

(**Note :** where C is the constant of integration)

85. If $\int e^x g(x)dx = e^x(Ax^2 + Bx + C) + D$, where D is constant of integration, then $(A + B + C)$ is equal to

(1) 2 (2) $\frac{14}{5}$

(3) 5 (4) $\frac{19}{5}$

86. If $\int\frac{3\cot 3x - \cot x}{\tan x - 3\tan 3x}dx = pf(x) + qg(x) + c$ where c is a constant of integration, then

(1) $p = 1;\ q = \frac{1}{\sqrt{3}};\ f(x) = x;\ g(x) = \ln\left|\frac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x}\right|$

(2) $p = 1;\ q = -\frac{1}{\sqrt{3}};\ f(x) = x;\ g(x) = \ln\left|\frac{\sqrt{3} - \tan x}{\sqrt{3} + \tan x}\right|$

(3) $p = 1;\ q = -\frac{2}{\sqrt{3}};\ f(x) = x;\ g(x) = \ln\left|\frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x}\right|$

(4) $p = 1;\ q = -\frac{1}{\sqrt{3}};\ f(x) = x;\ g(x) = \ln\left|\frac{\sqrt{3} + \tan x}{\sqrt{3} - \tan x}\right|$

87. Let $\int \frac{(1+x^4)dx}{(1-x^4)^{\frac{3}{2}}} = f(x)+C_1$ where $f(0) = 0$ and $\int f(x)dx = g(x)+C_2$ with $g(0) = 0$. If $g\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{k}$, find k.

88. If $\int \frac{(x-1)dx}{(x+x\sqrt{x}+\sqrt{x})\sqrt{\sqrt{x}(x+1)}} = 4\tan^{-1}[g(x)]+C,$ where C is an arbitrary constant of integration. Find $g^2(1)$.

89. The absolute value of $\frac{\int_0^{\pi/2}(x\cos x+1)e^{\sin x}dx}{\int_0^{\pi/2}(x\sin x-1)e^{\cos x}dx}$ is equal to

(1) e (2) πe

(3) $e/2$ (4) π/e

90. The interval [0, 4] is divided into n equal sub-intervals by the points $x_0, x_1, x_2, ..., x_{n-1}, x_n$ where $0 = x_0 < x_1 < x_2 < x_3 ..... < x_n = 4$. If $\delta x = x_i - x_{i-1}$ for $i = 1, 2, 3, ... n$ then $\lim_{\delta x\to 0}\sum_{i=1}^{n} x_i\delta x$ is equal to

(1) 4 (2) 8

(3) $\frac{32}{3}$ (4) 16

91. The value of the definite integral $\int_e^{e^{2010}} \frac{1}{x}\left(1+\frac{1-\ln x}{\ln x \ln\left(\frac{x}{\ln x}\right)}\right)dx,$ equals

(1) 2009 – ln (2010 – ln 2010) (2) 2010 – ln (2009 – ln 2009)

(3) 2009 – ln (2010 – ln 2009) (4) 2010 – ln (2010 – ln 2010)

92. If $\sum_{i=1}^{6}(\sin^{-1}x_i+\cos^{-1}y_i) = 9\pi,$ then $\int_{\sum_{i=1}^{6}x_i}^{\sum_{i=1}^{6}y_i} x\ln(1+x^2)\left(\frac{e^x}{1+e^{2x}}\right)dx$ is equal to

(1) 0 (2) $e^6 + e^{-6}$

(3) $\ln\left(\frac{37}{12}\right)$ (4) $e^6 - e^{-6}$

93. Consider a parabola $y = \frac{x^2}{4}$ and the point $F(0, 1)$.

Let $A_1(x_1, y_1), A_2(x_2, y_2), A_3(x_3, y_3), ..., A_n(x_n, y_n)$ are n points on the parabola such as $x_k > 0$ and $\angle OFA_k = \frac{k\pi}{2n}$ $(k = 1, 2, 3, ..., n)$. Then the value of $\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n} FA_k$, is equal to

(1) $\frac{2}{\pi}$ (2) $\frac{4}{\pi}$

(3) $\frac{8}{\pi}$ (4) None of these

94. The value of the definite integral $\int_0^1((e-1)\sqrt{\ln(1+(e-1)x}+e^{x^2})dx$ is equal to

(1) 0 (2) 1

(3) e (4) e^2

95. For real numbers a, b with $0 \le a \le \pi$, $a < b$. Let $I(a, b)=\int_a^b e^{-x}\sin x\,dx$.

If $\lim_{b\to\infty} I(a, b)=0$, and the value of a is $k\pi$ ($k > 0$), then the value of k, is

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$

(3) $\frac{3}{4}$ (4) 1

96. Let $f(x)$ be a continuous and periodic function such that $f(x) = f(x + T)$ for all $x \in R$, $T > 0$. If $\int_{-2T}^{a+5T} f(x)dx = 19\ (a > 0)$ and $\int_0^T f(x)dx = 2$, then $\int_0^a f(x)dx$ is equal to

(1) 3 (2) 5

(3) 7 (4) 9

97. If x and y are independent variables and $f(x)=\left(\int_0^x e^{x-y}f'(y)dy\right)-(x^2-x+1)e^x$, where $f(x)$ is a differentiable function, then $f\left(\frac{-1}{2}\right)$ equals

(1) $2\sqrt{e}$ (2) $-2\sqrt{e}$

(3) $\frac{-2}{\sqrt{e}}$ (4) $\frac{2}{\sqrt{e}}$

98. $\int_0^1(\sqrt[4]{1-x^7}-\sqrt[7]{1-x^4})dx$ is equal to

(1) $\frac{1}{2}$ (2) 1

(3) 0 (4) None of these

99. A function $f(x)$ satisfies $f(x)=f\left(\frac{c}{x}\right)$ for some real number c ($c > 1$) and $\forall\ x > 0$. If $\int_1^{\sqrt{c}}\frac{f(x)}{x}dx=3$, then the value of $\int_1^c\frac{f(x)}{x}dx$ is

(1) 0 (2) 3

(3) –3 (4) 6

Paragraph for Q. Nos. 100 and 101

Let a polynomial f satisfies the relation $f(f(f(x)))+(1-p)f(x)=3\ \forall x\in R$ where $p\in R$.

100. If leading coefficient of $f(x)$ is 2 then the value of $\frac{d}{dx}(f(f(x)))$ at $x = p$ is

(1) 2

(2) 4

(3) 9

(4) 17

101. If leading coefficient of $f(x)$ is negative and $f(0) = 4$ then $\int_{-1}^{1} f^{-1}(x)dx$ is equal to

(1) 8 (2) 16

(3) 32 (4) 64

102. Let $S_n=\sum_{k=1}^{n}\frac{k^2+n^2}{n^3}$ and $T_n=\sum_{k=0}^{n-1}\frac{k^2+n^2}{n^3}$ for n = 1, 2, 3, ... then

(1) $S_n<\frac{4}{3}$ (2) $T_n>\frac{4}{3}$

(3) $S_n>\frac{4}{3}$ (4) $T_n<\frac{4}{3}$

103. Let $f : R \to (0, \infty)$ be a real valued function satisfying $\int_0^x t f(x-t)dt = e^{2x}-1$, then which of the following is(are) correct?

(1) The value of $(f^{-1})'(4)$ equals $\frac{1}{8}$

(2) Derivative of $f(x)$ with respect to e^x at $x = 0$ is equal to 8

(3) The value of $\lim_{x\to 0}\frac{f(x)-4}{x}$ equals 4

(4) The value of $f(0)$ is equal to 4

104. Let $J=\int_0^{1/2}\left(\frac{1}{4}-x^2\right)^4 dx$ and $K=\int_0^{1/2}x^4(1-x)^4\,dx$, then

(1) $\frac{J}{K}=2$ (2) $J - K = 0$

(3) $J=\frac{1}{2}\int_0^1 x^4(1-x)^4\,dx$ (4) $K=\frac{1}{1260}$

105. Let $f_n(x)+f_n(y)=\frac{x^n+y^n}{x^n y^n}$ for all $x, y \in R-\{0\}$ where $n \in N$ and $g(x) = \max\{f_2(x), f_3(x), \frac{1}{2}\}$, $\forall\, x \in R-\{0\}$.

Column-I	Column-II
(A) The possible value(s) of $\sum_{k=1}^{\infty} f_{2k}(\text{cosec}\,\theta)+\sum_{k=1}^{\infty} f_{2k}(\sec\theta)$ where $\theta \neq \frac{k\pi}{2}$, $k \in I$ is/are	(p) 1
(B) The value(s) of n for which $h(x) = f_n(x)$ sgn x is even, is/are	(q) 2
(C) Values of consecutive integers between which, $\int_{1/2}^{2} g(x)dx$ lie are	(r) 3
(D) Number of values of x for which the function $y = g(x)\ \forall\ x \in R-\{0\}$ is non-differentiable, is/are	(s) 4

106. Given that $U_n = \{x(1-x)\}^n$ and $n \geq 2$ and $\frac{d^2U_n}{dx^2} = n(n-1)U_{n-2} - 2n(2n-1)U_{n-1}$, further if $V_n = \int_0^1 e^x \cdot U_n\, dx$, then for $n \geq 2$, we have $V_n + K_1 n(2n-1).\ V_{n-1} + K_2 n(n-1)\ V_{n-2} = 0$. Find $(K_1 + K_2)$.

107. Given a function g, continuous everywhere such that $g(1) = 5$ and $\int_0^1 g(t)dt = 2$. If $f(x) = \frac{1}{2}\int_0^x (x-t)^2\, g(t)dt$, then compute the value of $f'''(1) - f''(1)$.

108. Given $\lim_{n\to\infty}\left(\frac{{}^{3n}C_n}{{}^{2n}C_n}\right)^{1/n} = \frac{a}{b}$ where a and b are relatively prime, find the value of $(a + b)$.

109. If $\lim_{t\to\infty}\left(\int_1^t \frac{dx}{1+\sqrt[3]{x^2}} - at^b\right)$ exists and equals to non-zero finite number L, where a and b are positive real numbers, then find the value of $(ab - 4L - 3\pi)$.

110. Let $f(x)$ be a continuous function defined from $[0, 2] \to R$ and satisfying the equation $\int_0^2 f(x)(x - f(x))dx = \frac{2}{3}$. Find the value of $2f(1)$.

111. Let f be a differentiable function defined such that $f : [0, 27] \to \left[\frac{1}{3}, 6\right]$ and $f'(x) < 0\ \forall\ x \in D_f$. If $\int_0^{27} x f'(x)dx = \lambda - 3\int_0^3 x^2 f(x^3)dx$ then find the value of λ.

(**Note :** D_f denotes the domain of the function)

112. If $P(x) = (2013)\,x^{2012} - (2012)\,x^{2011} - 16x + 8$, then $P(x) = 0$ for $x \in \left[0,\ 8^{\frac{1}{2011}}\right]$ has

(1) Exactly one real root

(2) No real root

(3) At least one and at most two real roots

(4) At least two real roots

113. The value of t for which $\dfrac{\int_0^{\pi/2} (\sin x + t\cos x)\,dx}{\sqrt{\int_0^{\pi/2} (\sin x + t\cos x)^2\,dx}}$ is maximum lies in the interval

(1) $\left(0,\ \frac{1}{2}\right)$

(2) $\left(\frac{1}{2},\ \frac{3}{4}\right)$

(3) $\left(\frac{3}{4},\ \frac{3}{2}\right)$

(4) $\left(\frac{3}{2},\ \frac{5}{2}\right)$

114. Let $f(x) = \left(1+\frac{1}{x}\right)^x$ and $g(x) = \left(1+\frac{1}{x}\right)^{x+1}$, both f and g being defined for $x > 0$, then $f(x)$ and $g(x)$ will have nature that

(1) Both $f(x)$ and $g(x)$ are increasing

(2) $f(x)$ is increasing and $g(x)$ is decreasing

(3) $f(x)$ is decreasing and $g(x)$ is increasing

(4) Both $f(x)$ and $g(x)$ are decreasing

115. If the equation $x^3 - 3x + 1 = 0$ has three real roots x_1, x_2, x_3, where $x_1 < x_2 < x_3$, then the value of $(\{x_1\} + \{x_2\} + \{x_3\})$ is equal to

(**Note :** $\{x\}$ denotes the fractional part of x)

(1) $\frac{3}{2}$ (2) 1

(3) 2 (4) $\frac{5}{2}$

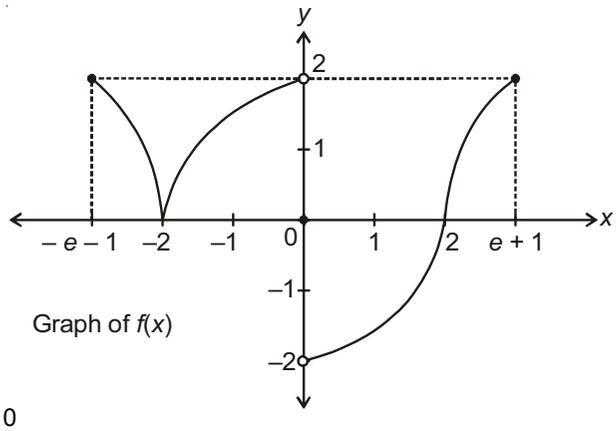
116. Let $f : R \to R$ be a function defined by $f(x) = 2x^3 - 21x^2 + 78x + 24$. Number of integers in the solutions set of x satisfying the inequality $f(f(f(x) - 2x^3)) \geq f(f(2x^3 - f(x)))$ is

(1) 3 (2) 4

(3) 5 (4) 6

Paragraph for Q. Nos. 117 to 119

Let $f(x)=\begin{cases}2\ln(-x-1), & -e-1\le x\le -2\\ \frac{1}{2}(4-x^2), & -2<x<0\\ 0, & x=0\\ \frac{1}{2}(x^2-4), & 0<x<2\\ 2\ln(x-1), & 2\le x\le e+1\end{cases}$



Graph of $f(x)$

and graph of $f(x)$ is as shown.

Also, $g(x)=\begin{bmatrix}\min\{f(t):-e-1\le t\le x\}, & -e-1\le x<0\\ \max\{f(t):0\le t\le x\}, & 0\le x\le e+1\end{bmatrix}$

117. Which one of the following statement does not hold good?

(1) Range of $g(x)$ is [0, 2]

(2) $g(x)$ is non-monotonic in [–2, 2]

(3) $g(x)$ is a continuous function

(4) $g(x)$ is an odd function

118. If $x = \alpha$ is the point of non-differentiability of $g(x)$ in $(-e-1, e+1)$ then the values of α is

(1) –2, 1

(2) –2, –1

(3) –2, 2

(4) –2, 0

119. If the equation $g(x) = k$ has exactly two distinct solution in $[-e-1, e+1]$ then the sum of all possible integral values of k is

(1) 0

(2) 1

(3) 2

(4) 3

Paragraph for Q. Nos. 120 to 122

Let $f(x) = a(x-2)(x-b)$, where $a, b \in R$ and $a \neq 0$.

Also, $f(f(x))=a^3\left(x^2-(2+b)x+2b-\frac{2}{a}\right)\left(x^2-(2+b)x+2b-\frac{b}{a}\right),\ a\neq 0,$

has exactly one real zeroes 5.

120. Which of the following is incorrect?

(1) The minimum value of $f(x)$ is 5 and attained at $x = –2$

(2) The maximum value of $f(x)$ is 2 and attained at $x = 5$

(3) The minimum value of $f(x)$ is 10 and attained at $x = 0$

(4) The maximum value of $f(x)$ is 24 and attained at $x = 6$

121. The slope of the straight line passing through $O(0, 0)$ and tangent to $y = f(x)$, can be

(1) 4

(2) 2

(3) $\frac{4}{9}$

(4) $\frac{2}{9}$

122. The value of definite integral $\int_2^5 f(x)dx$ is equal to

(1) 4 (2) 8

(3) 10 (4) 14

123. Let the function $f(x)$ be thrice differentiable and satisfies $f(f(x)) = 1 - x$ for all $x \in [0, 1]$. If $J = \int_0^1 f(x)dx$ and $f''\left(\frac{4}{5}\right) = 0$, then which of the following is(are) true?

(1) $f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) = 1$ (2) $J = \frac{1}{2}$

(3) $f'(x) = 0$ has at least one root in $x \in \left(\frac{1}{4}, \frac{3}{4}\right)$ (4) $f''(x) = 0$ has at least one root in $x \in \left(\frac{1}{2}, \frac{4}{5}\right)$

124. Let $f(x) = (x - 1)(x - 2)(x - 3).....(x - n)$, $n \in N$ and $\int \frac{f(x)f''(x) - [f'(x)]^2}{f^2(x)}dx = g(x) + C$, where C is arbitrary constant.

Column-I	Column-II
(A) If $f'(n) = 5040$, then n is divisible by	(p) 4
(B) If $g(x)$ is discontinuous at 9 points, $\forall\ x \in R$ then n is greater than	(q) 6
(C) If $g(x) = 5$ has 8 solutions, then n may be equal to	(r) 8
(D) If the number of roots of equation $f'(x) = 0$, be $(n - 5)^2 (n - 1)$, $(n > 1)$ then possible values of n is/are	(s) 9

125. Let f be a continuous function and satisfies $f'(x) > 0$ on $(-\infty, \infty)$ and the value of $f''(x)\ \forall\ x \in (0, \infty)$ is equal to minimum value of $\min_{x \in R}\{e^{-|x|} + 2,\ |x| + 2\}$.

If $L = \lim_{x \to \infty} \frac{3x^2 - \frac{3}{x^2 + 1} - 4f'(x)}{f(x)}$ then find the value of $[L^2]$.

(**Note :** $[k]$ denotes greatest integer less than or equal to k]

126. Let $f(x) = \frac{\pi}{2} + \left|\text{sgn}\left(\tan^{-1}\left(\frac{x}{1 + x^2}\right)\right)\right| \tan^{-1} x$, where sgn (y) denotes signum function of y, and $g(x)$ is the inverse of $f(x)$. If the complete set of values of k for which the equation $2g(x) + k(\pi - 2x) = 0$ has three distinct solutions is (a, ∞) then find a.

127. Find the length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2 + y^2 + 12x - 20y + 120 = 0$.

128. Let $g(a) = \int_0^{\pi/2} |\sin 2x - a\cos x|\,dx,\ 0 \le a \le 1$. If m and M is minimum and maximum value of $g(a)\ \forall\ a \in [0, 1]$ then find the value of $2(M + m)$.

129. Orthogonal trajectories of family of the curve $x^{2/3} + y^{2/3} = a^{2/3}$, where a is any arbitrary constant, is

(1) $x^{2/3} - y^{2/3} = c$ (2) $x^{4/3} - y^{4/3} = c$

(3) $x^{4/3} + y^{4/3} = c$ (4) $x^{1/3} - y^{1/3} = c$

130. A function $y = f(x)$ satisfies the condition $f'(x)\sin x + f(x)\cos x = 1$, $f(x)$ being bounded when $x \to 0$. If $I = \int_0^{\pi/2} f(x)dx$ then

(1) $\frac{\pi}{2} < I < \frac{\pi^2}{4}$ (2) $\frac{\pi}{4} < I < \frac{\pi^2}{2}$

(3) $1 < I < \frac{\pi}{2}$ (4) $0 < I < 1$

131. The substitution $y = z^{\alpha}$ transforms the differential equation $(x^2y^2 - 1)dy + 2xy^3dx = 0$ into a homogeneous differential equation for

(1) $\alpha = -1$ (2) 0

(3) $\alpha = 1$ (4) No value of α

132. Let $y'(x) + \frac{g'(x)}{g(x)} \cdot y(x) = \frac{g'(x)}{1 + g^2(x)}$ where $f'(x)$ denotes $\frac{df(x)}{dx}$ and $g(x)$ is a given non-constant differentiable function on R. If $g(1) = y(1) = 1$ and $g(e) = \sqrt{(2e-1)}$ then $y(e)$ equals

(1) $\frac{3}{2g(e)}$ (2) $\frac{1}{2g(e)}$

(3) $\frac{2}{3g(e)}$ (4) $\frac{1}{3g(e)}$

133. Let $f(x)(f(x) > 0)$ be a differentiable function satisfying $f^2(x) = \int_0^x (f^2(t) - f^4(t) + (f'(t))^2 dt + 100$, where $f^2(0) = 100$, then $\lim_{x\to\infty} f(x)$ can be

(1) 0 (2) 1

(3) $\frac{10}{9}$ (4) 10

Paragraph for Q. Nos. 134 to 136

Let f be a differentiable function satisfying

$\int_0^{f(x)} f^{-1}(t)dt - \int_0^x (\cos t - f(t))dt = 0$ and $f(0) = 1$

134. The number of solution(s) of the equation $\begin{vmatrix} f(2x) & f(x) \\ \sin x & 2 \end{vmatrix} = 0$ in $(0, 2\pi)$ is

(1) 2 (2) 3

(3) 4 (4) 5

135. The value of $\int_0^{\pi/2} f(x)dx$ lies in the interval

(1) $\left(\frac{2}{\pi}, 1\right)$ (2) $\left(1, \frac{\pi}{2}\right)$

(3) $\left(\frac{3}{2}, \frac{\pi}{2}\right)$ (4) $\left(0, \frac{2}{\pi}\right)$

136. The value of $\lim_{x\to 0}\left(\left[\frac{\cos x}{f(x)}\right]+\left[\frac{\cos 2x}{f(2x)}\right]+\left[\frac{\cos 3x}{f(3x)}\right]+.....+\left[\frac{\cos(100x)}{f(100x)}\right]\right)$ is equal to

(**Note :** where [*k*] denotes greatest integer less than or equal to *k*)

(1) 0 (2) 4950

(3) 5049 (4) 5050

137. A curve $y = f(x)$ has the property that the perpendicular distance of the origin from the normal at any point *P* of the curve is equal to the distance of the point *P* from the *x*-axis. Then the differential equation of the curve

(1) Is homogeneous

(2) Can be converted into linear differential equation with some suitable substitution

(3) Is the family of circles touching the *x*-axis at the origin

(4) The family of circles touching the *y*-axis at the origin

138. A function $y = f(x)$ satisfying the differential equation $\frac{dy}{dx}\cdot\sin x - y\cos x + \frac{\sin^2 x}{x^2} = 0$ is such that, $y \to 0$ as $x \to \infty$ then the statement which is correct is

(1) $\lim_{x\to 0} f(x) = 1$ (2) $\int_0^{\pi/2} f(x)dx$ is less than $\frac{\pi}{2}$

(3) $\int_0^{\pi/2} f(x)dx$ is greater than unity (4) $f(x)$ is an odd function

139. Let *f* be a continuous function satisfying the equation $\int_0^x f(t)dt + \int_0^x t f(x-t)dt = e^{-x} - 1$, then find the value of $e^{10} f(10)$.

140. Let A_n be the area of region bounded by a curve $y = x^3 (1 - x^2)^n$, $0 \le x \le 1$ and the *x*-axis, then the value of $\sum_{n=1}^{\infty} A_n$ is equal to

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{1}{4}$ (4) 1

141. Let Z be a complex number such that $\text{Re}(Z) = \sqrt{x^2+4}$ and $\text{Im}(Z) = \sqrt{y-4}$ satisfying $|Z| = \sqrt{10}$. Area enclosed by the set of points (x, y) on the complex plane, is

(1) $8\sqrt{6}$ (2) $4\sqrt{6}$

(3) $\frac{20\sqrt{10}}{3}$ (4) $\frac{40\sqrt{10}}{3}$

142. The graphs of $f(x) = x^2$ and $g(x) = cx^3$ $(c > 0)$ intersect at the points (0, 0) and $\left(\frac{1}{c}, \frac{1}{c^2}\right)$. If the region which lies between these graphs and over the interval $\left(0, \frac{1}{c}\right)$ has the area equal to $\frac{2}{3}$ then the value of c is

(1) 1 (2) $\frac{1}{3}$

(3) $\frac{1}{2}$ (4) 2

143. Infinite rectangles each of width 1 unit and height $\left(\frac{1}{n} - \frac{1}{n+1}\right)(n \in N)$ are constructed such that ends of exactly one diagonal of every rectangle lies along the curve $y = \frac{1}{x}$. The sum of areas of all such rectangles, is

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$

(3) $\frac{3}{4}$ (4) 1

Paragraph for Q. Nos. 144 to 146

Let $y = f(x)$ be a solution of the differential equation $\frac{dy}{dx} = -4x(y-2)$ with $f(0) = 2$ and

$g(x) = \max.\left\{1 + \frac{x}{2} - \left[\frac{x}{2}\right], \sin x, |x-1|\right\}$, $h(x) = \ln x$, $k(x) = \min.\{x^2 - 3x + 2, 2 + \sin x\}$

(**Note :** $[y]$ denotes greatest integer function less than or equal to y]

144. Area enclosed by the curves $y = f(x)$, $y = g(x)$ and the y-axis, is

(1) 1 (2) 2

(3) 3 (4) 4

145. Area bounded by the curve $y = f(x)$, $y = h(x)$ and the y-axis, is

(1) e (2) e^2

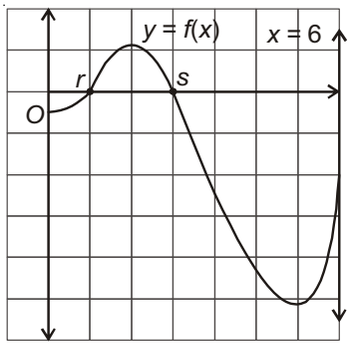
(3) $\frac{3e}{2}$ (4) $e^2 - 1$

146. Number of solutions of the equation $f(x) = k(x)$ in $[0, 2\pi]$, is

(1) 2 (2) 3

(3) 4 (4) 5

147. The graph of $y = f(x)$ is shown with roots r and s ($r < s$). Area bounded by the graph of $f(x)$, x-axis, $x = 0$ and $x = 6$ over the intervals $[0, r]$ and $[r, s]$ and $[s, 6]$ are $\frac{2}{5}$, 2 and 12 respectively. If $m = \int_s^0 f(x)dx$, $p = \int_r^6 f(x)dx$, $q = \left|\int_0^6 f(x)dx\right|$, $n = \int_0^6 |f(x)|dx$, then



(1) $p < m$ (2) $m < 1$

(3) $q < 11$ (4) $n > 11$

148. Let $f(x)$ be a polynomial of degree 3. If the curve $y = f(x)$ has relative extrema at $x = \frac{\pm 2}{\sqrt{3}}$ and passes through (0, 0) and (1, –2) dividing the circle $x^2 + y^2 = 4$ in two parts, then the area bounded by $x^2 + y^2 = 4$ and $y \geq f(x)$ is $\frac{k\pi}{2}$. Find the value of k.

149. Let $f(x) = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$, $g(x) = \cot^{-1}\left(\frac{2x}{x^2-1}\right)$ where $x \in (-1, 1)$. If area bounded by the curves $y = f(x) + g(x)$ and $y = \pi x^2$ is A, then find the value of $[A]$.

(**Note :** $[K]$ denotes greatest integer less than or equal to K]

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